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WORK INTEGRATED LEARNING PROGRAMMES

XR Simulation Learning Platform

Functional Requirement Document

Consolidated Updated Draft — Version 5.0

Detailed Requirements and Traceability Reference

Based on Platform FRD_v4, PM-Confirmed Clarifications and Review 006

Status	**DRAFT FOR REVIEW — NOT YET APPROVED**
Date	19 August 2026

The original Platform FRD_v4 remains unchanged. This consolidated draft preserves the original requirement IDs, records approved PM clarifications and additions, and identifies pending decisions for stakeholder review.

The v5 Document Set

You are reading the **Detailed Master**. Four documents make up the v5 requirement set. Each answers a different question, and each says so on its first page so that no one reads the wrong one.

Table 1 — The four documents and what each is for

Document	Purpose	What is in it
Core FRD Platform_FRD_v5_ CORE_DRAFT_FOR_ REVIEW	**Primary review document. Read this to understand what is being built.**	30 pages. Scope, architecture, roles, capabilities, key workflows, the stakeholder decisions, the changes from v4, high-priority open questions and the sign-off table.
Team Lead Review Summary Platform_FRD_v5_ TEAM_LEAD_REVIEW_ SUMMARY	**Quick discussion document. Read this to understand what changed and what needs decisions.**	11 pages, mostly diagrams. What the project is, what the earlier reading missed, what the stakeholder confirmed, and what needs a decision next.
Detailed Master Platform_FRD_v5_ DRAFT_FOR_REVIEW	**Complete requirement, traceability and audit reference. Use this when exact requirement provenance or detail is needed.**	147 pages — long by design. All 131 original requirements with faithful wording, the traceability matrix, every acceptance criterion, the complete open-question register, the dependency register and the full decision records. Not intended to be read front to back.
Original FRD v4 source/ Platform FRD_v4.pdf	**Historical issued source. Remains unchanged.**	The issued baseline. Nothing in the v5 work edits it. Where v5 departs from it, the departure is recorded as a deviation or an extension rather than applied to the original.

One rule that holds across all four

- The original Platform FRD_v4 is never edited. Every departure from it is recorded as a deviation, an extension, or a gap — with its source named — rather than absorbed into the requirement text.
- The compact editions do not narrow scope. Where they summarise, the full detail remains in the Detailed Master and is cross-referenced.

Document Control

Table 2 — Document control

Field	Value
Document title	XR Simulation Learning Platform
Document type	Functional Requirement Document (FRD)
Version	5.0 — Consolidated Updated Draft, revision 2 (accuracy-hardened)
Edition	**Detailed Requirements and Traceability Reference** — the complete requirement, traceability and audit reference. For review, read the Core FRD instead; see “The v5 Document Set” above.
Date	19 August 2026
Status	DRAFT FOR REVIEW — NOT YET APPROVED
Source document	source/Platform FRD_v4.pdf — 17 pages, 12 sections, 131 numbered requirements. Unchanged by this draft.
Supporting sources	PM-confirmed baseline of 18 August 2026 (PM-01 – PM-12); PM closure decisions of 19 August 2026 (PM-13 – PM-15); Review 006 full-platform reframe, documents 01 – 13; Review 005 material where compatible.
Prepared for	Project Manager and Team Lead, XR Simulation Learning Platform programme
Prepared by	[Prepared-by name and role to be completed before circulation]
Approval status	Not approved. This draft is issued for stakeholder review only.
Distribution and use	Internal programme distribution for review. This draft does not supersede Platform FRD_v4 and must not be quoted as an approved requirement baseline until the sign-off table below is completed.

Preservation notice

The original Platform FRD_v4 remains unchanged. This consolidated draft preserves the original requirement IDs, records approved PM clarifications and additions, and identifies pending decisions for stakeholder review.

Revision history

Table 3 — Revision history

Version	Date	Status	Summary of change	Author
v4	As issued	Baseline	Original Platform FRD — 12 sections, 131 numbered requirements, three phases. Held unchanged in source/Platform FRD_v4.pdf.	Original authors
v5 Draft rev 2	19 August 2026	Draft for review	Accuracy hardening and compact review editions. Seven overclaims corrected — direct course purchase, assessment hints, answer-key visibility, the authentication and role-routing model, the closure of the role set, the description of institutional SSO, and three inherited design positions. Three new open questions recorded (QP-27, QP-28, QX-10) and one reclassified (QP-20). Two classification labels added (NOT-SPECIFIED, DESIGN-PENDING) plus OPTIONAL. Core FRD and Team Lead Review Summary editions created. No requirement removed and no scope changed.	[To be completed]
v5 Draft	19 August 2026	Draft for review	Full-platform consolidation based on PM clarification and Review 006. Records PM-01 – PM-15, the seven-role model, the two-plane architecture, entitlement-aware XR delivery, the configuration-driven assessment model, the confirmed learning progression, the SSO position, the incremental reporting strategy and the certification deferral. Adds architecture, end-to-end workflows, acceptance criteria, and open-question and dependency registers. All original requirement IDs preserved.	[To be completed]

Reviewers

Table 4 — Reviewers

Role	Name	Review focus	Status
Project Manager	[To be completed]	Scope, phasing, PM decision register, open product questions	Pending
Team Lead	[To be completed]	Architecture, workflows, dependency register, feasibility	Pending
Academic stakeholder	[To be completed]	Content governance, academic validation criteria, assessment model	Pending
Engineering lead	[To be completed]	Technical-pending items, integration boundaries, non-functional requirements	Pending
Design lead	[To be completed]	Role model, UI/UX requirements, acceptance criteria, accessibility	Pending

Approval and sign-off

This draft is not approved. Approval is recorded only when the table below is completed and signed. Until then, Platform FRD_v4 remains the issued baseline and this document is a review draft.

Table 5 — Approval and sign-off

Approver	Role	Signature	Date	Decision
[To be completed]	Project Manager			Approve / Approve with changes / Reject
[To be completed]	Team Lead			Approve / Approve with changes / Reject
[To be completed]	Academic stakeholder			Approve / Approve with changes / Reject
[To be completed]	Engineering lead			Approve / Approve with changes / Reject

Table of Contents

The v5 Document Set . . . 2

Document Control . . . 3

Revision history . . . 3

Reviewers . . . 4

Approval and sign-off . . . 4

Table of Contents . . . 5

List of Diagrams . . . 12

List of Tables . . . 12

Glossary . . . 17

1. Executive Summary . . . 18

1.1 What the platform is . . . 18

1.2 Why this draft was required . . . 18

1.3 The complete Phase-1 role model . . . 18

1.4 B2B and B2C . . . 19

1.5 XR delivery . . . 19

1.6 Assessment configuration . . . 19

1.7 SSO, reporting and certification . . . 19

1.8 What remains unresolved . . . 19

2. How to Read This Document . . . 20

2.1 Authority order . . . 20

2.2 Change classification system . . . 20

2.3 Requirement identifiers . . . 21

2.5 Accuracy corrections in this revision . . . 21

2.4 Phase notation . . . 22

3. Stakeholder Clarification and Decision Register . . . 23

3.1 PM-01 — Current UI/UX and requirement scope . . . 24

3.2 PM-02 — Required design process . . . 24

3.3 PM-03 — Platform model . . . 24

3.4 PM-04 — B2B and B2C . . . 25

3.5 PM-05 — Web portal to XR device delivery . . . 25

3.6 PM-06 — XR simulation availability . . . 26
3.7 PM-07 — Vendor and Academic Reviewer . . . 26
3.8 PM-08 — Assessment configuration . . . 27
3.9 PM-09 — Learning progression . . . 27
3.10 PM-10 — Institutional SSO . . . 27
3.11 PM-11 — Reporting and analytics strategy . . . 28
3.12 PM-12 — Certification . . . 28
3.13 PM-13 — Platform and tenancy model . . . 29
3.14 PM-14 — Content rejection and resubmission . . . 29
3.15 PM-15 — Assessment ownership for B2B and B2C . . . 29

4. Product Scope and Phasing . . . 31
4.1 How the phase figures are derived . . . 31
4.2 Scope matrix . . . 32
4.3 Phase distribution by section . . . 32

5. XR Simulation Learning Platform Architecture . . . 33
5.1 System context . . . 33
5.2 Role and workspace architecture . . . 34
5.3 Two-plane model . . . 35
5.4 Shared platform capabilities . . . 37
5.5 Logical application and service architecture . . . 38
5.6 Security and data boundaries . . . 39

6. Confirmed Role Model . . . 41
6.1 Student / Learner . . . 42
6.1.1 Primary responsibilities . . . 42
6.1.2 Permissions and allowed actions . . . 42
6.1.3 Prohibited actions . . . 42
6.1.4 Acceptance criteria . . . 42
6.1.5 Unresolved details . . . 43
6.2 Faculty . . . 43
6.2.1 Primary responsibilities . . . 43
6.2.2 Permissions and allowed actions . . . 43
6.2.3 Prohibited actions . . . 43
6.2.4 Acceptance criteria . . . 43
6.2.5 Unresolved details . . . 44
6.3 Institute Admin . . . 44
6.3.1 Primary responsibilities . . . 44
6.3.2 Permissions and allowed actions . . . 44

6.3.3 Prohibited actions . . . 44
6.3.4 Acceptance criteria . . . 45
6.3.5 Unresolved details . . . 45
6.4 Vendor . . . 45
6.4.1 Primary responsibilities . . . 45
6.4.2 Permissions and allowed actions . . . 45
6.4.3 Prohibited actions . . . 46
6.4.4 Acceptance criteria . . . 46
6.4.5 Unresolved details . . . 46
6.5 Academic Reviewer . . . 46
6.5.1 Primary responsibilities . . . 46
6.5.2 Permissions and allowed actions . . . 46
6.5.3 Prohibited actions . . . 47
6.5.4 Acceptance criteria . . . 47
6.5.5 Unresolved details . . . 47
6.6 Content Manager . . . 47
6.6.1 Primary responsibilities . . . 47
6.6.2 Permissions and allowed actions . . . 48
6.6.3 Prohibited actions . . . 48
6.6.4 Acceptance criteria . . . 48
6.6.5 Unresolved details . . . 48
6.7 Super Admin . . . 48
6.7.1 Primary responsibilities . . . 48
6.7.2 Permissions and allowed actions . . . 49
6.7.3 Prohibited actions . . . 49
6.7.4 Acceptance criteria . . . 49
6.7.5 Unresolved details . . . 49

7. Functional Requirements . . . 50
7.1 Platform Overview . . . 51
7.2 XR Content Delivery Architecture . . . 54
7.3 Content Organization and Learning Structure . . . 57
7.4 User Management and Onboarding . . . 60
7.5 Learning Experience Features . . . 61
7.6 Licensing and Monetization . . . 64
7.7 Enterprise Integrations . . . 65
7.8 Administration and Role Management . . . 67
7.9 Payment and Billing . . . 70
7.10 Security and Compliance . . . 71

7.11 Analytics and Reporting . . . 72
7.12 Platform Infrastructure and Hosting . . . 73

8. New and Clarified Requirements . . . 74

8.1 Added requirements . . . 74
V5-ADD-001 — Seven-role platform . . . 74
V5-ADD-002 — Vendor workspace . . . 74
V5-ADD-003 — Academic Reviewer workspace . . . 75
V5-ADD-004 — Two-plane platform model . . . 75
V5-ADD-005 — Content return with comments and reason . . . 76
V5-ADD-006 — Vendor correction and resubmission . . . 76
V5-ADD-007 — Review history and status traceability . . . 76
V5-ADD-008 — Entitlement-aware XR device delivery and access . . . 77
V5-ADD-009 — Delivery mechanism neutrality . . . 77
V5-ADD-010 — Platform architecture before detailed design . . . 78
8.2 Clarified requirements . . . 78
V5-CLR-001 — Existing XR simulations are available . . . 78
V5-CLR-002 — Configuration-driven assessment model . . . 79
V5-CLR-003 — Assessment consumption through flexible APIs . . . 79
V5-CLR-004 — Assessment ownership for B2B and B2C . . . 80
V5-CLR-005 — Learners do not author or configure assessments . . . 80
V5-CLR-006 — Approved learning progression . . . 80
V5-CLR-007 — Institutional SSO is required . . . 81
V5-CLR-008 — Incremental reporting strategy . . . 81
V5-CLR-009 — Certification deferred to Phase 2 . . . 81
V5-CLR-010 — B2B and B2C are both Phase 1 . . . 82
V5-CLR-011 — One portal, not one runtime . . . 82
V5-CLR-012 — Sample assessment configuration . . . 83

9. End-to-End Workflows . . . 84

9.1 FLOW 1 — Content to production . . . 85
9.2 FLOW 2 — B2B institutional learning . . . 87
9.3 FLOW 3 — B2C purchase to learning . . . 89
9.4 FLOW 4 — Assessment configuration to result . . . 90
9.5 FLOW 5 — XR entitlement and device delivery . . . 91
9.6 FLOW 6 — Institutional SSO . . . 93
9.7 FLOW 7 — ERP synchronisation . . . 94
9.8 FLOW 8 — Device binding and replacement . . . 96
9.9 FLOW 9 — Licence cap exhaustion . . . 96
9.10 FLOW 10 — Version release to live content . . . 96

10. XR Delivery Requirements . . . 98
10.1 The four runtime targets . . . 98
10.2 Supporting delivery requirements the FRD does define . . . 98
10.3 The entitled-experiment lifecycle . . . 99
10.4 Required user and system states . . . 99
10.5 Design rules for XR delivery . . . 100
10.6 Recorded gaps . . . 101

11. Assessment Requirements . . . 102
11.1 Why this is a correction of interpretation, not of requirement . . . 102
11.2 Ownership . . . 102
11.3 Question types . . . 102
11.4 Question sourcing . . . 103
11.5 Configuration fields . . . 103
11.5.1 Fields clearly implied by Platform FRD_v4 . . . 103
11.5.2 Fields required by the model, with no FRD value . . . 103
11.5.3 Questions requiring product clarification before they become fields . . . 104
11.6 Result, visibility and progress . . . 105
11.7 Consequences for the interface . . . 105
11.8 The API contract . . . 105

12. Learning Requirements . . . 106
12.1 The content hierarchy . . . 106
12.2 Sequencing and progression . . . 106
12.3 Learning resources . . . 106
12.4 Access, entitlement and error states . . . 107
12.5 Resuming learning . . . 107

13. Commerce, Licensing and Entitlements . . . 108
13.1 B2C individual commerce . . . 108
13.2 B2B institutional licensing . . . 108
13.3 Entitlement validation . . . 109

14. SSO, ERP and Integration Requirements . . . 110
14.1 Institutional single sign-on . . . 110
14.2 ERP integration . . . 110
14.3 Integration boundaries . . . 111

15. Reporting and Analytics . . . 112

15.1 The operational / analytical test . . . 112
15.2 Phase-1 recording requirements . . . 112
15.3 Phase-1 operational status by role . . . 113
15.4 What is deliberately not specified . . . 113

16. Non-Functional Requirements . . . 114

17. UI/UX Requirements . . . 116
17.1 The approved design process . . . 116
17.2 Navigation and shell . . . 116
17.3 Required interface states . . . 117
17.4 Accessibility . . . 117
17.5 Relationship to the screen inventory . . . 117

18. Acceptance Criteria . . . 118
18.1 Authentication, roles and tenancy . . . 118
18.2 Content governance . . . 119
18.3 Entitlement, licensing and payment . . . 119
18.4 XR delivery . . . 120
18.5 Assessment . . . 120
18.6 Learning and progression . . . 121
18.7 Integration and operations . . . 121

19. Questions and Decision Register . . . 122
19.1 Resolved stakeholder questions . . . 122
19.2 Open questions — summary . . . 123
19.3 High-priority open questions . . . 123

20. Dependency Register . . . 125

21. Summary of Changes from Platform FRD_v4 . . . 128
21.1 Deviations and extensions on record . . . 131

Appendix A. Traceability Matrix . . . 132

Appendix B. Complete Open-Question Register . . . 140
B.1 Product questions . . . 141
B.2 Technical questions . . . 142
B.3 Data / Integration questions . . . 143
B.4 Design questions . . . 144
B.5 Future-phase items . . . 145

Appendix C. Verification of This Draft 147

List of Diagrams

Figure 1 — System context — actors, external systems and integration boundaries . . . 33
Figure 2 — Role and workspace architecture — one portal, seven workspaces . . . 35
Figure 3 — Two-plane platform architecture confirmed by PM-13 . . . 36
Figure 4 — Shared capability architecture — fifteen logical domains . . . 37
Figure 5 — Logical application and service architecture — technology-neutral . . . 38
Figure 6 — FLOW 1 — content to production, with PM-14 return loops . . . 86
Figure 7 — FLOW 2 — B2B institutional learning journey . . . 88
Figure 8 — FLOW 3 — B2C purchase-to-learning journey . . . 89
Figure 9 — FLOW 4 — assessment configuration to result . . . 90
Figure 10 — FLOW 5 — XR entitlement and device-delivery lifecycle . . . 92
Figure 11 — FLOW 6 — institutional SSO, tenant and role resolution . . . 93
Figure 12 — FLOW 7 — ERP synchronisation . . . 95

List of Tables

Table 1 — The four documents and what each is for . . . 2
Table 2 — Document control . . . 3
Table 3 — Revision history . . . 3
Table 4 — Reviewers . . . 4
Table 5 — Approval and sign-off . . . 4
Table 6 — Glossary of terms . . . 17
Table 7 — Authority order applied throughout this document . . . 20
Table 8 — Change classification labels . . . 20
Table 9 — Accuracy corrections applied . . . 22
Table 10 — Summary of PM-confirmed decisions . . . 23
Table 11 — PM-01 decision record . . . 24
Table 12 — PM-02 decision record . . . 24
Table 13 — PM-03 decision record . . . 25
Table 14 — PM-04 decision record . . . 25
Table 15 — PM-05 decision record . . . 26
Table 16 — PM-06 decision record . . . 26
Table 17 — PM-07 decision record . . . 26
Table 18 — PM-08 decision record . . . 27
Table 19 — PM-09 decision record . . . 27
Table 20 — PM-10 decision record . . . 28
Table 21 — PM-11 decision record . . . 28

Table 22 — PM-12 decision record . . . 28
Table 23 — PM-13 decision record . . . 29
Table 24 — PM-14 decision record . . . 29
Table 25 — PM-15 decision record . . . 30
Table 26 — Derivation of the Phase-1 working set . . . 31
Table 27 — Phase scope matrix . . . 32
Table 28 — Original FRD requirement distribution by section and phase . . . 32
Table 29 — External actors and systems . . . 33
Table 30 — The two planes and the personal context . . . 36
Table 31 — The fifteen capability domains . . . 37
Table 32 — Logical components . . . 38
Table 33 — Security and data boundary requirements . . . 39
Table 34 — The seven confirmed roles . . . 41
Table 35 — Student / Learner — role definition . . . 42
Table 36 — Faculty — role definition . . . 43
Table 37 — Institute Admin — role definition . . . 44
Table 38 — Vendor — role definition . . . 45
Table 39 — Academic Reviewer — role definition . . . 46
Table 40 — Content Manager — role definition . . . 47
Table 41 — Super Admin — role definition . . . 48
Table 42 — How to read the requirement tables in this chapter . . . 50
Table 43 — Section 1 requirements — Platform Overview (24 requirements, 18 in Phase 1) . . . 51
Table 44 — Section 2 requirements — XR Content Delivery Architecture (17 requirements, 11 in Phase 1) . . . 54
Table 45 — Section 3 requirements — Content Organization and Learning Structure (11 requirements, 6 in Phase 1) 57
Table 46 — Section 4 requirements — User Management and Onboarding (9 requirements, 5 in Phase 1) . . . 60
Table 47 — Section 5 requirements — Learning Experience Features (11 requirements, 7 in Phase 1) . . . 61
Table 48 — Section 6 requirements — Licensing and Monetization (10 requirements, 4 in Phase 1) . . . 64
Table 49 — Section 7 requirements — Enterprise Integrations (9 requirements, 5 in Phase 1) . . . 65
Table 50 — Section 8 requirements — Administration and Role Management (9 requirements, 5 in Phase 1) . . . 67
Table 51 — Section 9 requirements — Payment and Billing (8 requirements, 4 in Phase 1) . . . 70
Table 52 — Section 10 requirements — Security and Compliance (8 requirements, 6 in Phase 1) . . . 71
Table 53 — Section 11 requirements — Analytics and Reporting (8 requirements, 4 in Phase 1) . . . 72
Table 54 — Section 12 requirements — Platform Infrastructure and Hosting (7 requirements, 3 in Phase 1) . . . 73
Table 55 — V5-ADD-001 record . . . 74
Table 56 — V5-ADD-002 record . . . 74
Table 57 — V5-ADD-003 record . . . 75
Table 58 — V5-ADD-004 record . . . 75

Table 59 — V5-ADD-005 record . . . 76
Table 60 — V5-ADD-006 record . . . 76
Table 61 — V5-ADD-007 record . . . 77
Table 62 — V5-ADD-008 record . . . 77
Table 63 — V5-ADD-009 record . . . 77
Table 64 — V5-ADD-010 record . . . 78
Table 65 — V5-CLR-001 record . . . 79
Table 66 — V5-CLR-002 record . . . 79
Table 67 — V5-CLR-003 record . . . 79
Table 68 — V5-CLR-004 record . . . 80
Table 69 — V5-CLR-005 record . . . 80
Table 70 — V5-CLR-006 record . . . 81
Table 71 — V5-CLR-007 record . . . 81
Table 72 — V5-CLR-008 record . . . 81
Table 73 — V5-CLR-009 record . . . 82
Table 74 — V5-CLR-010 record . . . 82
Table 75 — V5-CLR-011 record . . . 83
Table 76 — V5-CLR-012 record . . . 83
Table 77 — Workflow index . . . 84
Table 78 — FLOW 1 steps . . . 87
Table 79 — FLOW 1 recorded gaps . . . 87
Table 80 — FLOW 2 steps . . . 88
Table 81 — FLOW 3 steps . . . 89
Table 82 — FLOW 4 steps . . . 90
Table 83 — FLOW 5 steps . . . 92
Table 84 — The two concepts recorded, neither selected . . . 93
Table 85 — FLOW 6 steps . . . 94
Table 86 — FLOW 7 steps . . . 95
Table 87 — FLOW 8 steps . . . 96
Table 88 — FLOW 9 steps . . . 96
Table 89 — FLOW 10 steps . . . 96
Table 90 — Runtime targets and their application architecture . . . 98
Table 91 — Delivery-related Phase-1 requirements . . . 98
Table 92 — Lifecycle stages . . . 99
Table 93 — Required XR delivery states . . . 99
Table 94 — XR delivery gaps carried forward . . . 101
Table 95 — Assessment ownership . . . 102

Table 96 — Supported question types . . . 102
Table 97 — Question sourcing requirements . . . 103
Table 98 — Configuration fields with explicit FRD basis . . . 103
Table 99 — Configuration fields requiring a value decision . . . 103
Table 100 — Result and progress requirements . . . 105
Table 101 — Content structure . . . 106
Table 102 — Sequencing requirements . . . 106
Table 103 — Learning access states . . . 107
Table 104 — B2C commerce requirements . . . 108
Table 105 — B2B licensing requirements . . . 108
Table 106 — Where entitlement is enforced . . . 109
Table 107 — SSO requirements . . . 110
Table 108 — ERP requirements . . . 110
Table 109 — Integration boundaries and their status . . . 111
Table 110 — What qualifies as Phase-1 reporting . . . 112
Table 111 — What Phase 1 records . . . 112
Table 112 — Permitted Phase-1 operational status . . . 113
Table 113 — Non-functional requirement categories . . . 114
Table 114 — Navigation requirements . . . 116
Table 115 — Required states across the platform . . . 117
Table 116 — Authentication, roles and tenancy . . . 118
Table 117 — Content governance . . . 119
Table 118 — Entitlement, licensing and payment . . . 119
Table 119 — XR delivery . . . 120
Table 120 — Assessment . . . 120
Table 121 — Learning and progression . . . 121
Table 122 — Integration and operations . . . 121
Table 123 — Resolved stakeholder questions . . . 122
Table 124 — Open questions by class and priority . . . 123
Table 125 — High-priority open questions . . . 123
Table 126 — Dependency register . . . 126
Table 127 — Summary of changes from Platform FRD_v4 . . . 129
Table 128 — Deviation, extension and conflict register . . . 131
Table 129 — Traceability matrix — all 131 Platform FRD_v4 requirements . . . 133
Table 130 — Product open questions . . . 141
Table 131 — Technical open questions . . . 142
Table 132 — Data / Integration open questions . . . 143

Table 133 — Design open questions 144
Table 134 — Future-phase items 145
Table 135 — Verification results 147

Glossary

Terms are defined as they are used in this document. Where a term originates in Platform FRD_v4, the originating requirement is cited.

Table 6 — Glossary of terms

Term	Definition
AR	Augmented reality. Simulations running on Android and iOS mobile devices (FRD 1.2.4).
B2B	The institutional model, in which an organisation holds the licence and its students and faculty access the platform under a shared institutional subscription (FRD 1.1.2).
B2C	The individual model, in which a learner independently purchases modules or bundles and accesses them through a personal account (FRD 1.1.3).
Bundle	A purchasable group of simulation modules (FRD 6.1.2).
Capability domain	A logical grouping of platform behaviour defined once and consumed by many roles in different scopes. Fifteen domains are used in this document.
Central content plane	The platform-owned area where module builds, assets, metadata, validation, approval and publication live. Home of Vendor, Academic Reviewer, Content Manager and Super Admin (PM-13).
Course	A collection of modules grouped around an engineering topic or learning objective (FRD 3.2.1). Phase-1 purchase requirements name modules and bundles; direct course purchase is not specified.
Entitlement	The record that a given account is permitted to access a given module or package, whether created by institutional licensing or by individual purchase.
ERP	The institution's enterprise resource planning system, the authoritative source of student and academic structure data (FRD 7.1.1).
Gate	One of the four authority points in the content governance chain: submission, academic validation, content readiness approval and production publication (FRD 1.4.1–1.4.4).
Module	The smallest learning unit, containing a single XR simulation experience (FRD 3.1.1).
MR	Mixed reality. Simulations running on MR hardware through a sideloaded application (FRD 1.2.3, 2.1.2).
Operational status	A view answering “what is the state of the thing I operate right now?”, permitted in Phase 1 by PM-11, as distinct from analytics.
PENDING DECISION	A product or business question this document deliberately leaves open because no approved answer exists.
Sample configuration	A value shown only as an illustrative example. It is not a platform constant, a default or a recommendation.
Stream	An academic stream or department within an institution, used to map licensed packages to students (FRD 6.2.1, 7.2.3).
SSO	Single sign-on. Institutional authentication through an identity provider (FRD 4.2.2, PM-10).
Tenant plane	One isolated institutional environment. Home of Institute Admin, Faculty and Student/Learner (FRD 1.3.1, PM-13).
TECHNICAL DETAIL TO BE DEFINED	An engineering or integration question this document deliberately leaves open because no approved answer exists.
VR	Virtual reality. Simulations deployed through VR headsets (FRD 1.2.2, 2.1.1).
Workspace	The role-specific area of the single platform portal that a signed-in identity resolves into. Seven workspaces exist.
XR	Collective term for the web, VR, MR and AR simulation formats the platform supports (FRD 1.2.1–1.2.4).

1. Executive Summary

1.1 What the platform is

The XR Simulation Learning Platform is a multi-tenant SaaS learning management system that distributes XR-based engineering simulation modules to institutions and to individual learners (FRD 1.1.1). It is realised as one central web portal with role-based access and workspaces (PM-03). The portal is the control, entitlement, governance and administration plane; the simulations themselves run in a browser for the web format and on dedicated devices for the VR, MR and AR formats (FRD 1.2.1 – 1.2.4).

Content originates with authorised vendors, is academically validated, is approved for readiness and is published centrally (FRD 1.4.1 – 1.4.5). Institutions license packages mapped to academic streams; individual learners purchase modules and bundles. Both routes produce an entitlement, and the entitlement is what a learner actually consumes.

1.2 Why this draft was required

Platform FRD_v4 defines the product. The design work that followed it was executed under a narrower reading in which the detailed Phase-1 scope was taken to be the Student/Learner desktop experience. A direct clarification from the Project Manager on 18 August 2026 established that this reading was too narrow: **all Phase-1 requirements and all Phase-1 roles are in scope** (PM-01).

That single clarification changed the shape of the work from one workspace to seven. A further set of confirmations on 19 August 2026 settled the platform's tenancy model, the content rejection and resubmission path, and assessment ownership (PM-13, PM-14, PM-15). This draft consolidates the original requirements, the fifteen confirmed decisions, and the reconciled architecture into one reviewable document.

What changed from the earlier interpretation

- Requirement breadth is unchanged. The FRD still contains 78 Phase-1 requirements.
- Role depth changed from one role to seven. Six of the seven workspaces had never been designed.
- The earlier learner-facing work is not withdrawn. It is reclassified as one workspace within the platform, and most of it is reusable.

1.3 The complete Phase-1 role model

Seven roles are confirmed. Five are defined by the FRD's own role hierarchy (FRD 8.1.1 – 8.1.5). Two — Vendor and Academic Reviewer — appear in the FRD only as actors in the content governance workflow, with no role definition and no permission set; PM-07 promotes both to first-class roles with platform logins and workspaces. This is recorded as an extension of the FRD, not as something the FRD already said.

- **Central content and governance plane:** Vendor, Academic Reviewer, Content Manager, Super Admin.
- **Institution / tenant plane:** Institute Admin, Faculty, Student / Learner.
- **Spanning both planes:** Super Admin alone, through the controlled cross-tenant governance access the FRD grants at 1.3.5.

The B2C individual learner is a business context around the Learner role, not a separate role. **Seven formal roles are currently approved for Phase 1. Any additional Phase-1 formal role requires explicit stakeholder approval**, and FRD 8.3 — the FRD's own mechanism for defining custom roles — is Phase 3, so there is no Phase-1 mechanism to add one.

1.4 B2B and B2C

Both models are required and neither is sequenced ahead of the other (PM-04). The FRD places institutional and individual environments in Phase 1 with no phasing between them (FRD 1.1.2, 1.1.3). The two routes share one learning core and diverge only where the FRD makes them diverge: origin of the account, authentication path, origin of the entitlement, data environment, and whether an oversight role exists.

1.5 XR delivery

An entitled experiment must be able to become available on the device that runs it (PM-05). This is already an FRD requirement in substance: modules download only after licence validation (FRD 2.2.2) and require licence validation before execution (FRD 10.1.2). What was missing was any designed delivery experience.

The Project Manager described two implementation concepts and **selected neither**: direct delivery of the experiment, application or package to the device, or a centralized device application that exposes only the experiments the platform reports as entitled. Synchronisation, deep-link handoff and other appropriate solutions remain equally open. This document therefore specifies the lifecycle and the states a learner must be able to perceive, in language that survives any of those outcomes, and selects no mechanism.

1.6 Assessment configuration

Assessment behaviour is configured through the portal and consumed by the relevant Web or XR application through flexible APIs (PM-08). The FRD always framed assessment as configurable — it states no question count, no duration, no score weighting, no pass mark and no attempt limit anywhere. Any fixed values that appeared in earlier prototype work are recorded here as **sample configuration** and never as platform constants.

Ownership is settled: for both B2B and B2C, assessment setup and configuration belongs to Faculty or the authorized institution side, and the learner never authors or configures an assessment (PM-15).

1.7 SSO, reporting and certification

- **Institutional SSO is required** (PM-10). This resolves a contradiction inside the FRD, which places SSO configuration in Phase 1 and SSO authentication support in Phase 3. The provider, protocol and attribute mapping remain PENDING DECISION.
- **Reporting is incremental** (PM-11). Phase 1 delivers operational status — what is the state of the thing I operate right now — and the recording requirements the FRD places in Phase 1. Detailed metrics, KPI definitions and dashboards remain pending, consistent with the FRD's own phasing.
- **Certification is deferred to Phase 2** (PM-12). The FRD places certification program structure and completion criteria in Phase 1. This is recorded as a deviation from the FRD as written, not as a correction of it.

1.8 What remains unresolved

This draft records what is not yet decided rather than filling the gaps. The material open items are the XR delivery mechanism, the absence of any AR application architecture in the FRD, the assessment configuration API contract, the collision between device sign-in with LMS credentials and institutional SSO, ownership of descriptive-answer grading, how an institutional licence is transacted in Phase 1 before Phase-2 billing exists, which profile fields are user-editable rather than ERP-owned, and the Phase-1 origin of institutional role assignment. Each is carried in the open-question register with a priority and an owner.

Standing on the pending items

Four of these are pending by explicit stakeholder instruction rather than by omission: the XR delivery mechanism (PM-05), the SSO provider and protocol (PM-10), which institution supplies a pure-B2C assessment (PM-15), and the detailed reporting metrics (PM-11). They are not deficiencies in this document.

2. How to Read This Document

2.1 Authority order

Where sources disagree, this document resolves the disagreement in the following order and says which source it used.

Table 7 — Authority order applied throughout this document

Order	Source	How it is applied
1	Direct PM-confirmed decisions	PM-01 – PM-15. These take precedence over prior assumptions, recommendations and earlier design decisions.
2	Original FRD facts and requirement wording	Platform FRD_v4 requirement text is reproduced faithfully and never rewritten.
3	Review-006 reconciled architecture and mappings	The full-platform re-audit, role/permission matrix, capability map, workflows and screen inventory.
4	Review-005 decisions that remain compatible	Earlier learner-facing decisions are retained where the reframe does not supersede them.
5	Explicit assumptions and recommendations	Clearly labelled. Never presented as approved requirements.

Governance rules this document follows

- History is never silently rewritten.
- No PM-added rule is presented as something Platform FRD_v4 already contained.
- No original FRD requirement is removed. All 131 identifiers are present in this document.
- PM deviations and phase changes are recorded explicitly, with the original position shown alongside the new one.
- Unresolved technical and product choices are recorded as pending rather than answered.
- Recommendations made in the course of this work are not treated as approved requirements.

2.2 Change classification system

Every requirement and every material statement in this document carries one of the following classifications. Classifications are written as accessible text labels; no meaning depends on colour alone.

Table 8 — Change classification labels

Label	Meaning
UNCHANGED-FRD	The requirement stands as written in Platform FRD_v4. No stakeholder intervention changed its meaning, owner or phase.
PM-CLARIFIED	A PM statement changed the reading, the owning role or the surfaced scope of an existing FRD requirement. The requirement text itself is unchanged.
PM-ADDED	A requirement introduced by a PM decision that has no equivalent in Platform FRD_v4. Recorded as an addition, never presented as original FRD content.
PM-DEFERRED	A PM decision explicitly moved this requirement out of the current phase. The original phase is preserved and shown alongside the working phase.
FRD-INTERNAL-CONFLICT	Two statements in Platform FRD_v4 disagree with each other. The conflict is recorded, and any resolution is attributed to its source.
TECHNICAL-PENDING	An engineering, integration or architecture answer is required before this can be fully specified or designed.
PRODUCT-PENDING	A product, business or content-operations answer is required before this can be fully specified or designed.
FUTURE-PHASE	Placed in Phase 2 or Phase 3 by Platform FRD_v4 itself, and unchanged by this draft.

Label	Meaning
SAMPLE-CONFIGURATION	An illustrative value shown only as an example. Never a platform constant, a default or a recommendation.
NOT-SPECIFIED	Neither Platform FRD_v4 nor any stakeholder decision addresses this. It is neither required nor prohibited, and it is not included in the current design unless subsequently approved.
DESIGN-PENDING	A design decision that has not been taken. Any position shown is a reversible working design position, clearly labelled, and never an approved requirement.
OPTIONAL	A capability that may exist but that no source requires. It is carried so that it is not silently assumed, and it may be dropped without affecting any requirement.

2.3 Requirement identifiers

- **Original FRD identifiers are preserved exactly.** A requirement numbered 1.4.1 in Platform FRD_v4 is numbered 1.4.1 here, and its statement is reproduced faithfully.
- **Requirements added after the FRD use V5-ADD identifiers** — V5-ADD-001, V5-ADD-002 and so on. These are additions arising from PM decisions and are never renumbered into the FRD's own scheme.
- **Clarifications of existing FRD requirements use V5-CLR identifiers** — V5-CLR-001, V5-CLR-002 and so on. A clarification records how an existing requirement is now read; it does not replace the requirement.
- **Review-005 decision identifiers are not reused as requirement identifiers.** Where earlier design decisions are relevant they are described in words rather than cited as requirements.

2.5 Accuracy corrections in this revision

Seven claims in the first v5 draft asserted more than the sources support. They have been corrected in every edition of this document set. The corrections narrow what the document claims; none of them changes scope, and none removes a requirement.

Table 9 — Accuracy corrections applied

Topic	Previous wording	Corrected position	Why the original wording overclaimed
Direct course purchase	Stated as a permanent rule that a course is not a purchasable product.	Phase-1 purchase requirements explicitly define individual module and bundle purchasing. Direct course purchase is **not specified** and is not included in the current design unless subsequently approved. Classified NOT-SPECIFIED / PRODUCT-PENDING (`QP-28`).	No source prohibits course purchase. Absence of a requirement is not a prohibition.
Assessment hints	Hint behaviour listed among the values PM-08 confirms as configurable, and hint scoring treated as sample configuration.	Hints are classified **OPTIONAL / PRODUCT-PENDING**. No hint behaviour, hint scoring or hint limit is assumed in either direction (`QP-20`).	PM-08 confirms configurable assessment behaviour. It does not mention hints, and no FRD requirement defines them.
Answer-key visibility	A universal rule that the learner never sees the correct-answer key, presented as an approved requirement and as an API prohibition.	**Answer-key visibility and post-attempt feedback policy are PRODUCT-PENDING.** Learner-facing applications must receive only the assessment data permitted by the configured and approved feedback policy (`QP-27`).	Neither the FRD nor any stakeholder decision states a feedback policy. The rule was a design inference.
Authentication and role routing	A single authentication front door, a prohibition on pre-login role selection, and a workspace chooser presented as following from PM-03.	PM-03 confirms one central platform, role-based access and workspaces, and that institutional SSO is required. The sign-in model and the absence of a pre-login role choice are **DESIGN-PENDING** design positions (`QX-10`). A workspace chooser remains available only as a conditional pattern **if** multi-role identities are later approved (`QP-11`); none is mandated.	PM-03 does not specify a sign-in model, and no source addresses multi-role identities.
Role set	“No further role may be added.”	**Seven formal roles are currently approved for Phase 1. Any additional Phase-1 formal role requires explicit stakeholder approval.**	The original wording asserted a permanent closure that no source establishes.
Institutional SSO	Described in some flows as the “preferred” path, ordered first for institutional users.	**Institutional SSO is required and is available according to the institution's configured authentication model.** Provider, protocol, attribute mapping and fallback remain pending.	“Preferred” understated a confirmed requirement and implied a prominence decision no source makes.
Other inherited design positions	A cart-free checkout, standalone-module gating behaviour, and a caption-completion rule presented as requirements.	Each is relabelled **DESIGN-PENDING** and attributed to the reasoning behind it rather than to a requirement.	All three were legitimate design inferences from earlier work, but none is stated by a source.

The principle behind all seven

- A requirement document may record what a source says, and it may record a labelled design position. It must not present the second as the first.
- Absence of a requirement is not a prohibition. Where nothing is specified, the correct entry is NOT-SPECIFIED or PENDING — not a rule invented to fill the space.

2.4 Phase notation

Every requirement in Chapter 6 shows two phases. **FRD phase** is the phase Platform FRD_v4 assigns, and it never changes. **v5 working phase** is the phase this draft works to, which differs only where a PM decision or a recorded FRD internal conflict moves it. Where the two differ, the reason is stated in the same row.

3. Stakeholder Clarification and Decision Register

Fifteen decisions were confirmed directly by the Project Manager: PM-01 through PM-12 on 18 August 2026, and PM-13 through PM-15 on 19 August 2026. They are recorded here in full because several of them change the reading, the owner or the phase of requirements in Platform FRD_v4, and those changes must remain visible rather than being absorbed into the requirement text.

Table 10 — Summary of PM-confirmed decisions

ID	Decision	Relationship to Platform FRD_v4
PM-01	All Phase-1 requirements and all Phase-1 roles in scope	Alignment. Requirement breadth unchanged; role depth widened.
PM-02	Sitemap, permissions and workflows before detailed design	Process requirement. Not an FRD matter.
PM-03	One central portal with role-based workspaces	Alignment with FRD 1.1.1, with an explicit caveat that not every runtime is in the browser.
PM-04	Both B2B and B2C required; no sequencing	Alignment. FRD 1.1.2 and 1.1.3 are both Phase 1.
PM-05	Entitlement-aware XR delivery to the device	Alignment with FRD 2.2.2 and 10.1.2, plus a recorded gap: no AR application architecture exists.
PM-06	Required XR simulations are available	Alignment with FRD 1.4.1 and 1.1.4.
PM-07	Vendor and Academic Reviewer get logins and workspaces	EXTENSION. FRD 8.1 defines five roles; these two are actors only.
PM-08	Assessment is portal-configured and API-consumed	Alignment. The FRD states no assessment values anywhere.
PM-09	Approved learning progression	Alignment with FRD 3.1.1, 3.1.2 and 3.2.2.
PM-10	Institutional SSO required	RESOLVES an FRD internal conflict between 4.2.2 (Phase 1) and 7.4.1 (Phase 3).
PM-11	Core operations first; analytics incrementally	Alignment with the FRD's own phasing of dashboards into Phase 2 and Phase 3.
PM-12	Certification is Phase 2	DEVIATION. FRD 3.3.1 and 3.3.2 are Phase 1 as written.
PM-13	Two-plane content and tenant model	Fills an FRD silence. No FRD text is contradicted.
PM-14	Rejection, resubmission and review history	EXTENSION. FRD 1.4.1–1.4.4 describe a forward-only chain.
PM-15	Faculty / institution owns assessment configuration	Alignment with FRD 5.1.1, 5.1.2, 5.3.2 and 8.1.4, extended to the B2C case the FRD never addressed.

3.1 PM-01 — Current UI/UX and requirement scope

Confirmed statement

All Phase-1 requirements and all Phase-1 roles are in the current scope. The Student/Learner work remains reusable as one workspace within the complete platform.

Table 11 — PM-01 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	All twelve FRD sections; Section 8 role definitions in particular
Affected roles	All seven
Affected capability or workflow	All fifteen capability domains
Effect on the previous position	Supersedes the earlier interpretation that the detailed scope was the Student/Learner desktop LMS. Requirement breadth is unchanged at 78 Phase-1 requirements; role depth changes from one role to seven.
Implementation details still pending	None. The consequences are design and build work, not further decisions.

3.2 PM-02 — Required design process

Confirmed statement

Create the complete platform sitemap, role permissions and all-role workflows before taking the most important end-to-end flows into detailed Figma.

Table 12 — PM-02 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	Process requirement; not a functional section
Affected roles	All seven
Affected capability or workflow	Design process
Effect on the previous position	The previously planned detailed Figma transfer is placed on hold rather than cancelled. Nothing had been built in Figma, so no rework was created.
Implementation details still pending	Which end-to-end flows are taken into detailed design first remains a PM selection. A ranked recommendation exists but is not approved.

3.3 PM-03 — Platform model

Confirmed statement

Treat the solution as one central XR Simulation Learning Platform / web portal with role-based access and workspaces. This must not be read as saying every XR runtime executes inside the browser.

Table 13 — PM-03 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	1.1.1, 1.2.1–1.2.4, 2.1.1, 2.1.2
Affected roles	All seven
Affected capability or workflow	TEN, CFG, IAM, XRD
Effect on the previous position	Confirms one central platform with role-based access and workspaces, and confirms that the portal is the control, entitlement and administration plane rather than the execution plane for every simulation format. **PM-03 does not specify the sign-in model.** A single shared shell, a single sign-in entry point and one design system are design positions taken in this draft, not stakeholder-confirmed requirements.
Implementation details still pending	None at product level.

3.4 PM-04 — B2B and B2C

Confirmed statement

Both B2B institutional and B2C individual models are required. B2B-first / B2C-second is not a confirmed release sequence. Both must be represented in the platform architecture.

Table 14 — PM-04 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	1.1.2, 1.1.3, 1.3.1, 1.3.2, 6.1.1, 6.1.2, 6.2.1, 6.2.2, 9.1.x
Affected roles	Learner, Faculty, Institute Admin, Super Admin
Affected capability or workflow	TEN, ENT, COM
Effect on the previous position	Any B2B-first sequencing returns to pending. Both markets are represented throughout the architecture, workflows and screen planning.
Implementation details still pending	Release sequencing, if any, is a separate decision that has not been taken.

3.5 PM-05 — Web portal to XR device delivery

Confirmed statement

Entitled or subscribed XR experiments must have a mechanism to become available on the supported target device. Two concepts were described and neither was selected: (A) direct experiment, application or package delivery; (B) a centralized XR device application with entitlement-controlled experiments or modules.

Table 15 — PM-05 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	1.2.2–1.2.4, 2.1.1, 2.1.2, 2.2.1, 2.2.2, 2.3.x, 2.4.x, 10.1.2, 12.2.1
Affected roles	Learner, Super Admin, Institute Admin
Affected capability or workflow	XRD, ENT, SEC
Effect on the previous position	The product requirement is confirmed and is already supported by FRD 2.2.2 and 10.1.2. The earlier guidance-only XR model is incomplete. Delivery states must be designed in mechanism-neutral language.
Implementation details still pending	The technical mechanism is explicitly not selected: direct package delivery, a centralized application, synchronisation, deep-link handoff or another appropriate solution all remain open.

3.6 PM-06 — XR simulation availability

Confirmed statement

The required XR simulations and experiments are available. The current platform team is not assumed to build every simulation from scratch.

Table 16 — PM-06 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	1.4.1, 1.1.4
Affected roles	Vendor, Content Manager, Super Admin
Affected capability or workflow	GOV, CAT
Effect on the previous position	The platform must be designed to ingest, validate, publish, entitle and deliver existing simulations. Simulation authoring is not a Phase-1 platform surface.
Implementation details still pending	Technical discovery of existing builds, formats, versions, device compatibility, package types, integration interfaces and ownership contacts.

3.7 PM-07 — Vendor and Academic Reviewer

Confirmed statement

Vendor and Academic Reviewer require actual platform logins and workspaces. They are no longer merely external actors.

Table 17 — PM-07 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	1.4.1, 1.4.2, 1.4.5, 8.1 role hierarchy
Affected roles	Vendor, Academic Reviewer, and by consequence Content Manager and Super Admin
Affected capability or workflow	GOV, IAM
Effect on the previous position	EXTENSION RECORDED. FRD 8.1 defines exactly five roles. Vendor and Academic Reviewer appear in the FRD only as actors in the content governance workflow, with no role definition and no permission set. The confirmed role set becomes seven. FRD 8.3, the FRD's own mechanism for extending the role model, is Phase 3 and cannot justify this within Phase 1.
Implementation details still pending	None at product level. Their permission sets are derived in this document from FRD text and PM statements only.

3.8 PM-08 — Assessment configuration

Confirmed statement

Assessment setup must be configurable through the portal and linked to Web and XR applications through flexible APIs. Previous fixed prototype values are sample configuration only.

Table 18 — PM-08 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	5.1.1, 5.1.2, 5.2.1–5.2.3, 5.3.1, 5.3.2, 8.1.4, 11.2.1, 11.2.2
Affected roles	Faculty, Learner
Affected capability or workflow	ASM, PRG
Effect on the previous position	Fixed assessment values are reclassified as SAMPLE CONFIGURATION. The consumption screens survive structurally unchanged, but every displayed value must be read from a configuration rather than compiled in. The pre-attempt disclosure must be generated from the configuration.
Implementation details still pending	The configuration schema and the platform/application API contract are both TECHNICAL DETAIL TO BE DEFINED.

3.9 PM-09 — Learning progression

Confirmed statement

The approved learning sequence is: Learning Resources → Simulation → Assessment → Completion / progression.

Table 19 — PM-09 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	3.1.1, 3.1.2, 3.2.2, 11.1.1
Affected roles	Learner, Faculty
Affected capability or workflow	LRN, PRG
Effect on the previous position	The existing learner sequencing design is retained as the current approved learning-flow model. The approval covers the learner progression model only and is not generalised into governance sequencing or any other role's workflow.
Implementation details still pending	Whether the sequence is a platform constant or a per-course configurable setting is not stated.

3.10 PM-10 — Institutional SSO

Confirmed statement

Institutional SSO integration is required. It stays in the platform architecture and in the institutional authentication experience.

Table 20 — PM-10 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	4.2.2 (Phase 1), 7.4.1 (Phase 3), 7.4.2, 2.4.4
Affected roles	Institute Admin, Faculty, Learner, Super Admin
Affected capability or workflow	IAM, TEN, CFG
Effect on the previous position	Resolves an internal FRD phase contradiction in favour of the Phase-1 reading. Adds an Institute Admin SSO configuration surface that had never been designed: the requirement is a configuration capability, not merely a login button. SSO is **required** and is available according to the institution's configured authentication model; it is not described as merely preferred.
Implementation details still pending	Provider, protocol, attribute mapping and technical implementation all remain PENDING DECISION.

3.11 PM-11 — Reporting and analytics strategy

Confirmed statement

Build core functionality and operations first. Detailed reporting and analytics parameters will be added incrementally.

Table 21 — PM-11 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	11.1.x, 11.2.x (Phase 1); 8.2.x, 11.3.x (Phase 2); 11.4.x (Phase 3); bounded by 1.3.4 and 8.1.3
Affected roles	Faculty, Institute Admin, Super Admin, Content Manager
Affected capability or workflow	RPT, PRG
Effect on the previous position	No large analytics dashboards are designed now. Reporting is not removed from the long-term requirement set. Phase-1 role architecture may include operational summaries and status where required.
Implementation details still pending	Detailed report metrics, KPI definitions, formulas and filters all remain PENDING DECISION.

3.12 PM-12 — Certification

Confirmed statement

Certification is not part of the current Phase-1 product. Certification is a Phase-2 activity.

Table 22 — PM-12 decision record

Attribute	Record
Source	PM meeting, 18 August 2026
Affected FRD sections	3.3.1, 3.3.2 (both Phase 1 in the FRD); 3.5.1, 3.5.2, 4.5.2 already later phases
Affected roles	Learner, Faculty, Institute Admin
Affected capability or workflow	CAT, PRG
Effect on the previous position	DEVIATION RECORDED. FRD 3.3.1 and 3.3.2 are Phase 1 as written. The PM moves both to Phase 2. The FRD Phase-1 requirement count remains 78 as a document fact; the Phase-1 design working set becomes 76 plus 2 PM-deferred. No Phase-1 certification screens are built.
Implementation details still pending	Whether the structural certification-program node is still modelled in the content hierarchy, even with no Phase-1 experience, is deliberately not assumed either way.

3.13 PM-13 — Platform and tenancy model

Confirmed statement

Use the confirmed two-plane role model. Central / platform content and governance plane: Vendor, Academic Reviewer, Content Manager, Super Admin. Institution / tenant plane: Institute Admin, Faculty, Student / Learner. B2C learners use shared platform and learning capabilities outside the institutional tenant role model where appropriate.

Table 23 — PM-13 decision record

Attribute	Record
Source	PM confirmation, 19 August 2026
Affected FRD sections	1.1.4, 1.3.1, 1.3.2, 1.3.5, 1.4.1–1.4.5
Affected roles	All seven
Affected capability or workflow	TEN, GOV, SEC
Effect on the previous position	Moves the two-plane model from a working hypothesis to a confirmed architectural position. Vendor, Academic Reviewer and Content Manager operate only on the central content plane and therefore need no tenant access. Super Admin alone spans both planes and retains the FRD-defined cross-tenant governance capability. No additional cross-tenant permissions are granted to any other role.
Implementation details still pending	How the central content plane and tenant planes are physically separated (schema, service or logical boundary) is a technical detail. Whether central-plane roles need any read visibility into tenant usage of their content is a product question; no such permission is granted or assumed.

3.14 PM-14 — Content rejection and resubmission

Confirmed statement

The content workflow must support approval, changes required or rejection, comments and reasons, Vendor correction, resubmission, and review history and status traceability.

Table 24 — PM-14 decision record

Attribute	Record
Source	PM confirmation, 19 August 2026
Affected FRD sections	1.4.1, 1.4.2, 1.4.3, 1.4.4, 1.4.5
Affected roles	Vendor, Academic Reviewer, Content Manager, Super Admin
Affected capability or workflow	GOV
Effect on the previous position	EXTENSION RECORDED. FRD 1.4.1–1.4.4 describe a forward-only chain and contain no rejection, revision, resubmission or review-history requirement. PM-14 adds them. A corrected resubmission re-enters review; there is no automatic publication path after a rejection, and return authority never implies publish authority.
Implementation details still pending	Comment structure; whether a returned submission is versioned or corrected in place; whether a resubmission re-enters at the returning gate or restarts; whether the Vendor is notified of a return or discovers it on next sign-in; retention period and visibility scope of review history; whether Super Admin may also return a submission at the publication gate — PM-14 names Academic Reviewer and Content Manager only.

3.15 PM-15 — Assessment ownership for B2B and B2C

Confirmed statement

For both B2B and B2C, assessment setup and configuration is owned by Faculty or the authorized institution side. Learners do not author or configure assessments.

Table 25 — PM-15 decision record

Attribute	Record
Source	PM confirmation, 19 August 2026
Affected FRD sections	5.1.1, 5.1.2, 5.2.1–5.2.3, 5.3.1, 5.3.2, 8.1.4
Affected roles	Faculty, Institute Admin, Learner
Affected capability or workflow	ASM
Effect on the previous position	Assigns ownership to the PM-08 architecture without changing its shape. Assessment ownership ceases to be a B2B/B2C divergence — it converges on the same authorized side. The learner-side prohibition becomes a stated rule rather than an absence. Institute Admin is explicitly not granted blanket assessment-authoring permission; Faculty is the requirement-named role.
Implementation details still pending	Which Faculty, institution or content owner supplies the assessment for a pure B2C individual learner is explicitly deferred as a content and implementation detail, and is explicitly not a blocker. Whether Institute Admin holds any assessment authority at all remains open. Whether an assessment configuration is reusable across institutions or cohorts is not addressed.

4. Product Scope and Phasing

4.1 How the phase figures are derived

Platform FRD_v4 contains **131 numbered requirements** across twelve sections, distributed as **78 in Phase 1, 33 in Phase 2 and 20 in Phase 3**. These counts are document facts and are not changed by this draft.

Two adjustments produce the working set this draft is written to. Both are stated openly rather than absorbed into the counts.

Table 26 — Derivation of the Phase-1 working set

Step	Effect	Count	Basis
Original FRD Phase-1 requirement count	Unchanged document fact	78	Platform FRD_v4
Less PM-deferred certification requirements	3.3.1 and 3.3.2 move to Phase 2	−2	PM-12 — recorded deviation
Phase-1 design working set from the FRD		76	Derived
Plus one Phase-3 requirement given a Phase-1 working position	7.4.1 SSO authentication support	+1	PM-10 — resolves an FRD internal conflict with 4.2.2
Phase-1 working set used by this draft		77	Derived

Stating the certification position transparently

- Platform FRD_v4 places certification program structure (3.3.1) and certification completion criteria (3.3.2) in Phase 1.
- PM-12 moves both to Phase 2. This is a deviation from the FRD as written, not a correction of a mistake in it.
- This document does not claim that the FRD always said certification was Phase 2. The original phase is preserved in every table.

The same transparency applies to SSO in the opposite direction. FRD 4.2.2 places SSO configuration in Phase 1 and FRD 7.4.1 places SSO authentication support in Phase 3. That is an internal contradiction in the FRD. PM-10 confirms the Phase-1 reading, so 7.4.1 carries a Phase-1 working position here while its original Phase-3 designation is preserved and shown.

4.2 Scope matrix

Table 27 — Phase scope matrix

Phase	In scope	Basis
Current Phase 1	All original Phase-1 requirements except the two PM-deferred certification requirements · All seven confirmed roles · Both B2B and B2C · Core operations for every role · Content governance including rejection, resubmission and review history · Learning · Assessment configuration and consumption · XR entitlement and delivery · Licensing and entitlements · Payment capabilities defined by Phase 1 · Institutional SSO · ERP requirements defined by Phase 1 · Core progress tracking · Core operational reporting and status	Platform FRD_v4 Phase-1 subsections of all twelve sections; PM-01, PM-04, PM-05, PM-07, PM-08, PM-09, PM-10, PM-11, PM-13, PM-14, PM-15
Phase 2	Certification per PM-12 · All original FRD Phase-2 capabilities: content lifecycle states and version management (1.5.x, 1.6.x) · XR device registration and remote configuration (2.5.x, 2.6.x) · in-platform institutional course creation (3.4.x) · learning personalisation and recommendation (4.4.x, 4.5.x) · XR simulation-based assessment (5.4.x) · package, pricing and coupon management (6.3.x, 6.4.x) · real-time ERP provisioning (7.3.x) · institutional analytics dashboards (8.2.x) · institutional invoicing and subscription billing (9.3.x) · data protection compliance and retention (10.4.x) · expanded institutional reporting (11.3.x) · platform auto-scaling and load balancing (12.3.x)	Platform FRD_v4 Phase-2 subsections; PM-12 for certification
Phase 3	All original FRD Phase-3 capabilities: regional deployment and multi-region replication (1.7.x, 12.4.x) · XR telemetry and interaction analytics (2.7.x) · certification tracking and eligibility reporting (3.5.x) · AI learning assistant for learners (5.5.x) · licence lifecycle and grace periods (6.5.x) · enterprise identity integration and role mapping from identity-provider attributes (7.4.2) · dynamic and custom role configuration (8.3.x) · advanced billing controls (9.4.x) · advanced XR analytics (11.4.x)	Platform FRD_v4 Phase-3 subsections

Two phasing statements this draft does not make

- B2C is not moved to a later phase. Platform FRD_v4 places B2B (1.1.2) and B2C (1.1.3) in Phase 1 with no sequencing between them, and PM-04 confirms both are required.
- The earlier B2B-first recommendation is not an approved phase decision. It was a working interpretation, it is not FRD-derived, and it returns to pending.

4.3 Phase distribution by section

Table 28 — Original FRD requirement distribution by section and phase

§	Section	Phase 1	Phase 2	Phase 3	Total
1	Platform Overview	18	4	2	24
2	XR Content Delivery Architecture	11	4	2	17
3	Content Organization and Learning Structure	6	3	2	11
4	User Management and Onboarding	5	4	0	9
5	Learning Experience Features	7	2	2	11
6	Licensing and Monetization	4	4	2	10
7	Enterprise Integrations	5	2	2	9
8	Administration and Role Management	5	2	2	9
9	Payment and Billing	4	2	2	8
10	Security and Compliance	6	2	0	8
11	Analytics and Reporting	4	2	2	8
12	Platform Infrastructure and Hosting	3	2	2	7
	Total	**78**	**33**	**20**	**131**

Every one of the 131 identifiers above appears in Chapter 7 and in the traceability matrix at Appendix A.

5. XR Simulation Learning Platform Architecture

Technology neutrality

- This chapter is deliberately technology-neutral. No front-end framework, back-end language, runtime, cloud provider, database technology, message transport or deployment topology is selected here.
- Every box in every diagram is a LOGICAL capability. Whether a capability becomes a separate service, a module inside a larger application, or a managed product is an engineering decision this document does not take.
- Platform FRD_v4 12.1.1 names AWS, Azure and Google Cloud as examples of cloud infrastructure providers. That is an example in the FRD, not a selection, and it remains a PENDING DECISION.

5.1 System context

The platform sits between the people and organisations that use it and the external systems it depends on. The boundary matters because several of those external systems are undetermined: naming them as roles or as chosen products would assert decisions that have not been taken.

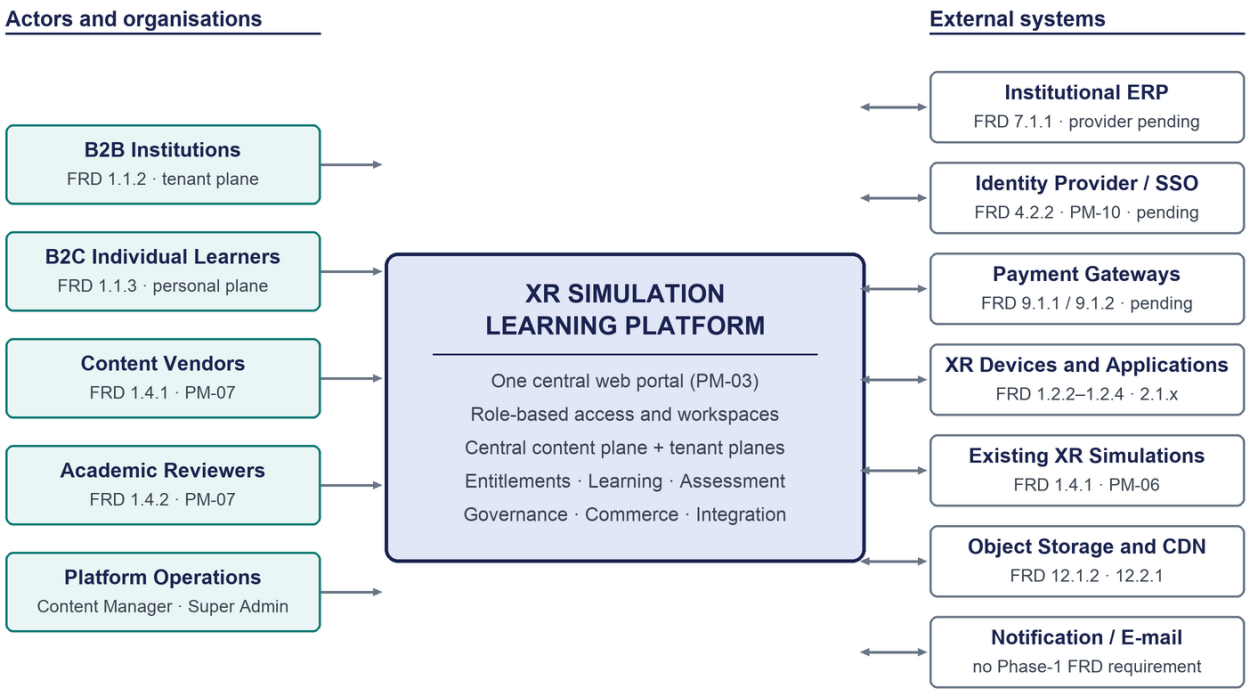


Figure 1 — System context — actors, external systems and integration boundaries

Table 29 — External actors and systems

External entity	Relationship to the platform	FRD basis	Status
B2B institutions	Hold licences, configure their tenant, provision users through ERP and authenticate through SSO	1.1.2, 4.2.1, 4.2.2, 7.x	Confirmed
B2C individual learners	Register, purchase, hold personal entitlements and learn	1.1.3, 4.1.1, 6.1.x	Confirmed
Content vendors	Submit XR module builds and assets; correct and resubmit after a return	1.4.1, 1.4.5, PM-07, PM-14	Confirmed
Academic reviewers	Validate submitted modules against engineering curriculum standards	1.4.2, PM-07	Confirmed

External entity	Relationship to the platform	FRD basis	Status
Institutional ERP systems	Push student enrolment, course mapping and academic structure data into the platform	7.1.1, 7.1.2, 7.2.x	Provider and contract PENDING
Identity providers	Authenticate institutional users through single sign-on	4.2.2, 7.4.1, PM-10	Provider and protocol PENDING
Payment gateways	Process domestic and international transactions for individual purchases	9.1.1, 9.1.2	Provider PENDING
XR devices and device applications	Execute VR, MR and AR simulations outside the browser	1.2.2–1.2.4, 2.1.1, 2.1.2	Delivery mechanism PENDING
Existing XR simulation content	Supplied rather than authored by the platform team	1.4.1, PM-06	Confirmed available; inventory pending discovery
Encrypted object storage and CDN	Hold and distribute protected asset bundles with acceptable latency	12.1.2, 12.2.1, 10.1.1	Concept confirmed; technology PENDING
E-mail and notification service	Would carry any outbound message the platform sends	No Phase-1 requirement exists	Not justified for Phase 1
Support and operations	Operate, monitor and support the platform	Implied by 10.3.1, 10.3.2	Model PENDING

Why notification is shown but not specified

- The word “notification” appears once in Platform FRD_v4, at requirement 9.4.2, which is a Phase-3 institution-facing renewal notification. There is no Phase-1 notification requirement of any kind.
- This draft therefore treats in-workflow status — a queue, a state on an object, an outcome on a submission — as implied by the workflow requirements themselves, and treats push notification as unjustified and unspecified.

5.2 Role and workspace architecture

PM-03 requires one central platform and portal with role-based access and workspaces — one platform, not seven products. This draft realises that as one shared shell and one design system, in which what differs between workspaces is the navigation content and the permitted actions rather than the surrounding chrome.

What is confirmed here, and what is a design position

- **Confirmed (PM-03):** one central platform and portal, with role-based access and role-based workspaces.
- **Design position, not confirmed:** a single shared shell, a single sign-in entry point for every role, and the absence of a role choice before sign-in. These are reasonable and internally consistent, but no source states them, so they are DESIGN-PENDING and remain open to change (`QX-10`).
- **Not assumed either way:** whether one person may hold more than one role in Phase 1 (`QP-11`).

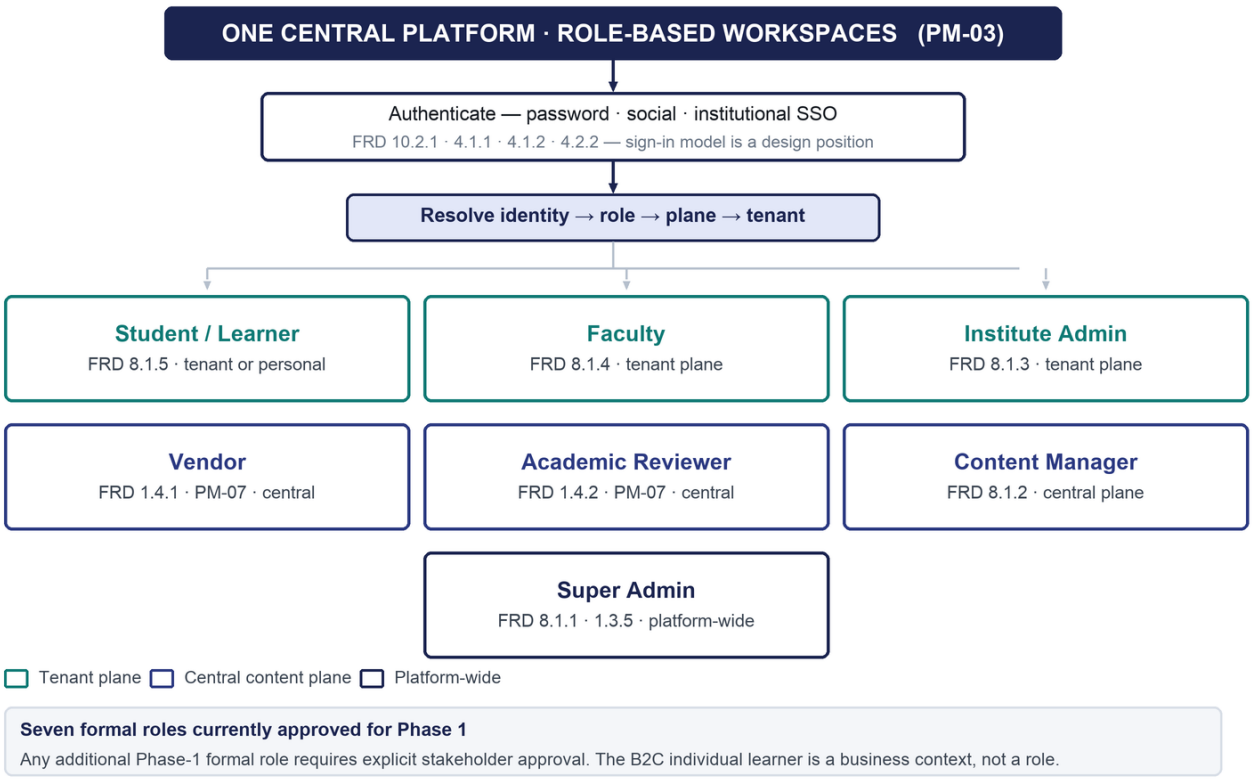


Figure 2 — Role and workspace architecture — one portal, seven workspaces

This draft resolves role **from the authenticated identity** rather than asking a person to declare it before signing in, because a pre-login role choice would expose the role structure. That is a design position and is recorded as DESIGN-PENDING (`QX-10`); no source requires or forbids either model. What is requirement-based is narrower: self-registration serves individual learners (FRD 4.1.1), while Faculty, administrators, vendors and reviewers are provisioned (FRD 7.2.2, 8.1.x, 1.4.1).

Whether one person may hold more than one role in Phase 1 is not stated by any source and is not assumed here (`QP-11`). If multi-role identities are later approved, a workspace chooser is available as a conditional design pattern, and roles would not be merged into one blended navigation because that would require permission composition the FRD does not define and FRD 8.3 is Phase 3. **No workspace chooser is mandated by this draft**, and nothing depends on one existing.

5.3 Two-plane model

PM-13 confirms the platform has two planes plus a personal context, and that Super Admin alone spans them. This fills a silence in the FRD rather than contradicting it: FRD 1.3.1 and 1.3.2 isolate institutional and B2C data, and FRD 1.3.5 grants cross-tenant access to Super Admin only, but the FRD never states where Vendor, Academic Reviewer and Content Manager sit.

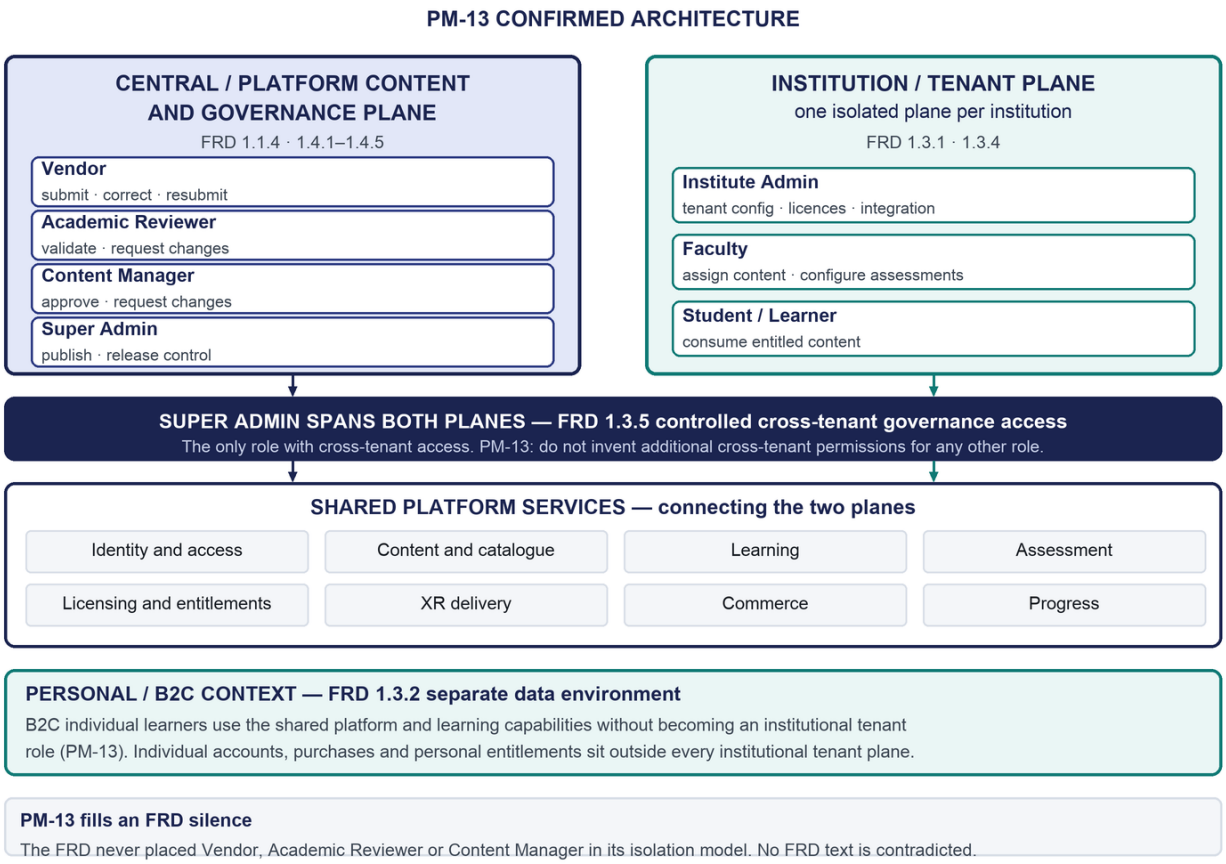


Figure 3 — Two-plane platform architecture confirmed by PM-13

Table 30 — The two planes and the personal context

Plane	FRD basis	Roles	What lives here
Central content and governance plane	1.1.4 central ownership; 1.4.1–1.4.5 governance chain	Vendor, Academic Reviewer, Content Manager, Super Admin	Module builds, assets, metadata, academic validation, readiness approval, publication, review history
Institution / tenant plane	1.3.1 tenant isolation; 1.3.4 tenant-scoped analytics	Institute Admin, Faculty, Student / Learner	Institutional users, entitlements, course structure, progress, licences, tenant-scoped operational status
Personal / B2C context	1.3.2 separate B2C data environment	Student / Learner in the individual context	Individual accounts, purchases, personal entitlements, personal device binding
Spanning all planes	1.3.5 controlled cross-tenant governance access	Super Admin only	Publishing, licensing, platform configuration, cross-tenant operational oversight

The boundary rule PM-13 attaches

- Vendor, Academic Reviewer and Content Manager operate only on the central content plane. They never operate on tenant data, so they need no tenant access and hold none.
- Super Admin retains the FRD-defined cross-tenant governance capability. No additional cross-tenant permission is granted to any other role, and none may be invented.
- How the planes are physically separated — schema, service or logical boundary — is TECHNICAL DETAIL TO BE DEFINED.

5.4 Shared platform capabilities

Seven roles do not mean seven products. If the platform were designed role-first, the same capability would be rebuilt seven times — seven ways to view a module, seven progress models, seven entitlement checks. Capability is therefore separated from role: a capability defines **what the platform can do**, and the role model defines **who may invoke it and in what scope**.

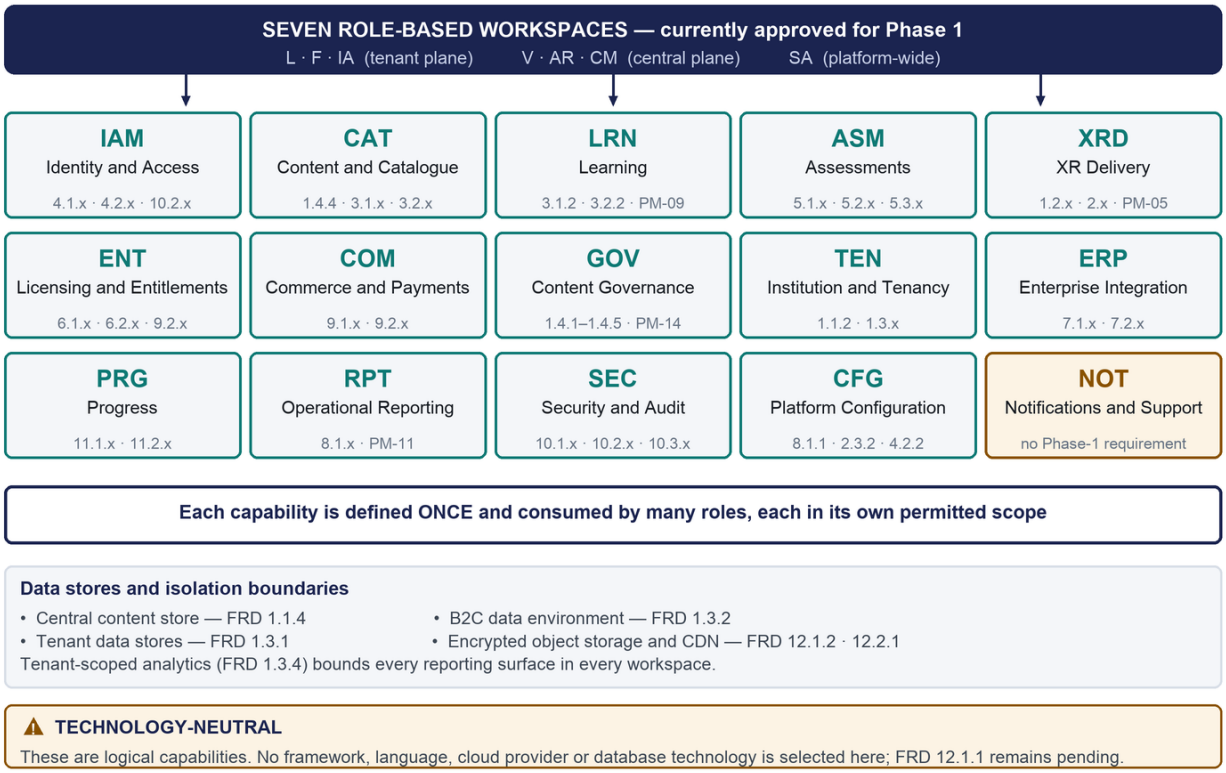


Figure 4 — Shared capability architecture — fifteen logical domains

Table 31 — The fifteen capability domains

Code	Domain	What it covers	Principal FRD basis
IAM	Identity and access	Authentication, session lifecycle, profile, workspace resolution, institutional SSO	4.1.x, 4.2.x, 4.3.1, 10.2.x
TEN	Institution and tenancy	Tenant identity, isolation, plane resolution, cross-tenant governance boundary	1.1.2, 1.3.1–1.3.5
CAT	Content and catalogue	Modules, courses, metadata, published-only visibility, central ownership	1.1.4, 1.4.4, 3.1.x, 3.2.x
LRN	Learning	Consuming entitled content in sequence toward completion	3.1.2, 3.2.2, 8.1.5, PM-09
ASM	Assessments	Question sourcing, configuration, delivery, scoring, result	5.1.x, 5.2.x, 5.3.x, PM-08, PM-15
XRD	XR delivery	Getting an entitled experience onto the device that runs it	1.2.x, 2.1.x–2.4.x, 12.2.1, PM-05
ENT	Licensing and entitlements	Who may access what, for how long, and how that is validated	6.1.x, 6.2.x, 2.2.2, 9.2.x, 10.1.2
COM	Commerce and payments	B2C discovery, purchase, payment and activation	1.1.3, 6.1.x, 9.1.x, 9.2.x
GOV	Content governance	Submission, validation, approval, publication, return, resubmission, history	1.4.1–1.4.5, 2.3.2, PM-14
ERP	Enterprise integration	Automated provisioning and academic structure synchronisation	7.1.x, 7.2.x
PRG	Progress	Recording and reflecting what a learner has done	11.1.x, 11.2.x
RPT	Operational status and reporting	Operational status in Phase 1; analytics incrementally thereafter	8.1.1, 8.1.3, 8.1.4, 1.3.4, PM-11
SEC	Security and audit	Protecting assets, sessions and governance decisions	10.1.x, 10.2.x, 10.3.x, 12.1.2
CFG	Platform configuration	Platform settings, release control, tenant and integration configuration	8.1.1, 2.3.2, 4.2.2, 7.2.1

Code	Domain	What it covers	Principal FRD basis
NOT	Notifications and support	In-workflow status where implied; no Phase-1 push notification requirement exists	9.4.2 is Phase 3 only

Two of these domains are structurally central. **Licensing and entitlements is the platform's spine** — learning reads it, XR delivery validates against it twice, commerce creates it, ERP populates it and tenancy scopes it. **Content governance is the most precisely specified domain in the FRD** and the one with the clearest authority ladder.

5.5 Logical application and service architecture

The following layering describes responsibilities, not deployment. It is drawn so that an engineering team can choose a technology stack without any statement in this document becoming false.

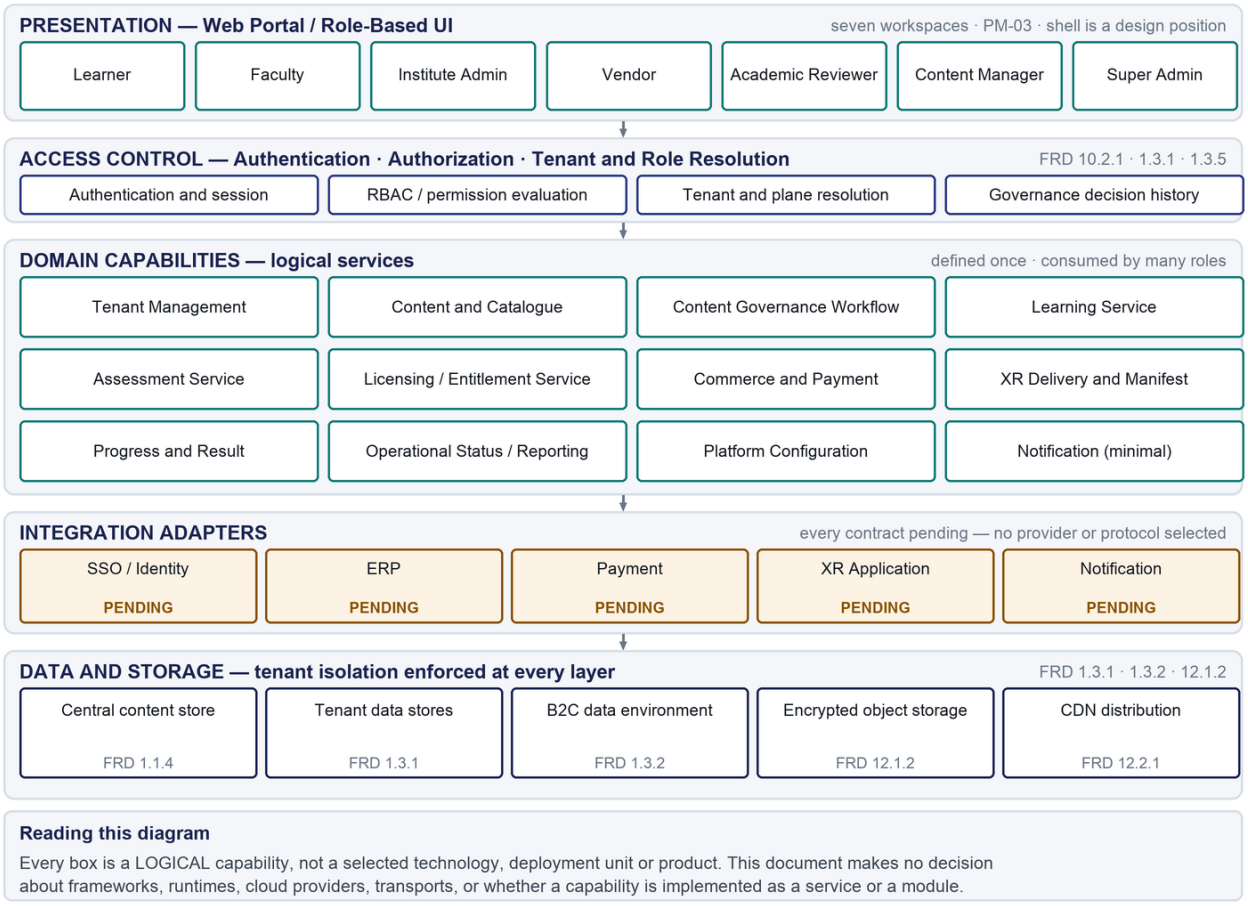


Figure 5 — Logical application and service architecture — technology-neutral

Table 32 — Logical components

Component	Responsibility	Notes
Web portal / role-based UI	One shell rendering seven workspaces from one design system	PM-03. Navigation content and permitted actions differ; chrome does not.
Authentication and RBAC	Verify identity, evaluate permissions, enforce prohibitions	FRD 10.2.1, 10.2.2. Password policy and multi-factor behaviour are PENDING.
Tenant management	Tenant identity, isolation boundary, plane and workspace resolution	FRD 1.3.1, 1.3.5, PM-13.
Content and catalogue service	Module and course model, metadata, published-only visibility	FRD 1.4.4, 3.1.x, 3.2.x.
Content governance workflow	Gate positions, decisions, returns, resubmission and review history	FRD 1.4.1–1.4.5, PM-14. Phase-1 status vocabulary is gate position, because lifecycle states are Phase 2.

Component	Responsibility	Notes
Assessment service	Question bank, assessment configuration, attempt delivery, scoring	PM-08, PM-15. Configuration schema and API contract are PENDING.
Licensing and entitlement service	Entitlement records, validation at browse and at execution, seat caps	FRD 2.2.2, 6.2.2, 10.1.2.
Commerce and payment service	Catalogue purchase, payment handoff, activation	FRD 6.1.x, 9.1.x, 9.2.x. Gateway is PENDING.
XR delivery and manifest capability	Make an entitled experiment available on the supported device	PM-05. The mechanism is deliberately unselected.
Progress and result service	Completion, time spent, scores and attempts	FRD 11.1.x, 11.2.x.
Operational status and reporting	Role-appropriate operational views, tenant-bounded	FRD 1.3.4, 8.1.x, PM-11. Detailed metrics are PENDING.
Integration adapters	SSO, ERP, payment, XR application and notification boundaries	Every contract is PENDING; no provider is named.
Secure storage and distribution	Encrypted asset storage and latency-appropriate distribution	FRD 10.1.1, 12.1.2, 12.2.1.
Data stores and tenant isolation	Central content store, per-tenant stores, separate B2C environment	FRD 1.1.4, 1.3.1, 1.3.2, 1.3.3.

5.6 Security and data boundaries

Four boundaries are stated by Platform FRD_v4 in unambiguous terms. They are not design preferences and they must survive into every screen and every interface.

1. **Only Super Admin publishes** (FRD 1.4.4). No other role holds publication authority in any state.
2. **Vendors can never publish** (FRD 1.4.5). This is an explicit prohibition, not merely an absent permission, and it is expressed positively — a vendor sees where a submission is, never a disabled publish control.
3. **Analytics never cross a tenant** (FRD 1.3.4), except through the Super Admin governance access the FRD grants at 1.3.5.
4. **Institutions cannot modify or redistribute content** (FRD 1.1.4).

Table 33 — Security and data boundary requirements

Boundary	Requirement	FRD basis
Tenant isolation	Each institution's user records, analytics, licensing data and course mapping are held in a logically separated environment with no cross-access.	1.3.1
B2C separation	Individual-learner data is held in a data environment independent of institutional databases.	1.3.2
Cross-tenant governance	Super Admin alone may view and manage across tenants, strictly for governance and operational oversight.	1.3.5, PM-13
Least privilege	Every permission in this document is derived from explicit FRD text or an explicit PM statement. Where neither defines a permission it is recorded as pending, never inferred.	8.1.x, PM-13, PM-14, PM-15
Licence validation	Entitlement is checked twice — before delivery and before execution — and a learner may pass the first and fail the second.	2.2.2, 10.1.2
Role-based permissions	Access is determined by the resolved role and scope, and prohibitions are enforced rather than hidden.	8.1.1–8.1.5, 1.4.5
Auditability	Content governance decisions, comments, returns and resubmissions are recorded and traceable.	PM-14
Protected XR assets	Asset bundles are encrypted before distribution and stored encrypted; no workspace offers a route to an unprotected copy.	10.1.1, 12.1.2
Data ownership	All hosted modules remain owned and governed by the platform owner and are not modifiable or redistributable by institutions.	1.1.4

Audit scope is deliberately bounded

- PM-14 requires review history and status traceability for the content governance chain. That is the audit trail this document specifies.
- Audit trails for licensing changes, entitlement grants, platform configuration and device-replacement approvals are not defined by any requirement and are not invented here. They are recorded as an open question.

6. Confirmed Role Model

Seven roles are confirmed. Five come from the FRD's own role hierarchy at 8.1.1 – 8.1.5. Two — Vendor and Academic Reviewer — appear in the FRD only as actors in the content governance workflow and are promoted to first-class roles by PM-07. That promotion is recorded here as an extension of the FRD, and the distinction between an FRD-defined role and a PM-added role is preserved in every table below.

Two rules that bound this chapter

- Every permission listed is derived from explicit FRD text or an explicit PM statement. Where neither source defines a permission, it is recorded as pending rather than inferred from what would be plausible.
- No permission is silently expanded. Where an earlier reading granted something the sources do not support, this document narrows it and says so.

Table 34 — The seven confirmed roles

Role	FRD basis	Plane	Status
Student / Learner	8.1.5	Tenant plane or personal B2C context	FRD-defined; reframed as one workspace
Faculty	8.1.4	Tenant plane	FRD-defined; newly in design scope
Institute Admin	8.1.3	Tenant plane	FRD-defined; newly in design scope
Vendor	1.4.1, 1.4.5 — actor only, no FRD role definition	Central content plane, own submissions	**PM-added role (PM-07)** — recorded extension
Academic Reviewer	1.4.2 — actor only, no FRD role definition	Central content plane	**PM-added role (PM-07)** — recorded extension
Content Manager	8.1.2	Central content plane	FRD-defined; newly in design scope
Super Admin	8.1.1, 1.3.5	Spans all planes	FRD-defined; newly in design scope

The B2C individual learner is a business context around the Learner role, not a separate role. External systems — ERP, identity providers, payment gateways and device application stores — are integrations, not roles. **Seven formal roles are currently approved for Phase 1. Any additional Phase-1 formal role requires explicit stakeholder approval**, and FRD 8.3 is Phase 3, so no Phase-1 mechanism exists to define one.

6.1 Student / Learner

Table 35 — Student / Learner — role definition

Attribute	Definition
Requirement basis	FRD 8.1.5
Platform plane and context	Tenant plane in the institutional context; personal / B2C context in the individual context
Purpose	Consume entitled learning content, complete simulations and assessments, and see personal progress.
B2B / B2C applicability	Both. The learning core is identical; the entitlement origin, authentication path and data environment differ.
Main navigation and capability areas	Home · My Learning · Progress · Account. Explore appears in the B2C context only.
Key workflows	FLOW 2 (B2B learning), FLOW 3 (B2C purchase to learning), FLOW 4 (assessment attempt), FLOW 5 (XR delivery), FLOW 6 (SSO sign-in), FLOW 8 (device replacement request)
Data visibility	Own records only. Progress is visible to Faculty and Institute Admin within the same tenant, and to nobody outside it.
Tenant boundary	Institutional learner is bound to one tenant with no tenant switcher. B2C learner sits outside every tenant.
Reporting needs in Phase 1	Personal progress: module completion, time spent, assessment scores and attempt history.
FRD traceability	8.1.5, 1.2.1–1.2.4, 2.2.2, 2.4.1, 2.4.2, 3.1.1, 3.1.2, 3.2.1, 3.2.2, 4.1.1, 4.1.2, 4.3.1, 5.2.1–5.2.3, 6.1.1, 6.1.2, 9.1.x, 9.2.x, 10.1.2, 10.2.x, 11.1.x, 11.2.x, 12.2.1

6.1.1 Primary responsibilities

- Work through entitled modules in the sequence the course defines
- Complete learning resources, the simulation and the assessment for each module
- Reach XR simulations on the supported device where the format requires one
- In the B2C context: discover, purchase and manage personal entitlements and device binding

6.1.2 Permissions and allowed actions

- View and consume entitled content only (8.1.5)
- Attempt assessments and view own results and attempt history (5.2.x, 11.2.1, 11.2.2)
- View own progress (8.1.5, 11.1.x)
- View and edit own profile, subject to ERP-owned fields (4.3.1)
- B2C only: purchase modules and bundles (6.1.1, 6.1.2), view bound devices (2.4.1), request device replacement (2.4.2)

6.1.3 Prohibited actions

- Never authors or configures an assessment — stated explicitly by PM-15
- Sees only the post-attempt feedback the approved feedback policy permits — **that policy is PRODUCT-PENDING (`QP-27`)**, so no answer-key rule is asserted in either direction
- Never sees another learner's data
- No commerce surface is offered to an institutional learner; a licence-cap wall directs them to their institution
- No content modification of any kind (1.1.4)

6.1.4 Acceptance criteria

- Given an entitled module, when the learner opens it, then its components are available in the defined sequence and no later module is reachable until it is complete.
- Given an unentitled module, when the learner attempts access, then an access-restricted state explains the cause and the next step without exposing internal codes.
- Given any assessment, when the learner views it, then no interface offers any means of editing its configuration.

6.1.5 Unresolved details

- Whether the learning sequence is per-course configurable or a platform constant
- Which profile fields are user-editable rather than ERP-owned
- Who grades descriptive answers, which can otherwise strand a learner mid-progression

6.2 Faculty

Table 36 — Faculty — role definition

Attribute	Definition
Requirement basis	FRD 8.1.4
Platform plane and context	Tenant plane
Purpose	Manage course assignments, own assessment configuration, and monitor learner operational status.
B2B / B2C applicability	Primarily B2B. PM-15 extends assessment ownership to the B2C case, on the Faculty or authorized institution side.
Main navigation and capability areas	Overview · Courses · Assessments · Learners · Account.
Key workflows	FLOW 2 (assignment and assessment setup), FLOW 4 (assessment configuration to result)
Data visibility	Own tenant only, and within it the learners and courses the Faculty member is responsible for.
Tenant boundary	Strictly tenant-scoped. Every reporting surface is additionally bounded by FRD 1.3.4.
Reporting needs in Phase 1	Operational learner status only in Phase 1: who has completed, who is blocked, what attempts exist.
FRD traceability	8.1.4, 1.3.4, 3.2.1, 3.2.2, 5.1.1, 5.1.2, 5.2.1–5.2.3, 5.3.1, 5.3.2, 11.1.1, 11.1.2, 11.2.1, 11.2.2, 7.2.3

6.2.1 Primary responsibilities

- Assign courses and modules to learners and cohorts within the tenant
- Source questions by structured spreadsheet upload and by AI-assisted generation
- Define and configure assessments, and assign or publish them
- Monitor which learners have completed, which are blocked and what attempts exist

6.2.2 Permissions and allowed actions

- Assign course or module content to learners and cohorts (8.1.4)
- Upload question banks through structured spreadsheet templates (5.1.1)
- Generate assessment questions using integrated AI tools (5.1.2)
- Configure assessment types, rules and parameters, including negative marking (5.3.2) and randomization (5.3.1)
- View learner results and attempt history within the tenant (11.2.1, 11.2.2)
- View operational learner status within the tenant (8.1.4, bounded by 1.3.4)

6.2.3 Prohibited actions

- No course creation — in-platform institutional course creation (3.4.1, 3.4.2) is Phase 2
- No content editing — content is centrally owned (1.1.4)
- No class-level performance reports — those are Phase 2 (11.3.1)
- Never crosses the tenant boundary (1.3.4)
- No publication authority in the content governance chain

6.2.4 Acceptance criteria

- Given a configured assessment, when a learner opens it, then the pre-attempt disclosure is generated from that configuration rather than from fixed copy.

- Given a question bank upload with invalid rows, when it is processed, then valid rows are accepted and each invalid row is reported individually.
- Given a Faculty member of institution A, when they search for learners, then no learner of institution B is returned in any state.

6.2.5 Unresolved details

- Whether Faculty holds any entitlement-granting authority
- Whether Faculty sees ERP synchronisation status
- Whether Faculty owns descriptive-answer grading — plausible but not stated by any requirement
- Whether an assessment configuration is reusable across courses or cohorts

6.3 Institute Admin

Table 37 — Institute Admin — role definition

Attribute	Definition
Requirement basis	FRD 8.1.3
Platform plane and context	Tenant plane
Purpose	Manage institutional configuration, student access, integrations and institutional operational status.
B2B / B2C applicability	B2B only.
Main navigation and capability areas	Overview · People · Courses and Streams · Licences · Integrations · Account.
Key workflows	FLOW 2 (institutional configuration and provisioning), FLOW 6 (SSO configuration), FLOW 7 (ERP synchronisation), FLOW 9 (licence cap exhaustion)
Data visibility	Own tenant only, across users, entitlements, course structure, licences and operational status.
Tenant boundary	Strictly tenant-scoped, with no tenant switcher.
Reporting needs in Phase 1	Institutional operational status only in Phase 1: seat utilisation, active learners, ERP synchronisation state, SSO configuration state.
FRD traceability	8.1.3, 1.1.2, 1.3.1, 1.3.4, 4.2.1, 4.2.2, 6.2.1, 6.2.2, 7.1.1, 7.1.2, 7.2.1, 7.2.2, 7.2.3, 2.4.3

6.3.1 Primary responsibilities

- Register and configure the institution and its tenant settings
- Configure institutional SSO and ERP integration, including the synchronisation schedule
- Manage student access and the mapping from academic streams to licensed packages
- Monitor licence seat utilisation before it is exhausted

6.3.2 Permissions and allowed actions

- Manage institutional configuration (8.1.3)
- Configure institutional SSO for the tenant (4.2.2, PM-10)
- Configure ERP integration and synchronisation interval (7.1.1, 7.1.2, 7.2.1)
- Manage student accounts and access within the tenant (8.1.3, 7.2.2)
- Map academic streams to licensed packages (6.2.1, 7.2.3)
- View institutional licences and seat utilisation (6.2.1, 6.2.2)
- View tenant-scoped institutional operational status (8.1.3, bounded by 1.3.4)

6.3.3 Prohibited actions

- No device registration or remote device management — Phase 2 (2.5.x, 2.6.x)
- No usage dashboards or adoption reports — Phase 2 (8.2.x, 11.3.2)
- No course authoring — Phase 2 (3.4.x)

- No content modification or redistribution (1.1.4)
- Never crosses the tenant boundary (1.3.1, 1.3.4)
- No blanket assessment-authoring permission — PM-15 explicitly declines to grant it

6.3.4 Acceptance criteria

- Given a configured identity provider, when the Institute Admin runs a connection test, then the outcome is reported explicitly and an enabled state is reached only after a successful test.
- Given a licence approaching its seat cap, when the Institute Admin views the institution overview, then seat utilisation is visible before learners are blocked.
- Given a failed ERP synchronisation run, when the Institute Admin views synchronisation history, then failed records are retained with a reason and a retry is available.

6.3.5 Unresolved details

- Who creates and manages Faculty accounts — the FRD names only student access
- How an institutional licence is transacted in Phase 1, given that institutional billing is Phase 2
- Whether Institute Admin holds any assessment authority at all
- Institute Admin's role, if any, in entitlement-aware device delivery
- Who authorises and performs MR sideloading

6.4 Vendor

Table 38 — Vendor — role definition

Attribute	Definition
Requirement basis	FRD 1.4.1 and 1.4.5 as an actor; role confirmed by PM-07
Platform plane and context	Central content plane, own submissions only
Purpose	Submit XR module builds, assets and metadata for review, and correct and resubmit work that is returned.
B2B / B2C applicability	Neither. The Vendor operates on the central content plane and is not part of either market context.
Main navigation and capability areas	My Submissions · New Submission · My Modules · Account. Submissions returned for correction appear within My Submissions.
Key workflows	FLOW 1 (content to production, including the correction and resubmission loop)
Data visibility	Own submissions and own published modules only.
Tenant boundary	No tenant scope. Central content plane only.
Reporting needs in Phase 1	Operational status of own submissions: gate position, outcome, and the reason for any return.
FRD traceability	1.4.1, 1.4.5, 1.1.4, 2.2.1, 1.4.3 (metadata), 4.3.1 (organisation profile), PM-07, PM-13, PM-14

6.4.1 Primary responsibilities

- Create submissions and upload XR module builds and supporting assets
- Supply module metadata
- Read reviewer comments on a returned submission and correct the work
- Resubmit corrected work so it re-enters review

6.4.2 Permissions and allowed actions

- Create a submission and submit it to the staging environment (1.4.1)
- Create and edit own submission metadata (1.4.1, 1.4.3)
- Edit and resubmit own returned submission (PM-14)
- View own submissions, their gate position and their review history (PM-14)
- View own published modules (1.4.4)

6.4.3 Prohibited actions

- **Never publishes to production — FRD 1.4.5 is the FRD's only explicit role prohibition**
- No visibility of any other vendor's submissions or modules
- No academic validation or readiness approval authority
- No tenant access of any kind — the Vendor operates only on the central content plane (PM-13)
- No route past a review gate; a resubmission re-enters review and never reaches publication directly (PM-14)

6.4.4 Acceptance criteria

- Given a submission at any gate, when the Vendor views it, then the current gate position is stated and no publish control exists in any state.
- Given a returned submission, when the Vendor opens it, then the reviewer's comments and stated reason are visible and the build, assets and metadata become editable.
- Given a corrected resubmission, when it is submitted, then it re-enters review at a review gate and does not appear in the publication queue.

6.4.5 Unresolved details

- The XR package and submission format — the FRD states only “encrypted asset bundles”
- Whether a returned submission is versioned or corrected in place
- Whether a resubmission re-enters at the returning gate or restarts at academic validation
- Whether the Vendor is notified of a return or discovers it on next sign-in

6.5 Academic Reviewer

Table 39 — Academic Reviewer — role definition

Attribute	Definition
Requirement basis	FRD 1.4.2 as an actor; role confirmed by PM-07
Platform plane and context	Central content plane
Purpose	Validate submitted modules against engineering curriculum standards, or return them for correction with a stated reason.
B2B / B2C applicability	Neither. Central content plane only.
Main navigation and capability areas	Validation Queue · Returned · Validated · Account.
Key workflows	FLOW 1 (academic validation gate, including the return path)
Data visibility	Submissions in the validation queue and the reviewer's own decision history.
Tenant boundary	No tenant scope.
Reporting needs in Phase 1	Operational status of the validation queue and of the reviewer's own decisions.
FRD traceability	1.4.2, 1.4.4, 1.1.4, 3.1.1, 3.1.2, 4.3.1 (profile), PM-07, PM-13, PM-14

6.5.1 Primary responsibilities

- Review submitted module content, metadata and documentation
- Assess alignment with engineering curriculum standards
- Record either academic validation or a request for changes with comments and a reason
- Maintain a traceable record of decisions

6.5.2 Permissions and allowed actions

- Review submitted modules in the validation queue (1.4.2)
- Record academic validation, advancing the submission to content readiness review (1.4.2)

- Request changes with comments and a stated reason, returning the submission to the Vendor (PM-14)
- View review history for submissions in scope (PM-14)

6.5.3 Prohibited actions

- Never publishes to production (1.4.4)
- Never approves content readiness — that is the Content Manager's gate (1.4.3)
- Cannot advance a submission past its own gate
- No tenant access of any kind (PM-13)
- No content editing — the reviewer decides, the vendor corrects (1.1.4)

6.5.4 Acceptance criteria

- Given a submission in the validation queue, when the reviewer records a decision, then it is either academic validation or a request for changes, and a request for changes cannot be recorded without a reason.
- Given any submission, when the reviewer views the workspace, then no publication control is present in any state.
- Given a decision has been recorded, when any authorised role views the submission, then that decision, its author and its comments are visible in the review history.

6.5.5 Unresolved details

- The curriculum standards validated against are not defined anywhere in Platform FRD_v4 — this is content someone must own
- Whether academic validation is ever surfaced to learners or buyers as a trust signal

6.6 Content Manager

Table 40 — Content Manager — role definition

Attribute	Definition
Requirement basis	FRD 8.1.2 and 1.4.3
Platform plane and context	Central content plane
Purpose	Review metadata, documentation and platform compatibility, and clear content for publication or return it for correction.
B2B / B2C applicability	Neither. Central content plane only.
Main navigation and capability areas	Review Queue · Returned · Content Library · Readiness · Account.
Key workflows	FLOW 1 (content readiness gate, including the return path)
Data visibility	All central content, at every gate position. No tenant data.
Tenant boundary	No tenant scope.
Reporting needs in Phase 1	Operational content pipeline status: what is in the queue, what has been returned, what is awaiting publication.
FRD traceability	8.1.2, 1.4.3, 1.4.4, 1.1.4, 3.1.1, 3.1.2, 2.2.1 (compatibility), PM-13, PM-14

6.6.1 Primary responsibilities

- Review module metadata, documentation and platform compatibility
- Read the academic validation record as read-only input
- Record either readiness approval or a request for changes with comments and a reason
- Maintain module metadata and hand cleared content to Super Admin for publication

6.6.2 Permissions and allowed actions

- Review submissions at the content readiness gate (1.4.3)
- Edit and maintain module metadata (8.1.2, 1.4.3)
- Approve content readiness, advancing the submission to the publication queue (1.4.3)
- Request changes with comments and a stated reason, returning the submission to the Vendor (PM-14)
- View the full content library by gate position (8.1.2)
- View review history (PM-14)

6.6.3 Prohibited actions

- Never publishes to production — that is Super Admin's exclusive authority (1.4.4)
- Cannot alter the academic validation record, which is read-only input from the Academic Reviewer
- No tenant access of any kind (PM-13)
- No entitlement, licensing or tenant configuration authority

6.6.4 Acceptance criteria

- Given a submission at the readiness gate, when the Content Manager records a decision, then it is either approval or a request for changes, and a request for changes cannot be recorded without a reason.
- Given an approved submission, when the Content Manager views readiness, then it appears as awaiting publication and no publish control is offered.
- Given a submission, when the Content Manager views the academic validation record, then it is presented read-only.

6.6.5 Unresolved details

- The module metadata schema reviewed at this gate is not enumerated in Platform FRD_v4
- The platform compatibility criteria are not defined
- Whether the Content Manager holds any version-release authority, or whether that is Super Admin's alone

6.7 Super Admin

Table 41 — Super Admin — role definition

Attribute	Definition
Requirement basis	FRD 8.1.1 and 1.3.5
Platform plane and context	Platform-wide; spans the central content plane and every tenant plane
Purpose	Hold full administrative control over platform configuration, module publishing, licensing management and cross-tenant operational oversight.
B2B / B2C applicability	Both, plus the central content plane.
Main navigation and capability areas	Overview · Publishing · Content · Institutions · Licensing · Platform · Account.
Key workflows	FLOW 1 (publication gate), FLOW 2 (institutional licensing), FLOW 5 (delivery oversight), FLOW 8 (device replacement approval)
Data visibility	Platform-wide, exercised as governance and operational oversight rather than as unrestricted data access.
Tenant boundary	Spans all tenants. The only role that does.
Reporting needs in Phase 1	Cross-tenant operational oversight and content pipeline status. No detailed analytics in Phase 1.
FRD traceability	8.1.1, 1.3.5, 1.4.4, 1.4.5, 2.3.2, 2.4.2, 6.2.1, 6.2.2, 9.2.1, 10.2.1, 12.1.1, PM-07, PM-13, PM-14

6.7.1 Primary responsibilities

- Publish approved modules to production and control version releases

- Manage institutional and individual licensing
- Assign platform roles and manage platform configuration
- Approve device-replacement licence transfers
- Exercise controlled cross-tenant governance and operational oversight

6.7.2 Permissions and allowed actions

- Publish approved modules to production — exclusive authority (1.4.4)
- Control when new module versions release to production (2.3.2)
- Manage licensing across institutional and individual entitlements (8.1.1, 6.2.x, 9.2.1)
- Approve device replacement and licence transfer (2.4.2)
- Manage platform configuration and platform role assignment (8.1.1)
- Controlled cross-tenant access for governance and operational oversight (1.3.5)

6.7.3 Prohibited actions

- No custom role creation — Phase 3 (8.3.x)
- No cross-tenant analytics dashboards — detailed metrics are deferred by PM-11, and 8.2.x and 11.3.x are Phase 2
- No bundle creation or pricing configuration — Phase 2 (6.3.x)
- Cross-tenant access is bounded to governance and operational oversight by the FRD's own wording (1.3.5), not unrestricted

6.7.4 Acceptance criteria

- Given a submission that has not passed both prior gates, when Super Admin views the publication queue, then it is absent from that queue.
- Given a device-replacement request, when Super Admin approves it, then the licence rebinds to the new device and the learner can see that it has.
- Given cross-tenant access, when Super Admin views an institution, then tenant-scoped data remains presented within its institutional context rather than aggregated across institutions.

6.7.5 Unresolved details

- Whether Super Admin may also return a submission at the publication gate — PM-14 names Academic Reviewer and Content Manager only
- How an institutional licence is transacted in Phase 1
- Whether an audit trail exists outside content governance — for licensing, entitlements, configuration or device replacement
- Whether Super Admin may see assessment answer keys

7. Functional Requirements

This chapter preserves the section structure and every requirement identifier of Platform FRD_v4. Requirement statements are reproduced faithfully. Nothing is removed, renumbered or rewritten.

Table 42 — How to read the requirement tables in this chapter

Column	What it records
ID	The original Platform FRD_v4 requirement identifier, unchanged.
Requirement	The original requirement statement, reproduced faithfully.
Phase	Two values: the phase Platform FRD_v4 assigns, then the phase this draft works to. Where they differ, the reason appears in the clarification column.
Classification	One of the labels defined in section 2.2.
Roles	The role or roles that own, invoke or are constrained by the requirement.
Market	B2B, B2C, Shared, or the note that no role-facing surface exists.
Capability	The logical capability domain or domains defined in section 5.4.
Clarification, dependencies and open questions	What a PM decision changed, what this requirement depends on, and any recorded gap, tension or pending decision.
Acceptance criterion	A single testable statement summarising what must be true. Fuller Given / When / Then criteria appear in Chapter 18.

Error and permission states

- Where a requirement can fail or be refused, the required behaviour is the same throughout this document: every state says what happened and what to do next, and no state exposes an internal authorization code, stack trace or identifier to a user.
- The specific states are enumerated per capability area in Chapters 10 to 15 and in the UI/UX requirements at Chapter 17.

7.1 Platform Overview

This section establishes the core structural rules of the platform: the platform model, the supported simulation formats, multi-tenant isolation, and the content validation and publishing workflow. PM-03 clarifies the platform model, PM-04 confirms both markets, PM-13 places every role in the isolation model, and PM-14 adds the return paths the governance chain needs in order to function as a set of gates.

Table 43 — Section 1 requirements — Platform Overview (24 requirements, 18 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
1.1.1	**Multi-Tenant SaaS Platform.** The platform shall operate as a multi-tenant SaaS-based Learning Management System (LMS) designed to distribute XR-based engineering simulation modules to both institutions and individual learners.	1 → 1	UNCHANGED-FRD	All seven	Shared	TEN, CFG	PM-03 clarifies that this single platform is realised as one central web portal with role-based workspaces. It must now host seven workspaces rather than one.	A single authenticated portal resolves the signed-in identity to exactly one workspace per held role.
1.1.2	**B2B Institutional Learning Environment.** The system shall support institutional deployments where organizations purchase licenses and allow students and faculty within the institution to access the platform under a shared institutional subscription.	1 → 1	UNCHANGED-FRD	Learner, Faculty, Institute Admin, Super Admin	B2B	TEN, ENT	The FRD names faculty here. Faculty access under an institutional subscription had no designed surface before this draft. Depends on 6.2.1, 7.2.2.	A faculty member and a student of the same institution both access the platform under the institution's licence, within one tenant.
1.1.3	**B2C Individual Learning Environment.** The platform shall support individual learners who independently purchase modules or learning packages and access simulations directly through personal accounts.	1 → 1	UNCHANGED-FRD	Learner, Super Admin	B2C	COM, ENT	PM-04 confirms B2B and B2C are both required and neither is sequenced ahead of the other. Depends on 6.1.1, 6.1.2, 9.1.x.	An individual learner registers, purchases a module and reaches it without any institutional affiliation.
1.1.4	**Centralized Content Ownership.** All XR modules hosted on the platform shall remain under the ownership and governance of the platform owner and shall not be modifiable or redistributable by institutions.	1 → 1	UNCHANGED-FRD	Super Admin, Content Manager, Institute Admin, Vendor	Shared	GOV, SEC	PM-06 confirms the platform ingests and governs existing simulations rather than authoring them. This requirement bounds what an Institute Admin may do to content and what a Vendor retains after submission.	No institutional role is offered any control that edits, exports or redistributes module content.
1.2.1	**Web Simulation Support.** The platform shall support browser-based simulations that can run directly within standard web environments without requiring XR hardware.	1 → 1	UNCHANGED-FRD	Learner	Shared	XRD, LRN	The only format executable inside the portal itself. Whether web simulations use the same encrypted-bundle and licence-validation path as XR formats is not stated in the FRD — TECHNICAL DETAIL TO BE DEFINED.	An entitled web simulation launches inside the portal with determinate asset loading and no XR hardware.
1.2.2	**VR Simulation Support.** The platform shall support virtual reality simulations deployed through VR headsets such as Meta Quest devices.	1 → 1	TECHNICAL-PENDING	Learner, Institute Admin, Super Admin	Shared	XRD	PM-05 requires an entitlement-aware route from the platform to the headset. The delivery mechanism is a PENDING DECISION. Depends on 2.1.1, 2.2.2.	An entitled VR simulation reaches the learner's supported headset and its state is visible in the portal, without the portal claiming to launch it.
1.2.3	**MR Simulation Support.** The platform shall support mixed reality simulations designed for MR hardware such as Ajna Lens.	1 → 1	TECHNICAL-PENDING	Learner, Institute Admin, Super Admin	Shared	XRD	FRD 2.1.2 makes MR a sideloaded application, which is administratively gated by definition. Who authorises and performs sideloading is a PENDING DECISION.	An entitled MR simulation is presented with an honest device requirement and an accurate availability state.
1.2.4	**AR Simulation Support.** The platform shall support augmented reality simulations that can run on mobile devices using Android and iOS operating systems.	1 → 1	TECHNICAL-PENDING	Learner	Shared	XRD	GAP: FRD Section 2 defines an application architecture for VR (2.1.1) and MR (2.1.2) only. There is no equivalent requirement for AR — no container, distribution channel, packaging requirement or update path. AR must not be assumed to behave like VR.	An entitled AR simulation is presented with an honest device requirement; no delivery behaviour is asserted that the FRD does not define.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
1.3.1	**Tenant-Wise Database Isolation.** Each institution's data shall be stored in logically separated tenant environments to prevent cross-access between institutions. This isolation shall apply to user records, analytics data, licensing data, and institution-specific course mapping.	1 → 1	UNCHANGED-FRD	Institute Admin, Super Admin	B2B	TEN, SEC	Must now scope five administrative workspaces, not one learner view. PM-13 places Institute Admin, Faculty and Learner on the tenant plane.	No tenant-plane role can see, name or reach data belonging to another institution, and no tenant switcher exists for them.
1.3.2	**B2C Database Isolation.** B2C users shall be maintained in a separate database environment that is independent of institutional databases to ensure clean separation between consumer and institutional data.	1 → 1	UNCHANGED-FRD	Super Admin	B2C	TEN, SEC	PM-13 confirms B2C individual learners use shared platform capabilities without becoming an institutional tenant role. Whether one person may hold both a B2B and a B2C identity, and whether those link, is a PENDING DECISION.	A B2C account's data is held apart from every institutional tenant environment.
1.3.3	**B2C Database Expansion.** The platform shall support deployment of additional B2C database instances across different geographic regions to ensure system performance and reduce latency for large user clusters.	1 → 1	UNCHANGED-FRD	No role-facing surface	B2C	TEN	Infrastructure behaviour with no role-facing surface. Region selection and deployment topology are TECHNICAL DETAIL TO BE DEFINED.	The data architecture permits additional regional B2C instances without a change to the role or permission model.
1.3.4	**Tenant-Scoped Analytics.** Analytics and reporting generated for each institution shall be limited to that tenant's dataset and shall not include data from other institutions.	1 → 1	UNCHANGED-FRD	Institute Admin, Faculty, Super Admin	B2B	RPT, TEN	A hard bound on every reporting surface in every workspace, including the operational status permitted by PM-11.	Every Faculty and Institute Admin reporting surface returns only that institution's data, with no cross-tenant aggregate anywhere.
1.3.5	**Cross-Tenant Governance Access.** A Super Admin role shall have controlled access to view and manage data across tenants strictly for governance and operational oversight.	1 → 1	UNCHANGED-FRD	Super Admin	Shared	TEN, SEC, RPT	The only cross-tenant access the FRD grants. PM-13 explicitly preserves it and forbids inventing additional cross-tenant permissions for any other role. No designed surface existed before this draft.	Only Super Admin can reach more than one tenant, and that access is presented as governance and operational oversight rather than unrestricted data access.
1.4.1	**Vendor Module Submission.** Authorized vendors shall submit XR module builds and associated assets through the platform's staging environment for review.	1 → 1	PM-CLARIFIED	Vendor, Content Manager	Shared	GOV	PM-07 promotes Vendor from an FRD actor to a confirmed role with a login and workspace. PM-13 places the Vendor on the central content plane. PM-14 adds correction and resubmission after a return. The submission package format is a PENDING DECISION (see 2.2.1).	An authorised vendor creates a submission, uploads a build and assets, supplies metadata and submits it; the submission appears in the Academic Reviewer's queue.
1.4.2	**Academic Validation by BITS Pilani.** All submitted modules shall undergo academic validation by the BITS Pilani academic team to ensure alignment with engineering curriculum standards.	1 → 1	PM-CLARIFIED	Academic Reviewer	Shared	GOV	PM-07 promotes Academic Reviewer to a confirmed role with a login and workspace. PM-14 adds an explicit request-changes decision with comments. GAP: the curriculum standards validated against are not defined anywhere in the FRD — this is content someone must own.	An Academic Reviewer records either academic validation or a request for changes with a stated reason; no module advances past this gate without a recorded decision.
1.4.3	**Content Manager Approval.** Following academic validation, the Content Manager shall review metadata, documentation, and platform compatibility before approving the module for publication.	1 → 1	PM-CLARIFIED	Content Manager	Shared	GOV, CAT	Defines the Content Manager workspace's core job and aligns exactly with 8.1.2. PM-14 adds a request-changes decision with comments. The module metadata schema reviewed here is not enumerated in the FRD.	A Content Manager reviews metadata, documentation and platform compatibility, reads the academic validation record read-only, and records either approval or a request for changes.
1.4.4	**Production Publishing by Super Admin.** Only the Super Admin shall have permission to publish approved modules to the production environment.	1 → 1	UNCHANGED-FRD	Super Admin	Shared	GOV	Publishing authority is exclusive. PM-14 forbids any automatic publication path after a return: publication is reachable only through recorded academic validation and content readiness approval.	No role other than Super Admin can publish, and a module reaches the publish queue only after both prior gates have recorded approval.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
1.4.5	**Restricted Vendor Publishing.** Third-party vendors shall not have direct publishing rights to the production platform.	1 → 1	UNCHANGED-FRD	Vendor, Super Admin	Shared	GOV, SEC	The FRD's only explicit role prohibition. It must be visibly respected rather than silently absent: the Vendor workspace shows where a submission is, never a disabled publish control.	No publish action exists in the Vendor workspace in any state, and no Vendor-initiated path reaches production.
1.5.1	**Module Lifecycle States.** Each XR module shall maintain lifecycle states including Draft, Under Review, Approved, Published, and Archived.	2 → 2	FUTURE-PHASE	Content Manager, Super Admin	Shared	CAT	Phase 2 in the original FRD and unchanged. Phase 1 governance therefore runs on gate position (submitted, validated, approved, published) derived from 1.4.1-1.4.4, not on this lifecycle vocabulary.	Not required in Phase 1.
1.5.2	**Academic Version Locking.** Institutions shall be able to lock a specific version of a module during an academic semester to ensure learning continuity.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	CAT, TEN	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
1.6.1	**Version Tracking.** The platform shall maintain version history for each XR module to track updates and revisions.	2 → 2	FUTURE-PHASE	Content Manager, Super Admin	Shared	CAT	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
1.6.2	**Version Rollback.** Administrators shall be able to revert modules to earlier versions in case of technical or academic issues.	2 → 2	FUTURE-PHASE	Super Admin	Shared	CAT	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
1.7.1	**Regional CDN Distribution.** XR asset bundles shall be distributed through regional CDN nodes to reduce download latency.	3 → 3	FUTURE-PHASE	No role-facing surface	Shared	XRD	Phase 3 in the original FRD and unchanged. Restated at 12.4.1.	Not required in Phase 1.
1.7.2	**Multi-Region Database Replication.** The platform shall support multi-region database replication to ensure availability and performance across global deployments.	3 → 3	FUTURE-PHASE	No role-facing surface	Shared	TEN	Phase 3 in the original FRD and unchanged. Restated at 12.4.2.	Not required in Phase 1.

7.2 XR Content Delivery Architecture

This section defines how XR simulation modules are packaged, delivered, updated and licence-controlled across supported devices. It is the section most affected by PM-05: the product requirement for entitlement-aware delivery is confirmed and already FRD-supported, while the technical mechanism is deliberately unselected. It is also the section carrying the largest recorded gap — no application architecture exists for AR.

Table 44 — Section 2 requirements — XR Content Delivery Architecture (17 requirements, 11 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
2.1.1	**Master VR Application.** A centralized VR application shall be deployed through the Meta Quest Store which will act as the container for downloading and launching XR simulation modules.	1 → 1	TECHNICAL-PENDING	Learner, Institute Admin, Super Admin	Shared	XRD	This FRD text describes a centralized container closely matching PM-05 Concept B, for VR only. PM-05 nevertheless made no selection between direct package delivery and a centralized application, so the mechanism remains a PENDING DECISION and this draft selects none.	The learner-facing delivery experience is described in mechanism-neutral terms that remain correct under either concept.
2.1.2	**MR Device Application.** Mixed Reality simulations shall run through a sideloaded application deployed on Ajna Lens devices.	1 → 1	TECHNICAL-PENDING	Learner, Institute Admin	B2B	XRD	Sideloading is administratively gated by definition; the FRD does not say who authorises or performs it. Recorded as a PENDING DECISION rather than assigned to Institute Admin.	MR access is presented as an institutionally provisioned route, with no self-service sideloading offered to a learner.
2.2.1	**XR Asset Bundle Packaging.** Simulation modules shall be packaged as encrypted asset bundles that are dynamically downloaded by the master XR application.	1 → 1	TECHNICAL-PENDING	No role-facing surface	Shared	XRD, SEC	The package format is stated only as 'encrypted asset bundles'. No file type, container or manifest format is defined — TECHNICAL DETAIL TO BE DEFINED. A vendor cannot yet be told precisely what to upload.	Delivery and upload surfaces describe the artefact generically until a format is defined; no format is asserted.
2.2.2	**License-Based Module Access.** Simulation modules shall only be downloaded after license validation is confirmed by the platform.	1 → 1	UNCHANGED-FRD	Learner, Super Admin	Shared	XRD, ENT	This is the existing FRD basis for PM-05's entitlement-aware delivery. Entitlement-aware delivery is therefore not a new requirement — it is an existing Phase-1 requirement that the earlier Student-only interpretation surfaced only as a restricted state.	No module is delivered to a device before the platform confirms a valid licence, and a failed validation produces an explicit, code-free explanation.
2.3.1	**Automatic Module Updates.** The XR application shall automatically download updated versions of simulation modules when available.	1 → 1	TECHNICAL-PENDING	Learner	Shared	XRD	Creates an 'update available' state the learner may see. Update behaviour depends on the pending delivery mechanism.	An updated module produces a visible 'update available' state; an unrequested update never blocks a learner from continuing.
2.3.2	**Controlled Update Releases.** Super Admin shall control when new versions of modules are released to production environments.	1 → 1	UNCHANGED-FRD	Super Admin	Shared	GOV, CFG	A Super Admin release-control surface with no prior design. Also the origin of mid-course content change; Phase 1 has no lifecycle states (1.5.1 is Phase 2) and no notification requirement.	A new module version reaches production only when Super Admin releases it; holding a release is an available and visible outcome.
2.3.3	**Module Update Without Application Reinstallation.** Simulation modules shall be updated independently of the master XR application to avoid requiring full application reinstallation.	1 → 1	TECHNICAL-PENDING	No role-facing surface	Shared	XRD	Constrains the eventual delivery model. No mechanism is selected by this draft.	The delivery model chosen must permit module-level update without reinstalling the device application.
2.4.1	**B2C Device Binding.** Individual user licenses shall be bound to a specific encrypted XR device ID and a user account.	1 → 1	PRODUCT-PENDING	Learner, Super Admin	B2C	ENT, XRD	A Phase-1 requirement with no designed surface in any workspace before this draft. Whether a mobile phone qualifies as an 'XR device ID' for AR is not stated.	A B2C learner can see which device a licence is bound to, and the binding is enforced at access and at execution.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
2.4.2	**Device Replacement Authorization.** Users requesting device replacement must receive approval from the Super Admin before transferring their license to a new device.	1 → 1	PRODUCT-PENDING	Learner, Super Admin	B2C	ENT, XRD	A two-role, cross-plane approval workflow; neither end existed before this draft. The FRD describes no rejection path, no evidence requirement and no limit on replacement frequency — all PENDING DECISION.	A learner submits a replacement request; the licence transfers only after Super Admin approval, and the request's state is visible to the learner throughout.
2.4.3	**Institutional Device Access.** Institutional deployments shall allow multiple students to access simulations on shared XR devices.	1 → 1	UNCHANGED-FRD	Learner, Institute Admin	B2B	XRD, TEN	Explains why B2B has no personal device binding: the institutional model is a shared device fleet.	An institutional learner reaches a simulation on a shared device without any personal device binding being asserted.
2.4.4	**Session-Based Device Login.** Students in institutional environments must log into the XR application using their LMS credentials to access licensed modules.	1 → 1	FRD-INTERNAL-CONFLICT	Learner, Institute Admin	B2B	IAM, XRD	TENSION: FRD 4.2.2 with PM-10 makes institutional SSO required, and an SSO-authenticated learner may hold no password to type into a headset. Both requirements are Phase 1 and they collide. Recorded, not designed around.	The device sign-in route for an institutional learner is stated only once a mechanism compatible with institutional SSO is defined.
2.5.1	**Device Registration.** Institutions shall be able to register XR devices on the platform to track device ownership and usage.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	XRD	Phase 2 in the original FRD and unchanged. Keeps device-management screens out of Phase 1.	Not required in Phase 1.
2.5.2	**Device Metadata Tracking.** The system shall record device metadata including device ID, device type, institution, and lab location.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	XRD	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
2.6.1	**Remote Module Control.** Administrators shall be able to enable or disable modules on specific XR devices remotely.	2 → 2	FUTURE-PHASE	Institute Admin, Super Admin	B2B	XRD	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
2.6.2	**Remote Application Updates.** XR application updates may be pushed remotely to registered devices.	2 → 2	FUTURE-PHASE	Super Admin	B2B	XRD	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
2.7.1	**Simulation Interaction Logs.** The platform shall capture interaction logs from XR sessions to analyze how learners interact with simulations.	3 → 3	FUTURE-PHASE	Faculty, Super Admin	Shared	RPT	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.
2.7.2	**Simulation Error Tracking.** The system shall record errors encountered by users during simulation scenarios to identify common learning difficulties.	3 → 3	FUTURE-PHASE	Faculty, Super Admin	Shared	RPT	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.

Recorded gap — no AR application architecture

- Platform FRD_v4 requires AR simulation support on Android and iOS at requirement 1.2.4.
- Section 2 then defines an application architecture for exactly two targets: a master VR application (2.1.1) and a sideloaded MR application (2.1.2). There is no equivalent requirement for AR.
- AR therefore has no container, no distribution channel, no packaging requirement and no update path anywhere in the document. Neither of the two delivery concepts the PM described is grounded for AR.
- This gap is flagged for technical resolution. It is not filled here, and AR is not assumed to behave like VR.

7.3 Content Organization and Learning Structure

This section defines the content hierarchy: modules, courses and certification programs. PM-09 confirms the learning progression built on this hierarchy. PM-12 moves the two certification requirements out of Phase 1, which is recorded here as a deviation from the FRD as written.

Table 45 — Section 3 requirements — Content Organization and Learning Structure (11 requirements, 6 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
3.1.1	**Simulation Module Definition.** A module shall represent the smallest learning unit and shall contain a single XR simulation experience.	1 → 1	UNCHANGED-FRD	Learner, Faculty, Content Manager	Shared	CAT, LRN	The atomic unit of the whole content model and the object every governance gate acts upon.	Every module presents exactly one simulation experience; no module offers a choice between simulations.
3.1.2	**Module Learning Components.** Modules may include supporting learning material such as theoretical explanations, simulation instructions, and assessments.	1 → 1	UNCHANGED-FRD	Learner, Faculty, Content Manager	Shared	CAT, LRN, ASM	'May' makes all three optional. Every module renderer must tolerate the absence of any component without appearing broken.	A module with no resources, no assessment, or neither, renders correctly and its completion model adjusts accordingly.
3.2.1	**Course Composition.** A course shall consist of a collection of modules grouped around a specific engineering topic or learning objective.	1 → 1	UNCHANGED-FRD	Learner, Faculty, Institute Admin	Shared	CAT, LRN	A structural grouping. Phase-1 purchase requirements explicitly define individual module and bundle purchasing (6.1.1, 6.1.2); direct course purchase is not specified and is not included in the current design unless subsequently approved.	A course lists its modules in their defined order. No purchase capability is asserted for a course, and none is prohibited.
3.2.2	**Course Module Sequencing.** Courses shall define the recommended sequence in which modules must be completed.	1 → 1	UNCHANGED-FRD	Learner, Faculty	Shared	LRN, PRG	Whether the sequence is a platform constant or a per-course setting a Faculty member may change is not stated by the FRD and is not settled by PM-09, which approved the model rather than its mutability. PENDING DECISION.	A learner cannot open a later module in a course before the prerequisite module is complete, and the reason is stated on the locked item.
3.3.1	**Certification Program Structure.** Certification programs shall consist of one or more courses grouped into a structured learning pathway.	1 → 2	PM-DEFERRED	No Phase-1 role-facing surface	Shared	CAT	DEVIATION RECORDED. The original FRD places this in Phase 1. PM-12 moves certification to Phase 2. Whether the structural program node is still modelled in the content hierarchy is deliberately not assumed either way.	No certification structure is presented in any Phase-1 workspace.
3.3.2	**Certification Completion Criteria.** Students shall receive certification upon completing all required courses and meeting minimum assessment performance thresholds.	1 → 2	PM-DEFERRED	No Phase-1 role-facing surface	Shared	CAT, PRG	DEVIATION RECORDED. The original FRD places this in Phase 1. PM-12 moves it to Phase 2. The FRD never quantifies the minimum thresholds anywhere.	No certificate artefact, issuance flow or eligibility surface is built in Phase 1.
3.4.1	**Institutional Course Creation.** Institute Admins shall be able to create courses by grouping available simulation modules.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	CAT, TEN	Phase 2 in the original FRD and unchanged. Creates tension T-1 with the Phase-1 Institute Admin role definition at 8.1.3, which names course structures.	Not required in Phase 1; in Phase 1 institutional course structure arrives through ERP.
3.4.2	**Course Module Mapping.** Institute Admins shall be able to add or remove modules from courses within their tenant environment.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	CAT, TEN	Phase 2 in the original FRD and unchanged. Part of tension T-1.	Not required in Phase 1.
3.4.3	**Course Stream Assignment.** Courses shall be mapped to specific academic streams or departments within an institution.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	CAT, TEN	Phase 2 in the original FRD and unchanged. Distinct from the Phase-1 stream-to-package mapping at 6.2.1 and 7.2.3.	Not required in Phase 1.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
3.5.1	**Student Certification Progress.** Students shall be able to track progress toward certification programs through their dashboard.	3 → 3	FUTURE-PHASE	Learner	Shared	PRG	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.
3.5.2	**Certification Eligibility Reports.** Faculty and institutional administrators shall be able to view reports identifying students who have met certification eligibility criteria.	3 → 3	FUTURE-PHASE	Faculty, Institute Admin	B2B	RPT	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.

Recorded deviation — certification

- Requirements 3.3.1 and 3.3.2 are Phase 1 in Platform FRD_v4 as written.
- PM-12 moves both to Phase 2. The original phase is preserved in the table above and the working phase is shown alongside it.
- The FRD Phase-1 requirement count remains 78 as a document fact. The Phase-1 design working set becomes 76 from the FRD, plus these 2 recorded as PM-deferred.
- Whether the structural certification-program node is still modelled in the content hierarchy, even though no Phase-1 certification experience is built, is deliberately not assumed either way.

7.4 User Management and Onboarding

This section defines how users join the platform and configure their profiles. Two Phase-1 requirements had no designed surface before this draft: institutional onboarding, and the SSO configuration capability that PM-10 confirms is required — a configuration surface, not merely a login button.

Table 46 — Section 4 requirements — User Management and Onboarding (9 requirements, 5 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
4.1.1	**Email Registration.** Users shall be able to create accounts using email registration.	1 → 1	UNCHANGED-FRD	Learner	B2C	IAM	Self-registration is the B2C origin path. Faculty, administrators, vendors and reviewers are provisioned, never self-registered.	Self-registration is offered only in the B2C context and never creates any role other than Learner.
4.1.2	**Social Login.** Users may register using supported social login providers.	1 → 1	PRODUCT-PENDING	Learner	B2C	IAM	The FRD never enumerates which providers are supported. Any provider shown in design material is a labelled sample, not an approved selection. PENDING DECISION.	Social sign-in is offered only in the B2C context; no provider is presented as confirmed until one is approved.
4.2.1	**Institutional Account Creation.** Institutions shall register their organization through the platform's institutional onboarding interface.	1 → 1	UNCHANGED-FRD	Institute Admin, Super Admin	B2B	IAM, TEN	Institutional onboarding had no designed surface before this draft; it was excluded purely by the earlier Student-only scope.	An institution can be registered and a tenant created, after which institutional configuration becomes available.
4.2.2	**SSO Integration.** Institutions shall be able to configure single sign-on authentication through supported identity providers.	1 → 1	FRD-INTERNAL-CONFLICT	Institute Admin, Super Admin	B2B	IAM, TEN, CFG	CONFLICT RESOLVED: the FRD places SSO configuration in Phase 1 here and SSO authentication support in Phase 3 at 7.4.1. PM-10 confirms the Phase-1 reading. The requirement is a configuration capability, not merely a login button; the Institute Admin configuration surface is new. Provider, protocol and attribute mapping are a PENDING DECISION.	An Institute Admin can configure, test and enable institutional SSO for their own tenant, and a failed test is reported explicitly.
4.3.1	**Profile Information Collection.** Users shall provide profile details including academic institution, specialization, and learning preferences.	1 → 1	PRODUCT-PENDING	Learner, Faculty	Shared	IAM	Now applies to Faculty profiles too, not only learners. Which fields are user-editable and which are ERP-owned is undefined and conflicts with the ERP authority at 7.1.1 — PENDING DECISION.	ERP-sourced fields are presented read-only with their provenance stated; user-editable fields are not silently overwritten without notice.
4.4.1	**Learning Progress Tracking.** The platform shall track modules accessed, modules completed, and time spent on simulations.	2 → 2	FUTURE-PHASE	Learner, Faculty, Institute Admin	Shared	PRG	Phase 2 in the original FRD. Note that equivalent Phase-1 recording requirements exist at 11.1.1 and 11.1.2.	Not required in Phase 1 beyond the recording required by 11.1.1 and 11.1.2.
4.4.2	**Assessment Performance Tracking.** Student assessment scores shall be recorded and linked to the learner profile.	2 → 2	FUTURE-PHASE	Learner, Faculty	Shared	PRG, ASM	Phase 2 in the original FRD. Equivalent Phase-1 recording exists at 11.2.1.	Not required in Phase 1 beyond the recording required by 11.2.1.
4.5.1	**Personalized Module Recommendations.** The platform shall recommend modules based on the user's learning history and performance.	2 → 2	FUTURE-PHASE	Learner	Shared	LRN	Phase 2 in the original FRD and unchanged. The only consumer of the Phase-1 'learning preferences' profile field.	Not required in Phase 1.
4.5.2	**Certification Path Recommendations.** Students shall receive recommendations for certification programs aligned with their learning progress.	2 → 2	FUTURE-PHASE	Learner	Shared	LRN	Phase 2 in the original FRD. Depends on certification, which PM-12 places in Phase 2.	Not required in Phase 1.

7.5 Learning Experience Features

This section defines assessment authoring, question types and quiz configuration. PM-08 confirms that assessment behaviour is configuration, not constants, and PM-15 assigns ownership to Faculty or the authorized institution side for both markets. A scope catch is recorded here: AI question generation is a Phase-1 Faculty requirement that had been treated as excluded on the basis of a different, Phase-3 requirement.

Table 47 — Section 5 requirements — Learning Experience Features (11 requirements, 7 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
5.1.1	**Question Bank Upload.** Faculty shall be able to upload quiz question banks through structured spreadsheet templates.	1 → 1	UNCHANGED-FRD	Faculty	B2B and B2C	ASM	The authoring half of the assessment model; no surface existed before this draft. PM-15 confirms Faculty or the authorized institution side owns this for both B2B and B2C. The template structure is not specified by the FRD.	A Faculty member uploads a structured question bank; invalid rows are reported individually rather than failing the whole upload silently.
5.1.2	**AI Question Generation.** Faculty shall be able to generate assessment questions using AI tools integrated into the platform.	1 → 1	PRODUCT-PENDING	Faculty	B2B and B2C	ASM	SCOPE CATCH RECORDED. This is a Phase-1 Faculty requirement. It was previously covered by an exclusion that cited FRD 5.5 (AI Learning Assistant), which is Phase 3 and learner-facing — a different requirement, phase and user. It is reinstated as in-scope and PRODUCT-PENDING. Human review of generated questions before use is a recommendation in this draft, not an approved requirement.	Generated questions enter the question bank in a reviewable state; no generated question reaches a learner without the Faculty owner accepting it.
5.2.1	**Multiple Choice Questions.** The system shall support standard MCQ-based assessments.	1 → 1	UNCHANGED-FRD	Learner, Faculty	Shared	ASM	Whether MCQs are single-select or multi-select is not stated; multi-select would require a partial-credit rule that no scoring model defines. PENDING DECISION.	An MCQ renders, is answerable and is scored automatically according to the assessment configuration.
5.2.2	**True/False Questions.** The system shall support binary true/false assessment questions.	1 → 1	UNCHANGED-FRD	Learner, Faculty	Shared	ASM	Automatically scorable, like MCQ.	A true/false question renders, is answerable and is scored automatically.
5.2.3	**Descriptive Questions.** The system shall support short descriptive answer questions.	1 → 1	PRODUCT-PENDING	Learner, Faculty	Shared	ASM	No Phase-1 requirement defines who grades descriptive answers, on what turnaround, or against what rubric. This is compounded by sequencing: progression gates on completion, so an ungraded answer can strand a learner indefinitely. Grading ownership must not be invented — PENDING DECISION.	A submitted descriptive answer enters an explicit awaiting-evaluation state that the learner can understand; no grading owner is asserted.
5.3.1	**Randomized Question Order.** Quiz questions may be randomized for each attempt.	1 → 1	UNCHANGED-FRD	Faculty	Shared	ASM	'May' makes this a configurable setting rather than a constant. PM-08 confirms it belongs in the assessment configuration.	Randomization is an explicit setting whose current value is disclosed to the learner before an attempt begins.
5.3.2	**Negative Marking.** Faculty shall be able to enable negative marking for incorrect responses.	1 → 1	UNCHANGED-FRD	Faculty, Learner	Shared	ASM	The FRD makes this explicitly Faculty-configurable. Any fixed penalty value is SAMPLE CONFIGURATION, never a platform constant. PM-15 confirms Faculty or the authorized institution side owns it for both B2B and B2C.	Negative marking is enabled or disabled per assessment, and the pre-attempt disclosure states the configured behaviour rather than fixed copy.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
5.4.1	**Simulation Performance Scoring.** Scores generated during XR simulations shall be mapped to student performance records.	2 → 2	FUTURE-PHASE	Learner, Faculty	Shared	ASM, PRG	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
5.4.2	**Simulation Attempt Tracking.** The system shall track the number of simulation attempts made by each learner.	2 → 2	FUTURE-PHASE	Learner, Faculty	Shared	PRG	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
5.5.1	**Concept Explanation Support.** The AI assistant shall provide explanations for engineering concepts related to simulation modules.	3 → 3	FUTURE-PHASE	Learner	Shared	LRN	Phase 3 in the original FRD and unchanged. Distinct from 5.1.2, which is a Phase-1 Faculty authoring capability.	Not required in Phase 1.
5.5.2	**Practice Question Generation.** The AI assistant shall generate practice questions for students preparing for assessments.	3 → 3	FUTURE-PHASE	Learner	Shared	ASM	Phase 3 in the original FRD and unchanged. Distinct from 5.1.2.	Not required in Phase 1.

Recorded scope catch — AI question generation

- Requirement 5.1.2, AI question generation, is a Phase-1 Faculty authoring capability.
- It had previously been treated as out of scope on the basis of an exclusion that cited requirement 5.5, the AI Learning Assistant — which is Phase 3 and learner-facing. Different requirement, different phase, different user.
- 5.1.2 is reinstated here as an in-scope Phase-1 requirement classified PRODUCT-PENDING: whether to build it in Phase 1 is a decision, not an inherited exclusion.
- Human review of AI-generated questions before use is a recommendation in this draft, not an approved requirement.

7.6 Licensing and Monetization

This section defines the licensing models for individual learners and institutional customers. Both are Phase 1 and PM-04 confirms neither is sequenced ahead of the other. Two Phase-1 tensions with Phase-2 requirements are recorded: bundles are purchasable before anything creates them, and institutional licences must exist before institutional billing exists.

Table 48 — Section 6 requirements — Licensing and Monetization (10 requirements, 4 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
6.1.1	**Individual Module Purchase.** Users shall be able to purchase individual simulation modules.	1 → 1	UNCHANGED-FRD	Learner, Super Admin	B2C	COM, ENT	Phase-1 purchase requirements explicitly define individual module and bundle purchasing. Direct course purchase is not specified and is not included in the current design unless subsequently approved (see QP-28).	An individual learner can purchase a single module and reach it immediately after activation.
6.1.2	**Bundle Purchases.** Users shall be able to purchase bundles of multiple modules.	1 → 1	UNCHANGED-FRD	Learner, Super Admin	B2C	COM, ENT	TENSION: bundles are purchasable in Phase 1, but bundle creation (6.3.1) and pricing configuration (6.3.2) are Phase 2. Who creates bundles and sets prices in Phase 1 is a PENDING DECISION; no Phase-1 bundle-authoring surface is invented.	A bundle purchase entitles the learner to every module the bundle contains, each reachable individually.
6.2.1	**Stream-Based Licensing.** Institutions shall purchase packages mapped to academic streams.	1 → 1	PRODUCT-PENDING	Institute Admin, Super Admin	B2B	ENT, TEN	The institutional purchasing and licence surface had no prior design. TENSION: institutional billing (9.3.1, 9.3.2) is Phase 2, so Phase 1 requires institutional licences to exist without defining how they are transacted in-platform. PENDING DECISION.	An institutional licence exists against a tenant and maps to academic streams; no in-platform purchase flow is asserted until the transaction mechanism is decided.
6.2.2	**Capped Institutional License.** Institutions may purchase capped licenses based on maximum student count.	1 → 1	UNCHANGED-FRD	Institute Admin, Super Admin	B2B	ENT, TEN	The learner meets the cap as an access wall. The Institute Admin must be able to see seat utilisation before learners reach it.	Seat utilisation against the cap is visible to the Institute Admin before exhaustion, and an exhausted cap produces an explicit learner-facing explanation without a purchase prompt.
6.3.1	**Bundle Creation.** Super Admin shall be able to create bundles of simulation modules.	2 → 2	FUTURE-PHASE	Super Admin	B2C	COM	Phase 2 in the original FRD and unchanged. Creates the Phase-1 tension recorded against 6.1.2.	Not required in Phase 1.
6.3.2	**Pricing Configuration.** Pricing rules shall be configurable through the admin dashboard.	2 → 2	FUTURE-PHASE	Super Admin	Shared	COM	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
6.4.1	**Coupon Generation.** Administrators shall be able to generate coupon codes for promotional campaigns.	2 → 2	FUTURE-PHASE	Super Admin	B2C	COM	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
6.4.2	**Coupon Expiry Configuration.** Coupons shall include configurable expiration dates.	2 → 2	FUTURE-PHASE	Super Admin	B2C	COM	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
6.5.1	**License Activation.** Licenses shall activate automatically after successful payment.	3 → 3	FUTURE-PHASE	Learner	B2C	ENT	Phase 3 in the original FRD as a licence-lifecycle requirement. Note that automated activation on payment is separately required in Phase 1 at 9.2.1.	Phase-1 behaviour is governed by 9.2.1.
6.5.2	**Grace Period.** Institutional licenses shall support a configurable grace period after expiry.	3 → 3	FUTURE-PHASE	Institute Admin, Super Admin	B2B	ENT	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.

7.7 Enterprise Integrations

This section defines integration with institutional ERP systems and identity providers. PM-10 resolves the FRD's own contradiction over the phase of SSO. The ERP requirements imply an Institute Admin configuration and status surface that had no prior design, and they raise a source-of-truth question against the profile requirement at 4.3.1.

Table 49 — Section 7 requirements — Enterprise Integrations (9 requirements, 5 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
7.1.1	**Supported ERP Systems.** The platform shall support integration with widely used academic ERP systems including Oracle ERP, Moodle-based LMS environments, and other ERP platforms through standardized API integration.	1 → 1	TECHNICAL-PENDING	Institute Admin, Super Admin	B2B	ERP	The specific provider and the integration contract are undetermined for any given institution — PENDING DECISION. The named systems are examples in the FRD, not a confirmed selection for this programme.	ERP integration is configured per tenant against a provider-neutral connection model; no provider is presented as selected.
7.1.2	**ERP API Integration.** The platform shall provide API endpoints that allow external ERP systems to push student enrollment data, course mappings, and academic structure information into the LMS.	1 → 1	TECHNICAL-PENDING	Institute Admin, Super Admin	B2B	ERP	An Institute Admin visible integration-status surface is implied by this requirement and had no prior design. In Phase 1 institutional course structure arrives from ERP rather than from in-platform authoring, because 3.4.1 and 3.4.2 are Phase 2.	Inbound ERP data is accepted, validated and reflected in tenant state, with the outcome of each run visible to the Institute Admin.
7.2.1	**Scheduled ERP Synchronization.** The system shall support scheduled synchronization with institutional ERP systems at configurable intervals to update student records and course assignments.	1 → 1	UNCHANGED-FRD	Institute Admin	B2B	ERP, CFG	'Configurable intervals' implies an Institute Admin configuration surface that had no prior design. Entitlements may be added or removed between sessions; no Phase-1 notification requirement exists anywhere in the FRD.	An Institute Admin sets the synchronisation interval and can see the last run, its outcome, its counts and its errors.
7.2.2	**ERP-Based Account Creation.** Student accounts shall be automatically created when student records are received from ERP systems during synchronization.	1 → 1	UNCHANGED-FRD	Learner, Institute Admin	B2B	ERP, IAM	A different account-origin path from 4.1.1, not a contradiction of it. Institutional learners are provisioned; they do not self-register.	A student record received from ERP produces an account the learner can sign into, with entitlements already in place.
7.2.3	**Stream-Based Access Assignment.** Students shall be automatically assigned simulation packages based on the academic stream information received from the ERP system.	1 → 1	UNCHANGED-FRD	Learner, Institute Admin, Faculty	B2B	ERP, ENT	The Institute Admin needs stream-to-package mapping visibility, which had no prior design. Depends on 6.2.1.	A student in a mapped stream receives the mapped package automatically; a student in an unmapped stream sees an empty entitlement state that explains what to do.
7.3.1	**First Login Provisioning.** When a student logs into the platform for the first time, the system shall trigger a real-time API request to the ERP system to retrieve the latest academic profile.	2 → 2	FUTURE-PHASE	Learner, Institute Admin	B2B	ERP	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
7.3.2	**Dynamic Entitlement Assignment.** Simulation module access shall be automatically assigned based on the course mapping received from the ERP system.	2 → 2	FUTURE-PHASE	Learner, Institute Admin	B2B	ERP, ENT	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
7.4.1	**SSO Authentication Support.** The platform shall support institutional single sign-on authentication through providers such as Google Workspace, Microsoft Azure Active Directory, and other identity providers.	3 → 1	FRD-INTERNAL-CONFLICT	Learner, Faculty, Institute Admin	B2B	IAM	CONFLICT RESOLVED. The original FRD places this in Phase 3 while placing SSO configuration in Phase 1 at 4.2.2. PM-10 confirms institutional SSO is required, so the working phase is Phase 1. The named providers are FRD examples, not a selection; provider and protocol remain a PENDING DECISION.	Institutional SSO is an available authentication path for a configured tenant, and a failed assertion produces an explicit failure state.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
7.4.2	**Role-Based Access Mapping.** User roles such as Student, Faculty, and Institute Admin shall be automatically mapped based on identity provider attributes.	3 → 3	FUTURE-PHASE	Institute Admin, Super Admin	B2B	IAM	Phase 3 in the original FRD and unchanged. Its absence in Phase 1 leaves the Phase-1 origin of institutional role assignment undefined — a recorded PENDING DECISION, not an invented mechanism.	Not required in Phase 1; Phase-1 role assignment origin is recorded as pending.

7.8 Administration and Role Management

This section defines the administrative control structure. PM-01 brings all five FRD-defined roles into scope and PM-07 adds two more. Two internal phase tensions are recorded here: the Phase-1 Institute Admin definition names course structures while course creation is Phase 2, and the Phase-1 Faculty definition names performance monitoring while class-level reports are Phase 2.

Table 50 — Section 8 requirements — Administration and Role Management (9 requirements, 5 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
8.1.1	**Super Admin Role.** The Super Admin role shall have full administrative control over platform configuration, module publishing, licensing management, and cross-tenant analytics.	1 → 1	UNCHANGED-FRD	Super Admin	Shared	CFG, GOV, ENT, RPT	Four distinct capability areas, none with a prior designed surface. Per PM-11 the analytics area is operational oversight only in Phase 1; detailed metrics remain pending. Cross-tenant access is bounded by 1.3.5 to governance and operational oversight.	A Super Admin can configure the platform, publish modules, manage licences and see cross-tenant operational status, with no detailed analytics dashboard asserted.
8.1.2	**Content Manager Role.** The Content Manager role shall manage simulation module uploads, module metadata updates, and content readiness for publishing.	1 → 1	UNCHANGED-FRD	Content Manager	Shared	GOV, CAT	Aligns exactly with 1.4.3. PM-13 places the Content Manager on the central content plane; the role holds no tenant access and needs none.	A Content Manager can review and edit module metadata and hand cleared content to Super Admin for publication, without any publish capability.
8.1.3	**Institute Admin Role.** The Institute Admin role shall manage institutional configurations including course structures, student access, and institutional reporting.	1 → 1	FRD-INTERNAL-CONFLICT	Institute Admin	B2B	TEN, IAM, CAT, RPT	TENSION T-1 RECORDED: this Phase-1 role definition names course structures, but the requirements that would let an Institute Admin author a course (3.4.1, 3.4.2) are Phase 2. Working interpretation: in Phase 1 the Institute Admin views, maps and assigns structure that arrives from ERP, and does not create courses. Institutional reporting is bounded to operational status by PM-11 and to the tenant by 1.3.4.	An Institute Admin manages tenant configuration, student access and operational institutional status, with no course-authoring capability offered.
8.1.4	**Faculty Role.** Faculty users shall manage course assignments, assessments, and student performance monitoring.	1 → 1	FRD-INTERNAL-CONFLICT	Faculty	B2B and B2C	LRN, ASM, RPT	TENSION T-2 RECORDED: this Phase-1 role definition names student performance monitoring, but class-level performance reports (11.3.1) are Phase 2. Working interpretation per PM-11: Phase-1 Faculty sees operational learner status, not aggregated analytics. PM-15 confirms Faculty or the authorized institution side owns assessment configuration for both B2B and B2C.	A Faculty member assigns content, configures assessments, and sees who has completed, who is blocked and what attempts exist — without any aggregated performance report.
8.1.5	**Student Role.** Students shall access licensed modules, complete simulations, and view personal learning progress.	1 → 1	UNCHANGED-FRD	Learner	Shared	LRN, XRD, PRG	Reclassified from the whole product to one workspace among seven. The learner-facing design produced under the earlier interpretation remains reusable, with assessment values read from configuration and XR access extended to entitlement-aware delivery.	A learner reaches only entitled modules, completes them in the approved sequence, and sees their own progress and no one else's.
8.2.1	**Institutional Usage Monitoring.** Institute Admins shall have access to dashboards displaying module usage, student engagement metrics, and completion statistics.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	RPT	Phase 2 in the original FRD and unchanged. PM-11 agrees with this phasing: dashboards are not a Phase-1 deliverable.	Not required in Phase 1.

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
8.2.2	**License Utilization Tracking.** Administrators shall be able to monitor license utilization to determine whether institutional licenses are fully utilized.	2 → 2	FUTURE-PHASE	Institute Admin, Super Admin	B2B	RPT, ENT	Phase 2 in the original FRD. The narrower Phase-1 operational seat-utilisation view permitted against 6.2.2 is not this requirement.	Not required in Phase 1 beyond operational seat utilisation under 6.2.2.
8.3.1	**Custom Role Creation.** Administrators shall be able to define custom roles with specific permissions tailored to institutional requirements.	3 → 3	FUTURE-PHASE	Super Admin	Shared	CFG, IAM	Phase 3 in the original FRD and unchanged. This is precisely why the seven-role set must be fixed in Phase 1: there is no Phase-1 mechanism for adding a role, which is why PM-07's confirmation was necessary.	Not required in Phase 1; no role may be added in Phase 1 without an explicit stakeholder confirmation.
8.3.2	**Permission-Level Access Control.** Custom roles shall support granular permission assignment across content management, reporting, and administrative actions.	3 → 3	FUTURE-PHASE	Super Admin	Shared	CFG, IAM	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.

Two recorded FRD internal phase tensions

- T-1 — Requirement 8.1.3 places “course structures” in the Phase-1 Institute Admin role definition, but the requirements that would let an Institute Admin author a course (3.4.1, 3.4.2) are Phase 2. Working interpretation: in Phase 1 institutional course structure arrives through ERP (7.1.2, 7.2.1, 7.2.3) and the Institute Admin views, maps and assigns rather than authors. This is the narrowest reading that satisfies both requirements without inventing authoring tools, and it is carried as an open question rather than settled.
- T-2 — Requirement 8.1.4 places “student performance monitoring” in the Phase-1 Faculty role definition, but class-level performance reports (11.3.1) are Phase 2. Working interpretation per PM-11: Phase-1 Faculty receives operational learner status — who has completed, who is blocked, what attempts exist — and not aggregated analytics.

7.9 Payment and Billing

This section defines payment processing and billing. Phase 1 covers individual payment, automated licence activation and entitlement assignment. No payment provider is selected, and no currency, tax, refund or invoice rule is invented by this draft.

Table 51 — Section 9 requirements — Payment and Billing (8 requirements, 4 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
9.1.1	**Indian Payment Gateway Support.** The platform shall support integration with major Indian payment gateways to allow domestic transactions.	1 → 1	TECHNICAL-PENDING	Learner	B2C	COM	No gateway is selected. Redirect versus embedded checkout is also undetermined. PENDING DECISION.	A domestic purchase completes through a provider-neutral payment step whose success, decline and indeterminate outcomes are all designed.
9.1.2	**International Payment Gateway Support.** The platform shall support international payment gateways to allow global purchases by individual learners.	1 → 1	PRODUCT-PENDING	Learner	B2C	COM	Currency handling, tax treatment and price presentation are compliance-adjacent and undefined. No tax or legal rule is invented by this draft. PENDING DECISION.	An international purchase completes without the interface asserting any currency, tax or refund rule that has not been approved.
9.2.1	**Automated License Activation.** Upon successful payment, licenses shall be automatically activated and linked to the user account.	1 → 1	UNCHANGED-FRD	Learner, Super Admin	B2C	ENT, COM	Activation is automatic; no manual step is required of the learner.	A successful payment activates the licence and links it to the account without any further learner action.
9.2.2	**Entitlement Assignment.** Purchased modules or packages shall be automatically assigned to the user's dashboard.	1 → 1	UNCHANGED-FRD	Learner	B2C	ENT	Fulfilled by the learner's My Learning surface. Receipt and invoice format for a B2C buyer is undefined; institutional GST invoicing (9.3.1) is Phase 2. PENDING DECISION.	A purchased module appears in My Learning immediately after activation and is reachable from there.
9.3.1	**GST-Compliant Invoice Generation.** Invoices generated for institutional purchases shall comply with GST regulations.	2 → 2	FUTURE-PHASE	Institute Admin, Super Admin	B2B	COM	Phase 2 in the original FRD and unchanged. Its phase is the direct cause of the Phase-1 institutional transaction question recorded against 6.2.1.	Not required in Phase 1.
9.3.2	**Institutional Subscription Billing.** The platform shall support annual institutional subscriptions with invoice generation and payment tracking.	2 → 2	FUTURE-PHASE	Institute Admin, Super Admin	B2B	COM	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
9.4.1	**Payment Status Tracking.** Administrators shall be able to track payment status for institutional accounts.	3 → 3	FUTURE-PHASE	Super Admin	B2B	COM	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.
9.4.2	**Grace Period Notifications.** Institutions shall receive automated notifications when license renewal deadlines approach.	3 → 3	FUTURE-PHASE	Institute Admin	B2B	NOT	Phase 3 in the original FRD and unchanged. This is the only place the word 'notification' appears in the FRD, which is why no Phase-1 notification system is designed.	Not required in Phase 1.

7.10 Security and Compliance

This section defines the security mechanisms protecting intellectual property, user data and system integrity. Its Phase-1 requirements now apply across seven workspaces rather than one. Password policy, multi-factor authentication and lockout behaviour are absent from the FRD entirely and are recorded as pending rather than invented.

Table 52 — Section 10 requirements — Security and Compliance (8 requirements, 6 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
10.1.1	**XR Asset Encryption.** All XR simulation asset bundles shall be encrypted before distribution to prevent unauthorized access.	1 → 1	UNCHANGED-FRD	No role-facing surface	Shared	SEC, XRD	Reinforces the absence of any download or export affordance for protected content. The encryption scheme is TECHNICAL DETAIL TO BE DEFINED.	No workspace offers a route to obtain an unprotected asset bundle.
10.1.2	**Secure Module Authorization.** Modules shall require license validation before execution within the XR application.	1 → 1	UNCHANGED-FRD	Learner	Shared	SEC, ENT, XRD	The second entitlement gate — at execution, not only at browse. A learner can pass the browse-time check at 2.2.2 and still fail this one; that must not read as a defect.	A module refuses to execute without a valid licence, and the resulting state explains what happened and what to do next without internal codes.
10.2.1	**Secure Login Authentication.** The platform shall enforce secure authentication protocols to verify user identities before granting access.	1 → 1	PRODUCT-PENDING	All seven	Shared	SEC, IAM	Now applies to all seven workspaces. Password policy, multi-factor authentication, lockout and reset behaviour are absent from the FRD entirely — PENDING DECISION rather than invented targets.	No workspace is reachable without authentication, and the authentication policy is stated once and applied to every role.
10.2.2	**Session Management.** User sessions shall automatically expire after inactivity to prevent unauthorized access.	1 → 1	UNCHANGED-FRD	All seven	Shared	SEC, IAM	The principle that expiry must not destroy unsaved work now extends to administrative workspaces — an Institute Admin mid-SSO-configuration or a Vendor mid-upload has equivalent unsaved state. Whether the expiry duration is per-tenant configurable is not stated.	An expiring session warns before it expires and does not silently discard unsaved administrative or learner work.
10.3.1	**VAPT Compliance.** The platform shall undergo periodic Vulnerability Assessment and Penetration Testing to identify security risks.	1 → 1	UNCHANGED-FRD	No role-facing surface	Shared	SEC	A process requirement with no interface. The testing period and scope are not stated — PENDING DECISION.	A VAPT schedule exists and is owned before production release.
10.3.2	**Security Patch Management.** Identified vulnerabilities shall be addressed through timely security patches.	1 → 1	UNCHANGED-FRD	No role-facing surface	Shared	SEC	A process requirement with no interface. 'Timely' is not quantified in the FRD; no numeric target is invented here.	A patch process exists with a stated owner and response expectation before production release.
10.4.1	**Data Access Controls.** Role-based access controls shall restrict access to sensitive user data.	2 → 2	FUTURE-PHASE	Super Admin	Shared	SEC	Phase 2 in the original FRD. Note that role-based permission enforcement is nevertheless required in Phase 1 by the role definitions at 8.1.x and the isolation requirements at 1.3.x.	Not required in Phase 1 beyond the enforcement required by 1.3.x and 8.1.x.
10.4.2	**Data Retention Policies.** The platform shall implement configurable data retention policies for user data and analytics records.	2 → 2	FUTURE-PHASE	Super Admin	Shared	SEC	Phase 2 in the original FRD and unchanged. Retention periods are not stated and are not invented here.	Not required in Phase 1.

7.11 Analytics and Reporting

This section defines analytics and reporting. Every Phase-1 requirement here is a **recording** requirement, not a dashboard requirement — the dashboards sit in Phase 2 and Phase 3 in the FRD's own numbering. PM-11 therefore agrees with the FRD's phasing rather than deviating from it, and every reporting surface remains bounded by the tenant-scoping rule at 1.3.4.

Table 53 — Section 11 requirements — Analytics and Reporting (8 requirements, 4 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
11.1.1	**Module Completion Tracking.** The system shall record module completion status for each learner.	1 → 1	UNCHANGED-FRD	Learner, Faculty, Institute Admin	Shared	PRG	A recording requirement, not a dashboard requirement. Completion is recorded against the account rather than a device, so a device-run simulation reports back to the account.	Completing every required component of a module records module completion against the learner's account and unlocks the next module.
11.1.2	**Time Spent Tracking.** The platform shall track the time spent by students interacting with simulations.	1 → 1	PRODUCT-PENDING	Learner, Faculty, Institute Admin	Shared	PRG	A recording requirement. The granularity at which time is presented to Faculty and Institute Admin is bounded by 1.3.4 and by PM-11's operational-only limit; the reporting parameters themselves are PENDING DECISION.	Time spent is recorded per learner and is visible within the tenant boundary, without any aggregated analytical view.
11.2.1	**Assessment Score Tracking.** Assessment scores shall be stored and associated with the learner's academic profile.	1 → 1	UNCHANGED-FRD	Learner, Faculty	Shared	PRG, ASM	The shape of a stored score now depends on the assessment configuration (PM-08) rather than on fixed values.	A completed attempt stores a score against the learner and is visible to the learner and, within the tenant, to Faculty.
11.2.2	**Attempt Tracking.** The system shall record the number of attempts made for each simulation or assessment.	1 → 1	UNCHANGED-FRD	Learner, Faculty	Shared	PRG, ASM	Counting attempts is not the same as limiting them; any attempt limit is a configuration value, not a platform constant.	Every attempt is recorded and retrievable as attempt history, independently of any configured limit.
11.3.1	**Class-Level Performance Reports.** Faculty shall be able to view aggregated performance metrics for their classes.	2 → 2	FUTURE-PHASE	Faculty	B2B	RPT	Phase 2 in the original FRD and unchanged. Creates tension T-2 with the Phase-1 Faculty role definition at 8.1.4.	Not required in Phase 1.
11.3.2	**Institutional Adoption Reports.** Institute Admins shall receive reports on student engagement and module adoption.	2 → 2	FUTURE-PHASE	Institute Admin	B2B	RPT	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
11.4.1	**Simulation Heatmaps.** The system shall generate heatmaps identifying areas where learners frequently encounter errors.	3 → 3	FUTURE-PHASE	Faculty, Super Admin	Shared	RPT	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.
11.4.2	**Decision Path Analysis.** Simulation decision paths shall be analyzed to identify common mistakes and learning patterns.	3 → 3	FUTURE-PHASE	Faculty, Super Admin	Shared	RPT	Phase 3 in the original FRD and unchanged.	Not required in Phase 1.

7.12 Platform Infrastructure and Hosting

This section defines the infrastructure that hosts and scales the platform. No cloud provider, region, deployment topology or storage technology is selected by this draft. The named providers in FRD 12.1.1 are examples in the FRD, not a selection.

Table 54 — Section 12 requirements — Platform Infrastructure and Hosting (7 requirements, 3 in Phase 1)

ID	Requirement	Phase FRD → v5	Classification	Roles	Market	Capability	Clarification, dependencies and open questions	Acceptance criterion
12.1.1	**Cloud Infrastructure Hosting.** The platform shall be deployed on cloud infrastructure providers such as AWS, Azure, or Google Cloud.	1 → 1	TECHNICAL-PENDING	No role-facing surface	Shared	CFG	The named providers are FRD examples, not a selection. No cloud platform, region or deployment topology is selected by this draft — PENDING DECISION.	A hosting decision is recorded before build, and no architectural statement in this document depends on which provider is chosen.
12.1.2	**Encrypted Asset Storage.** XR asset bundles shall be stored in encrypted cloud storage systems.	1 → 1	UNCHANGED-FRD	No role-facing surface	Shared	SEC, XRD	Reinforces 10.1.1. The storage technology and encryption scheme are TECHNICAL DETAIL TO BE DEFINED.	Asset bundles are stored encrypted at rest, with no route to an unprotected copy from any workspace.
12.2.1	**CDN-Based Asset Delivery.** Simulation assets shall be delivered through a Content Delivery Network to reduce download latency.	1 → 1	UNCHANGED-FRD	Learner	Shared	XRD	Latency is learner-visible, so waiting is a designed experience rather than an incidental one. Determinate progress is used wherever the delivery mechanism can supply it.	Asset loading and device preparation present honest progress; an indeterminate wait is presented as such rather than as a stalled interface.
12.3.1	**Dynamic Server Scaling.** Application servers shall automatically scale based on platform traffic.	2 → 2	FUTURE-PHASE	No role-facing surface	Shared	CFG	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
12.3.2	**Load Balancing.** Traffic shall be distributed across multiple servers to maintain platform stability.	2 → 2	FUTURE-PHASE	No role-facing surface	Shared	CFG	Phase 2 in the original FRD and unchanged.	Not required in Phase 1.
12.4.1	**Regional CDN Nodes.** Regional CDN nodes shall be deployed to support global user access.	3 → 3	FUTURE-PHASE	No role-facing surface	Shared	XRD	Phase 3 in the original FRD and unchanged. Restates 1.7.1.	Not required in Phase 1.
12.4.2	**Multi-Region Database Replication.** Database replication shall ensure high availability across geographic regions.	3 → 3	FUTURE-PHASE	No role-facing surface	Shared	TEN	Phase 3 in the original FRD and unchanged. Restates 1.7.2.	Not required in Phase 1.

8. New and Clarified Requirements

This chapter records requirements that arise from stakeholder decisions rather than from Platform FRD_v4. They are given their own identifiers so that they can never be mistaken for original FRD content. **V5-ADD** identifies an addition; **V5-CLR** identifies a clarification of how an existing requirement is now read.

Reading these identifiers correctly

- A V5-ADD requirement did not exist in Platform FRD_v4. Where it extends the FRD, the extension is stated in the relationship row.
- A V5-CLR requirement does not replace an FRD requirement. It records the confirmed reading of one that already exists.
- Neither kind is renumbered into the FRD's own scheme, and neither reuses a design-decision identifier from earlier reviews.

8.1 Added requirements

V5-ADD-001 — Seven-role platform

Requirement

The platform shall provide role-based access and a distinct workspace for each of seven confirmed roles: Student / Learner, Faculty, Institute Admin, Vendor, Academic Reviewer, Content Manager and Super Admin.

Table 55 — V5-ADD-001 record

Attribute	Record
Source	PM-07 with PM-01
Classification	PM-ADDED
Role(s)	All seven
Market	Shared
Capability	IAM, TEN
Relationship to Platform FRD_v4	Extends FRD 8.1, which defines five roles. FRD 8.3 custom role creation is Phase 3 and cannot justify the extension within Phase 1.
Dependencies	Depends on FRD 10.2.1 authentication and on tenant and plane resolution.
Acceptance criterion	A signed-in identity reaches the workspace its approved role grants, and no role outside the seven currently approved appears in any interface.
Open question	QD-02 — the Phase-1 origin of institutional role assignment is undefined.

V5-ADD-002 — Vendor workspace

Requirement

The platform shall provide the Vendor with an authenticated workspace in which to create submissions, upload XR module builds and assets, supply metadata, view submission status and review history, and correct and resubmit returned work.

Table 56 — V5-ADD-002 record

Attribute	Record
Source	PM-07 with PM-14
Classification	PM-ADDED
Role(s)	Vendor
Market	Central content plane

Attribute	Record
Capability	GOV
Relationship to Platform FRD_v4	FRD 1.4.1 and 1.4.5 describe the Vendor as an actor with no role definition and no permission set.
Dependencies	Depends on the staging environment implied by FRD 1.4.1 and on the package format, which is pending.
Acceptance criterion	A Vendor can submit, track, correct and resubmit; no publish control exists anywhere in the workspace.
Open question	QT-05 — the XR package and submission format is undefined.

V5-ADD-003 — Academic Reviewer workspace

Requirement

The platform shall provide the Academic Reviewer with an authenticated workspace in which to review submitted modules, record academic validation, request changes with comments and a stated reason, and view decision history.

Table 57 — V5-ADD-003 record

Attribute	Record
Source	PM-07 with PM-14
Classification	PM-ADDED
Role(s)	Academic Reviewer
Market	Central content plane
Capability	GOV
Relationship to Platform FRD_v4	FRD 1.4.2 names the academic team as an actor with no role definition and no permission set.
Dependencies	Depends on the curriculum validation criteria, which are undefined.
Acceptance criterion	An Academic Reviewer records exactly one of two outcomes per submission, and a request for changes cannot be recorded without a reason.
Open question	QD-08 — the curriculum standards validated against are not defined.

V5-ADD-004 — Two-plane platform model

Requirement

The platform shall separate a central content and governance plane from institution tenant planes, with Vendor, Academic Reviewer, Content Manager and Super Admin operating on the central plane, and Institute Admin, Faculty and Student / Learner operating within a tenant plane. Super Admin alone spans both.

Table 58 — V5-ADD-004 record

Attribute	Record
Source	PM-13
Classification	PM-ADDED
Role(s)	All seven
Market	Shared
Capability	TEN, GOV, SEC
Relationship to Platform FRD_v4	Fills a silence in the FRD. FRD 1.3.1, 1.3.2 and 1.3.5 isolate data and grant cross-tenant access to Super Admin only, but never place Vendor, Academic Reviewer or Content Manager.
Dependencies	Depends on tenant and plane resolution at authentication.
Acceptance criterion	No central-plane role can reach tenant data, and no tenant-plane role other than none can reach another tenant. Only Super Admin spans planes.

Attribute	Record
Open question	How the planes are physically separated is TECHNICAL DETAIL TO BE DEFINED.

V5-ADD-005 — Content return with comments and reason

Requirement

The platform shall allow the Academic Reviewer and the Content Manager each to return a submission to the Vendor as changes required, and shall require reviewer comments and a stated reason on any such return.

Table 59 — V5-ADD-005 record

Attribute	Record
Source	PM-14
Classification	PM-ADDED
Role(s)	Academic Reviewer, Content Manager, Vendor
Market	Central content plane
Capability	GOV
Relationship to Platform FRD_v4	Extends FRD 1.4.1–1.4.4, which describe a forward-only chain with no rejection or revision requirement.
Dependencies	Depends on V5-ADD-007 review history.
Acceptance criterion	A return cannot be recorded without a reason, and the Vendor sees that reason on the returned submission.
Open question	QP-26 — whether Super Admin may also return at the publication gate is not stated and is not assumed.

V5-ADD-006 — Vendor correction and resubmission

Requirement

The platform shall allow the Vendor to correct a returned submission and resubmit it, and a resubmitted submission shall re-enter review. There shall be no automatic publication path following a return.

Table 60 — V5-ADD-006 record

Attribute	Record
Source	PM-14
Classification	PM-ADDED
Role(s)	Vendor
Market	Central content plane
Capability	GOV
Relationship to Platform FRD_v4	Extends FRD 1.4.1–1.4.4. Return authority never implies publish authority.
Dependencies	Depends on V5-ADD-005 and V5-ADD-007.
Acceptance criterion	A corrected resubmission enters a review gate and never appears in the publication queue without passing both prior gates.
Open question	Whether a resubmission re-enters at the returning gate or restarts is a working position, not a stakeholder statement.

V5-ADD-007 — Review history and status traceability

Requirement

The platform shall record every decision, comment, return and resubmission in the content governance chain, and shall make that history traceable to the Vendor for their own submissions and to the Academic Reviewer, Content Manager and Super Admin.

Table 61 — V5-ADD-007 record

Attribute	Record
Source	PM-14
Classification	PM-ADDED
Role(s)	Vendor, Academic Reviewer, Content Manager, Super Admin
Market	Central content plane
Capability	GOV, SEC
Relationship to Platform FRD_v4	Extends FRD 1.4.1–1.4.4, which contain no history or traceability requirement.
Dependencies	Depends on a decision store.
Acceptance criterion	For any submission, the ordered set of decisions, their authors and their comments is retrievable by every role permitted to see it.
Open question	Retention period and visibility scope of review history are undefined. Audit outside content governance is not covered.

V5-ADD-008 — Entitlement-aware XR device delivery and access

Requirement

The platform shall provide a mechanism by which an XR experiment to which a learner is entitled becomes available on the supported target device, and shall validate entitlement before delivery and again before execution.

Table 62 — V5-ADD-008 record

Attribute	Record
Source	PM-05
Classification	PM-ADDED
Role(s)	Learner, Super Admin
Market	Shared
Capability	XRD, ENT
Relationship to Platform FRD_v4	The product requirement is already supported by FRD 2.2.2 and 10.1.2. What is added is the requirement that this be a designed, perceivable experience rather than only a restricted state.
Dependencies	Depends on the delivery mechanism, which is pending, and on the package format, which is undefined.
Acceptance criterion	An entitled experiment reaches a compatible device and its state is perceivable to the learner throughout; an unentitled one is refused with a code-free explanation.
Open question	QT-01 — the delivery mechanism. QT-02 — no AR application architecture exists.

V5-ADD-009 — Delivery mechanism neutrality

Requirement

The platform architecture shall accommodate either direct delivery of the experiment, application or package to the target device, or a centralized device application exposing only entitled experiments, without this document selecting between them.

Table 63 — V5-ADD-009 record

Attribute	Record
Source	PM-05
Classification	PM-ADDED
Role(s)	Learner
Market	Shared
Capability	XRD
Relationship to Platform FRD_v4	FRD 2.1.1 describes a centralized master VR application, which resembles the second concept for VR only. That is a fact about the FRD, not a stakeholder selection.
Dependencies	Depends on technical discovery of the existing simulation inventory and its integration interfaces.
Acceptance criterion	Every learner-facing delivery state is named in language that remains accurate under either concept, and no state name presumes a mechanism.
Open question	QT-01 — pending by explicit instruction, not by omission.

V5-ADD-010 — Platform architecture before detailed design

Requirement

The complete platform sitemap, role permission model and all-role workflows shall be produced and reviewed before the most important end-to-end flows are taken into detailed interface design.

Table 64 — V5-ADD-010 record

Attribute	Record
Source	PM-02
Classification	PM-ADDED
Role(s)	All seven
Market	Shared
Capability	Design process
Relationship to Platform FRD_v4	A process requirement with no equivalent in the FRD.
Dependencies	Depends on the requirement re-audit, role model, capability map and workflow set produced in this draft.
Acceptance criterion	No detailed interface design begins for a flow before its sitemap position, permission set and workflow have been recorded and reviewed.
Open question	Which flows are taken first is a stakeholder selection that has not been made.

8.2 Clarified requirements

V5-CLR-001 — Existing XR simulations are available

Clarified position

The required XR simulations and experiments are available. The platform shall be designed to ingest, validate, publish, entitle and deliver existing simulations; simulation authoring is not a Phase-1 platform surface.

Table 65 — V5-CLR-001 record

Attribute	Record
Source	PM-06
Classification	PM-CLARIFIED
Existing FRD requirements affected	1.4.1, 1.1.4
Role(s)	Vendor, Content Manager, Super Admin
Effect on the previous position	Removes any assumption that the platform team must originate every simulation. Directs technical discovery toward existing builds, formats, versions, device compatibility, package types and integration interfaces.
Acceptance criterion	No workspace offers simulation-authoring capability, and the content model assumes supplied rather than originated content.

V5-CLR-002 — Configuration-driven assessment model

Clarified position

Assessment behaviour shall be configurable through the portal. Question count, timing, scoring weights, pass threshold, attempt limit, randomization and negative marking are configuration values, not platform constants.

Table 66 — V5-CLR-002 record

Attribute	Record
Source	PM-08
Classification	PM-CLARIFIED
Existing FRD requirements affected	5.1.1, 5.2.1–5.2.3, 5.3.1, 5.3.2
Role(s)	Faculty, Learner
Effect on the previous position	Corrects an interpretation rather than reversing a requirement. Platform FRD_v4 states no question count, duration, weighting, pass mark or attempt limit anywhere.
Acceptance criterion	Every assessment value displayed to a learner is read from the assessment's configuration, and no value is compiled into an interface.

V5-CLR-003 — Assessment consumption through flexible APIs

Clarified position

A configured assessment shall be consumable by the relevant Web or XR application through flexible platform API integration, and the result shall return to the platform for progress recording.

Table 67 — V5-CLR-003 record

Attribute	Record
Source	PM-08
Classification	PM-CLARIFIED
Existing FRD requirements affected	5.2.x, 11.2.1, 11.2.2
Role(s)	Faculty, Learner, external applications
Effect on the previous position	Confirms that applications consume the configuration rather than embedding the rules. The contract, schema, transport and versioning scheme are TECHNICAL DETAIL TO BE DEFINED.
Acceptance criterion	An application renders an assessment from the platform's configuration, and the result and attempt return to the platform. A learner-facing application receives only the assessment data the approved feedback policy permits.

V5-CLR-004 — Assessment ownership for B2B and B2C

Clarified position

For both B2B and B2C, assessment setup and configuration is owned by Faculty or the authorized institution side.

Table 68 — V5-CLR-004 record

Attribute	Record
Source	PM-15
Classification	PM-CLARIFIED
Existing FRD requirements affected	5.1.1, 5.1.2, 5.3.2, 8.1.4
Role(s)	Faculty, Institute Admin
Effect on the previous position	Removes assessment ownership as a point of B2B/B2C divergence — it converges on the same authorized side. Institute Admin is explicitly not granted blanket authoring permission; Faculty is the requirement-named role.
Acceptance criterion	Assessment authoring and configuration capability exists only on the Faculty or authorized institution side, in both market contexts.

V5-CLR-005 — Learners do not author or configure assessments

Clarified position

A learner shall never author or configure an assessment, in any market context.

Table 69 — V5-CLR-005 record

Attribute	Record
Source	PM-15
Classification	PM-CLARIFIED
Existing FRD requirements affected	No FRD text grants this; the prohibition is now stated
Role(s)	Learner
Effect on the previous position	Converts an absence into a stated rule. An absent permission can drift back in; a stated prohibition cannot.
Acceptance criterion	No learner-facing interface offers any assessment authoring or configuration control in any state.

V5-CLR-006 — Approved learning progression

Clarified position

The approved learning progression is Learning Resources, then Simulation, then Assessment, then Completion and progression.

Table 70 — V5-CLR-006 record

Attribute	Record
Source	PM-09
Classification	PM-CLARIFIED
Existing FRD requirements affected	3.1.1, 3.1.2, 3.2.2, 11.1.1
Role(s)	Learner, Faculty
Effect on the previous position	Confirms the existing learner sequencing design as the current approved model. The approval covers the learner progression model only and is not generalised into governance sequencing or any other role's workflow.
Acceptance criterion	A module presents its components in the approved order, and completion is evaluated against every required component.

V5-CLR-007 — Institutional SSO is required

Clarified position

Institutional single sign-on integration is required, including the institution-side capability to configure it.

Table 71 — V5-CLR-007 record

Attribute	Record
Source	PM-10
Classification	PM-CLARIFIED
Existing FRD requirements affected	4.2.2 (Phase 1) and 7.4.1 (Phase 3)
Role(s)	Institute Admin, Faculty, Learner, Super Admin
Effect on the previous position	Resolves an internal FRD phase contradiction in favour of the Phase-1 reading, and surfaces a configuration capability that had never been designed.
Acceptance criterion	An institution can configure, test and enable SSO for its own tenant, and institutional users can authenticate through it.

V5-CLR-008 — Incremental reporting strategy

Clarified position

Core functionality and operations are built first. Detailed reporting and analytics parameters are added incrementally once the underlying functionality and available data are understood.

Table 72 — V5-CLR-008 record

Attribute	Record
Source	PM-11
Classification	PM-CLARIFIED
Existing FRD requirements affected	11.1.x, 11.2.x Phase 1; 8.2.x, 11.3.x Phase 2; 11.4.x Phase 3; bounded by 1.3.4
Role(s)	Faculty, Institute Admin, Super Admin, Content Manager
Effect on the previous position	Agrees with the FRD's own phasing rather than deviating from it. Phase-1 role architecture may include operational summaries and status where required; no large analytics dashboard is designed now, and reporting is not removed from the long-term requirement set.
Acceptance criterion	Every Phase-1 reporting element answers a current-state operational question and is bounded to a single tenant.

V5-CLR-009 — Certification deferred to Phase 2

Clarified position

Certification is not part of the current Phase-1 product. Certification is a Phase-2 activity.

Table 73 — V5-CLR-009 record

Attribute	Record
Source	PM-12
Classification	PM-DEFERRED
Existing FRD requirements affected	3.3.1 and 3.3.2, both Phase 1 as written
Role(s)	Learner, Faculty, Institute Admin
Effect on the previous position	DEVIATION RECORDED. This is a stakeholder decision to move requirements out of the phase the FRD assigns them, not a correction of the FRD. The original phase is preserved everywhere in this document.
Acceptance criterion	No certification structure, certificate artefact, issuance flow or eligibility surface appears in any Phase-1 workspace.

V5-CLR-010 — B2B and B2C are both Phase 1

Clarified position

Both the B2B institutional model and the B2C individual model are required in Phase 1, and neither is sequenced ahead of the other.

Table 74 — V5-CLR-010 record

Attribute	Record
Source	PM-04
Classification	PM-CLARIFIED
Existing FRD requirements affected	1.1.2 and 1.1.3, both Phase 1
Role(s)	All customer-facing roles
Effect on the previous position	Any B2B-first or B2C-first sequencing returns to pending. It was never FRD-derived and is not a stakeholder decision.
Acceptance criterion	Both market contexts are represented in the architecture, the workflows and the screen planning, with no release order asserted.

V5-CLR-011 — One portal, not one runtime

Clarified position

The solution is one central platform and web portal with role-based access and workspaces. This shall not be read as requiring every XR runtime to execute inside the browser.

Table 75 — V5-CLR-011 record

Attribute	Record
Source	PM-03
Classification	PM-CLARIFIED
Existing FRD requirements affected	1.1.1 against 1.2.2, 1.2.3, 1.2.4, 2.1.1, 2.1.2
Role(s)	All seven
Effect on the previous position	Clarifies that the portal is the control, entitlement, governance and administration plane. Only the web format (1.2.1) executes in the browser; VR, MR and AR execute outside it.
Acceptance criterion	No interface claims to launch a VR, MR or AR simulation inside the desktop browser, and no non-functional launch control is presented.

V5-CLR-012 — Sample assessment configuration

Clarified position

Any specific assessment values shown in design or specification material are sample configuration only. They are not platform constants, defaults or recommendations.

Table 76 — V5-CLR-012 record

Attribute	Record
Source	PM-08
Classification	SAMPLE-CONFIGURATION
Existing FRD requirements affected	No FRD requirement states any assessment value
Role(s)	Faculty, Learner
Effect on the previous position	Reclassifies earlier prototype values. The capabilities those values exercised remain valid as behaviour the configuration model must be able to express.
Acceptance criterion	No specification, screen or handoff presents a specific assessment value as a platform default, a recommended value or an approved setting.

9. End-to-End Workflows

Ten cross-role workflows are required by Phase-1 requirements. Each is described as a sequence of steps with the requirement that drives it. Where a stakeholder left a mechanism pending, the workflow states **what must happen** and never **how it is implemented**; an abstract step is a record of an open decision, not a gap in the specification.

Table 77 — Workflow index

Flow	Name	Roles	Requirement basis	Status
FLOW 1	Content to production	Vendor, Academic Reviewer, Content Manager, Super Admin	1.4.1–1.4.5, 2.3.2, PM-14	Fully specified
FLOW 2	B2B institutional learning	Super Admin, Institute Admin, Faculty, Learner	6.2.x, 7.2.x, 8.1.x, 3.2.2, 11.1.x	Transaction step pending
FLOW 3	B2C purchase to learning	Learner	1.1.3, 6.1.x, 9.1.x, 9.2.x	Provider pending
FLOW 4	Assessment configuration to result	Faculty, platform, application, Learner	5.1.x–5.3.x, 11.2.x, PM-08, PM-15	API contract pending
FLOW 5	XR entitlement and device delivery	Learner, Super Admin	1.2.x, 2.x, 10.1.2, 12.2.1, PM-05	Mechanism pending by instruction
FLOW 6	Institutional SSO	Institute Admin, Faculty, Learner	4.2.1, 4.2.2, 7.2.2, 10.2.1, PM-10	Provider pending
FLOW 7	ERP synchronisation	Institute Admin, platform, Learner	7.1.1, 7.1.2, 7.2.1–7.2.3	Provider pending
FLOW 8	Device binding and replacement	Learner, Super Admin	2.4.1, 2.4.2	Rejection path undefined
FLOW 9	Licence cap exhaustion	Learner, Institute Admin	6.2.2, 7.2.3	Seat acquisition pending
FLOW 10	Version release to live content	Super Admin, Learner	2.3.1, 2.3.2, 2.3.3	Notification undefined

9.1 FLOW 1 — Content to production

This is the most precisely specified sequence in Platform FRD_v4 and the one with no prior designed surface at all: three of its four roles were outside the earlier scope. PM-14 adds the return paths that make its gates function as gates.

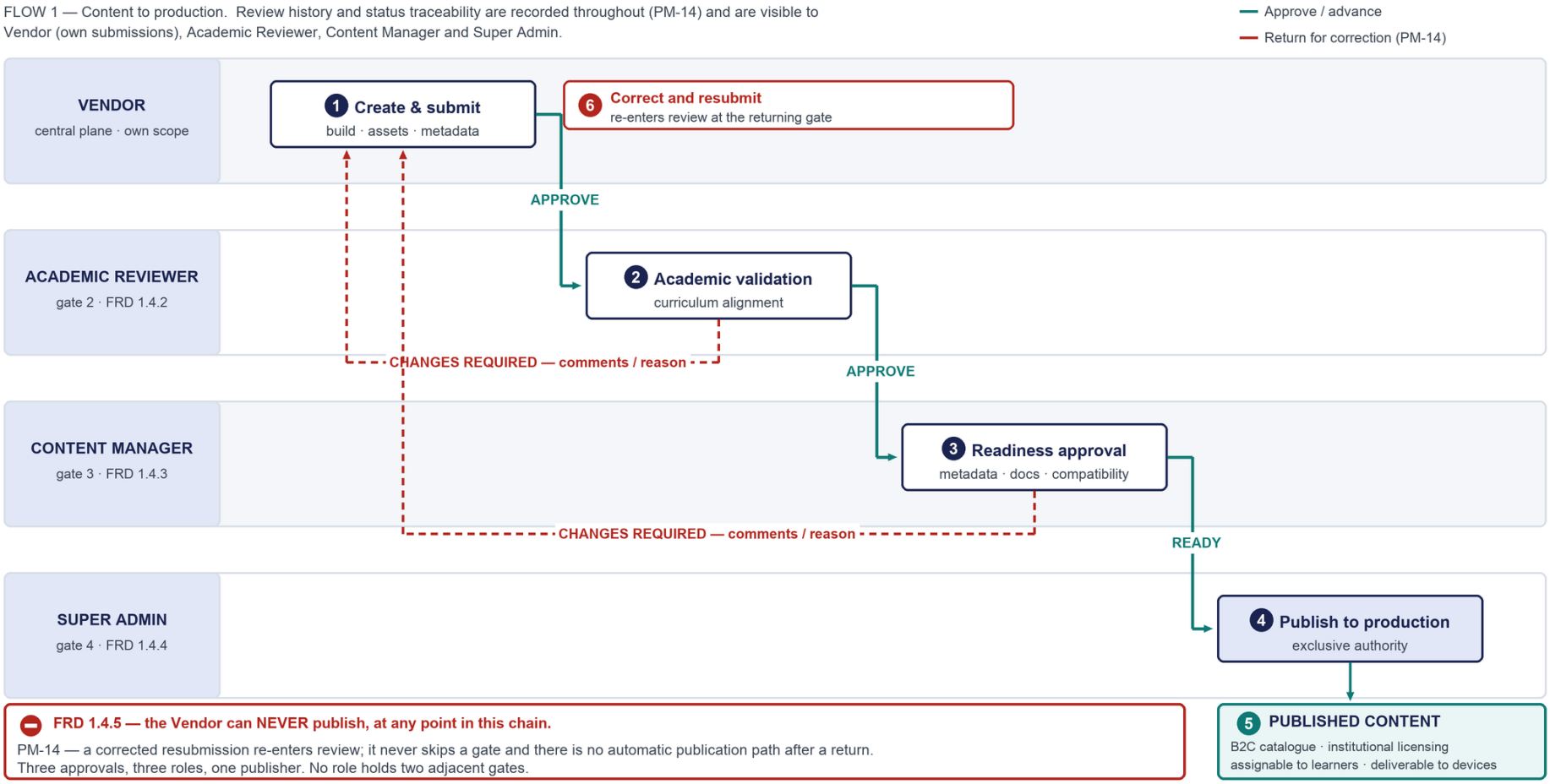


Figure 6 — FLOW 1 — content to production, with PM-14 return loops

Table 78 — FLOW 1 steps

Step	Actor	What happens	Requirement	Outcome or state
1	Vendor	Creates a submission, uploads the XR module build and supporting assets, and supplies module metadata, then submits to the staging environment.	1.4.1	With Academic Reviewer
2	Academic Reviewer	Reviews module content, metadata and documentation, and assesses alignment with engineering curriculum standards. Records one of two decisions.	1.4.2, PM-14	Validated → step 3, or Changes required → step 6
3	Content Manager	Reviews metadata, documentation and platform compatibility, and reads the academic validation record read-only. Records one of two decisions.	1.4.3, PM-14	Ready to publish → step 4, or Changes required → step 6
4	Super Admin	Publishes the module to the production environment. No other role may perform this step.	1.4.4, 1.4.5	Published
5	Platform	Published content enters the B2C catalogue, becomes licensable to institutions, becomes assignable to learners, and becomes deliverable to devices.	1.4.4, 6.1.x, 6.2.x	Available
6	Vendor	Reads the reviewer's comments and stated reason, corrects the build, assets or metadata, and resubmits. The submission re-enters review at a review gate.	PM-14	Re-enters review — never routes to publication
All	All four roles	Every decision, comment, return and resubmission is recorded and traceable. The Vendor always sees the gate position and the reason for any return.	PM-14	Review history

Constraints on this flow

- Three approvals, three roles, one publisher. No role holds two adjacent gates.
- A returning role can send work back but never forward past its own gate, and gains no publication authority.
- There is no automatic publication path after a return. Publication is reachable only through recorded academic validation and recorded content readiness approval.
- The Vendor prohibition at FRD 1.4.5 is expressed positively — the Vendor sees where a submission is, never a disabled publish control, because a disabled control implies a permission that will never exist.

Table 79 — FLOW 1 recorded gaps

Gap	Nature	Handling
The engineering curriculum standards validated against are not defined anywhere in the FRD.	Content ownership	The decision surface is specified; the rubric is recorded as content someone must own.
The XR package and submission format is stated only as “encrypted asset bundles”.	Technical	The upload step is specified generically; no format is asserted.
Whether Super Admin may also return a submission at the publication gate.	Product	PM-14 names Academic Reviewer and Content Manager only. Not assumed either way.
Whether a resubmission re-enters at the returning gate or restarts at academic validation.	Product	A working position is recorded — it re-enters at the returning gate, with both prior decisions preserved in history — and is labelled as a working position.
Whether the Vendor is notified of a return or discovers it on next sign-in.	Product	In-workflow status is specified; no push notification is designed, because no Phase-1 requirement asks for one.

9.2 FLOW 2 — B2B institutional learning

The learner half of this journey was substantially designed under the earlier interpretation. The entire institutional half — the machinery that produces the entitlement the learner consumes — was not.

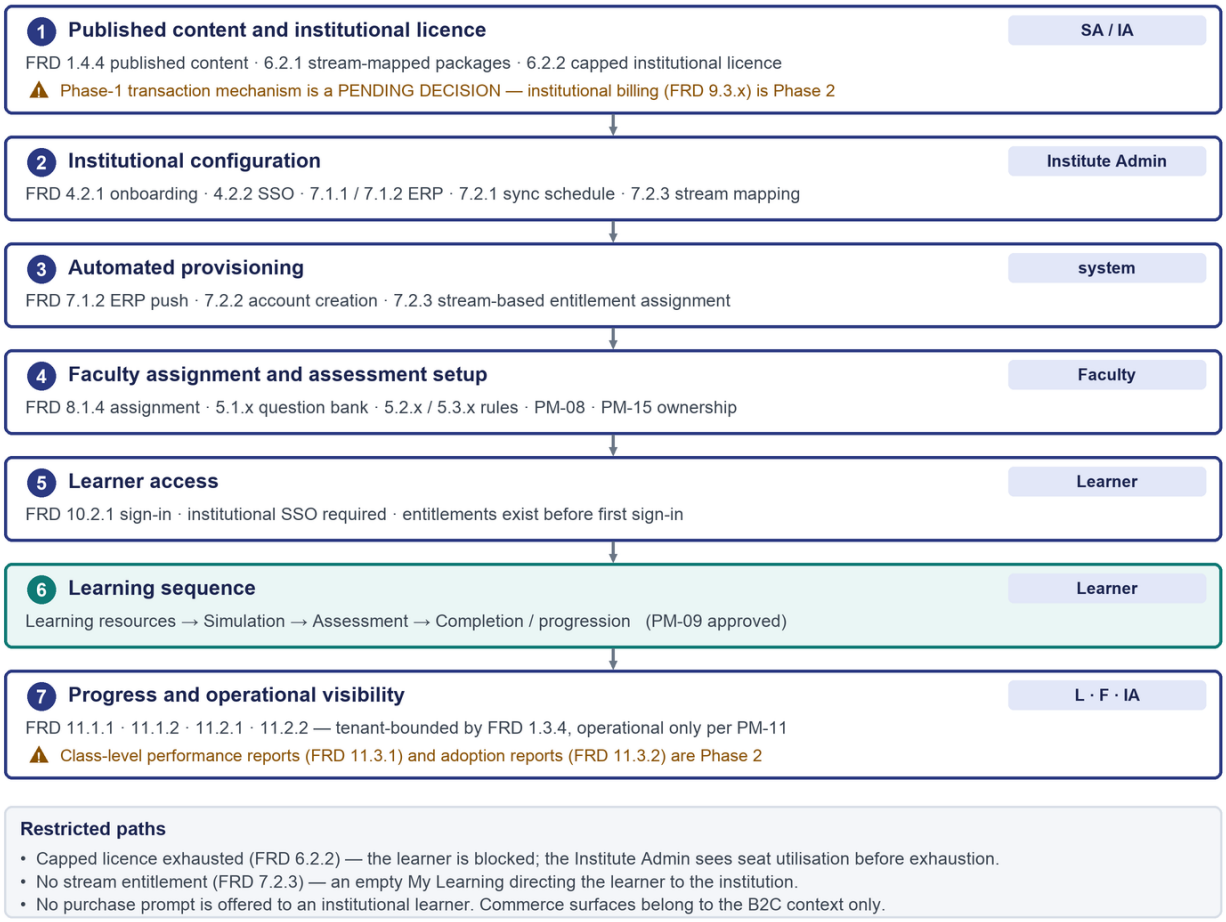


Figure 7 — FLOW 2 — B2B institutional learning journey

Table 80 — FLOW 2 steps

Step	Actor	What happens	Requirement
1	Super Admin / Institute Admin	An institutional licence exists against the tenant: packages mapped to academic streams, capped by maximum student count.	6.2.1, 6.2.2
2	Institute Admin	Configures ERP integration and the synchronisation schedule, configures institutional SSO, and maps academic streams to licensed packages.	7.1.1, 7.1.2, 7.2.1, 4.2.2, 7.2.3
3	Platform	Scheduled ERP synchronisation pushes student records; accounts are created automatically and stream-based packages are assigned. Entitlements exist before the learner first signs in.	7.1.2, 7.2.2, 7.2.3
4	Faculty	Views course structure received from ERP, assigns courses or modules to learners and cohorts, and configures assessments.	8.1.4, PM-08, PM-15
5	Learner	Signs in and reaches entitled content. Institutional SSO is required and is available according to the institution's configured authentication model.	10.2.1, 4.2.2
6	Learner	Works through learning resources, then the simulation, then the assessment, and the module completes.	3.1.2, 3.2.2, PM-09, 11.1.1
7	Learner, Faculty, Institute Admin	Progress is recorded and is visible: to the learner as personal progress, and to Faculty and Institute Admin as tenant-bounded operational status.	11.1.x, 11.2.x, 1.3.4, PM-11

- **Restricted path — licence cap exhausted.** The learner is blocked with an explanation directing them to their institution and no purchase prompt. The Institute Admin sees seat utilisation before exhaustion (FRD 6.2.2).
- **Restricted path — no stream entitlement.** An empty entitlement state explains that the institution assigns access, rather than presenting an empty list without explanation (FRD 7.2.3).
- **Recorded gap.** How an institutional licence is transacted in Phase 1 is undefined, because institutional billing (FRD 9.3.1, 9.3.2) is Phase 2. Licences are shown as existing; no in-platform purchase flow is invented.

9.3 FLOW 3 — B2C purchase to learning

This is the most complete journey in the platform. PM-04 changes nothing about it except its framing: it is one branch of a multi-role platform rather than half of a two-audience product.

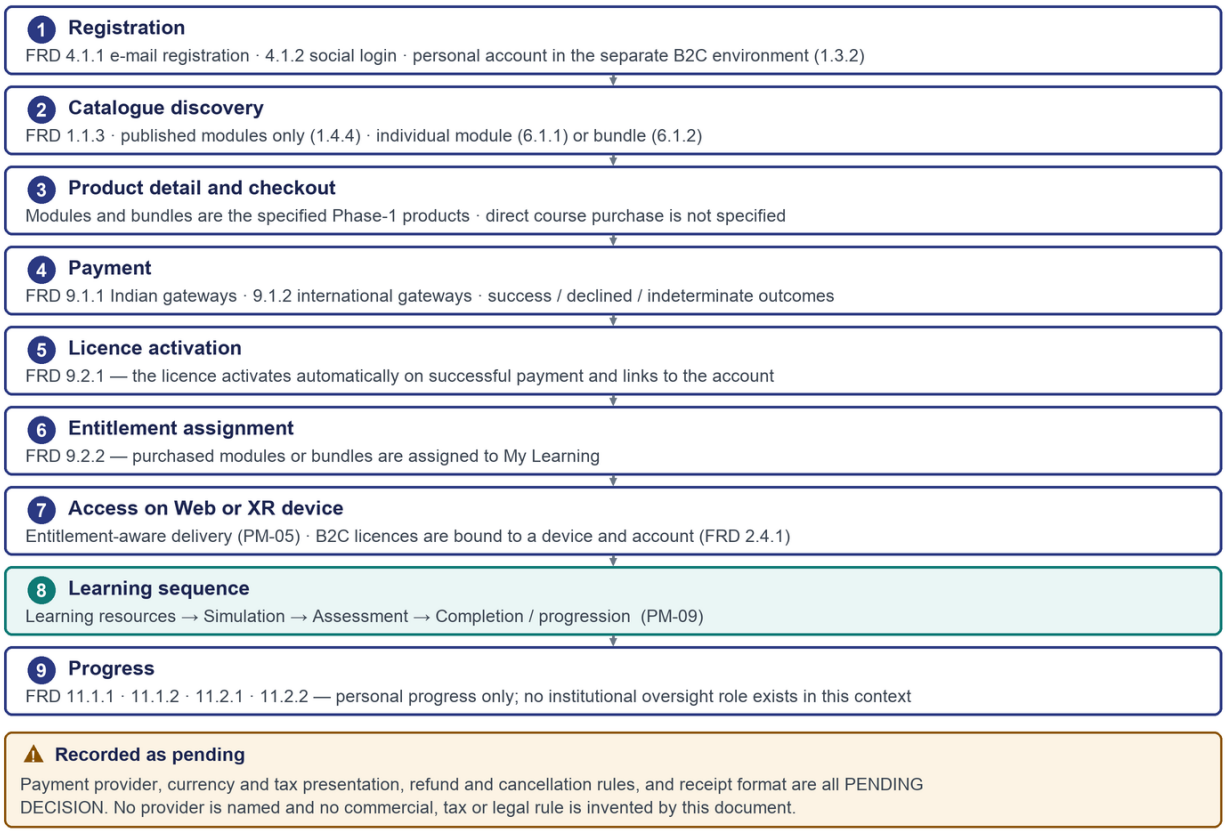


Figure 8 — FLOW 3 — B2C purchase-to-learning journey

Table 81 — FLOW 3 steps

Step	What happens	Requirement
1	The learner self-registers by e-mail or through a supported social login provider, creating a personal account in the separate B2C data environment.	4.1.1, 4.1.2, 1.3.2
2	The learner browses the catalogue of published modules and bundles. Phase-1 purchase requirements name modules and bundles; direct course purchase is not specified.	1.1.3, 1.4.4, 6.1.1, 6.1.2
3	The learner views a product and proceeds to checkout. Access duration is shown only where authoritative data exists.	6.1.1, 6.1.2
4	Payment is executed through a gateway. Success, decline and indeterminate outcomes are each designed.	9.1.1, 9.1.2
5	The licence activates automatically on successful payment and links to the account.	9.2.1
6	The purchased module or bundle is assigned to the learner's entitlements and appears in My Learning.	9.2.2
7	The learner reaches the simulation, in the browser for the web format or on the supported device otherwise. B2C licences are bound to a device and account.	1.2.x, 2.4.1, PM-05
8	The learner works through learning resources, then the simulation, then the assessment, and the module completes.	3.1.2, PM-09, 11.1.1
9	Progress is recorded and visible to the learner. No institutional oversight role exists in this context.	11.1.x, 11.2.x

Recorded as pending in this flow

- The payment provider is not selected, and whether checkout is a redirect or an embedded flow is undetermined.
- Currency handling, tax treatment and price presentation are compliance-adjacent and undefined. No tax or legal rule is invented here.
- Refund, cancellation and revocation rules are not stated by any requirement.
- The receipt or invoice a B2C buyer receives is undefined; GST-compliant institutional invoicing (FRD 9.3.1) is Phase 2 and institution-facing.

9.4 FLOW 4 — Assessment configuration to result

The consumption half of this flow was designed under the earlier interpretation and survives structurally unchanged. The authoring half did not exist. What changes in the consumption half is where its numbers come from: every displayed value is now read from the configuration.

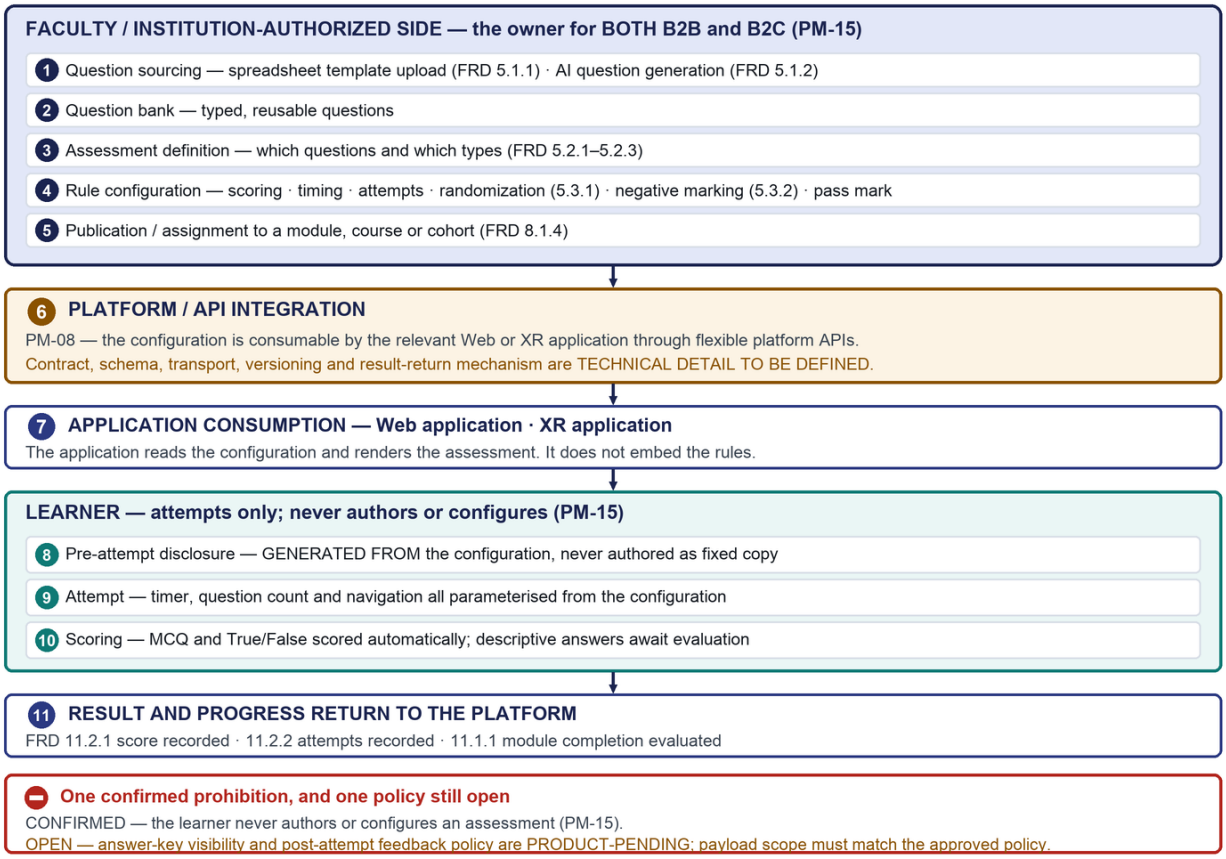


Figure 9 — FLOW 4 — assessment configuration to result

Table 82 — FLOW 4 steps

Step	Actor	What happens	Requirement
1	Faculty / authorized institution	Sources questions by structured spreadsheet upload or by AI-assisted generation.	5.1.1, 5.1.2
2	Faculty / authorized institution	Maintains a question bank of typed, reusable questions.	5.1.1
3	Faculty / authorized institution	Defines the assessment: which questions and which of the three supported types.	5.2.1, 5.2.2, 5.2.3
4	Faculty / authorized institution	Configures rules: scoring, timing, attempts, randomization, negative marking and pass threshold. Hints, if in scope, would be a further optional setting — no source defines them.	5.3.1, 5.3.2, PM-08
5	Faculty / authorized institution	Publishes or assigns the assessment to a module, course or cohort.	8.1.4
6	Platform	Exposes the configuration to consuming applications through flexible platform APIs.	PM-08

Step	Actor	What happens	Requirement
7	Web or XR application	Reads the configuration and renders the assessment. It does not embed the rules.	PM-08
8	Learner	Reads a pre-attempt disclosure generated from the configuration, attempts the assessment, and submits.	5.2.x, PM-08
9	Platform	Automatically scores MCQ and true/false questions. Descriptive answers enter an awaiting-evaluation state.	5.2.1, 5.2.2, 5.2.3
10	Platform	Records the score and the attempt, and evaluates module completion.	11.2.1, 11.2.2, 11.1.1

One confirmed prohibition, and one policy that is still open

- **Confirmed:** the learner never authors or configures an assessment (PM-15).
- **Open:** answer-key visibility and post-attempt feedback policy are PRODUCT-PENDING. No requirement states what feedback a learner may see. Whatever is approved, learner-facing applications must receive only the assessment data that policy permits — displayed scope and payload scope must match. See `QP-27`.

Recorded gap — descriptive-answer grading

- FRD 5.2.3 permits short descriptive answers. No Phase-1 requirement defines who grades them, on what turnaround, or against what rubric.
- This is compounded by the approved progression: completion gates the next module, so an ungraded answer can strand a learner indefinitely.
- Faculty is the plausible owner because FRD 8.1.4 names assessments among Faculty responsibilities, but no requirement says so. Grading ownership is recorded as a PENDING DECISION and is not invented here.

9.5 FLOW 5 — XR entitlement and device delivery

The product requirement is confirmed. The technical mechanism is deliberately abstract, by explicit stakeholder instruction. This flow therefore specifies the lifecycle and the states a learner must be able to perceive, and names no transport, protocol, package format or endpoint.

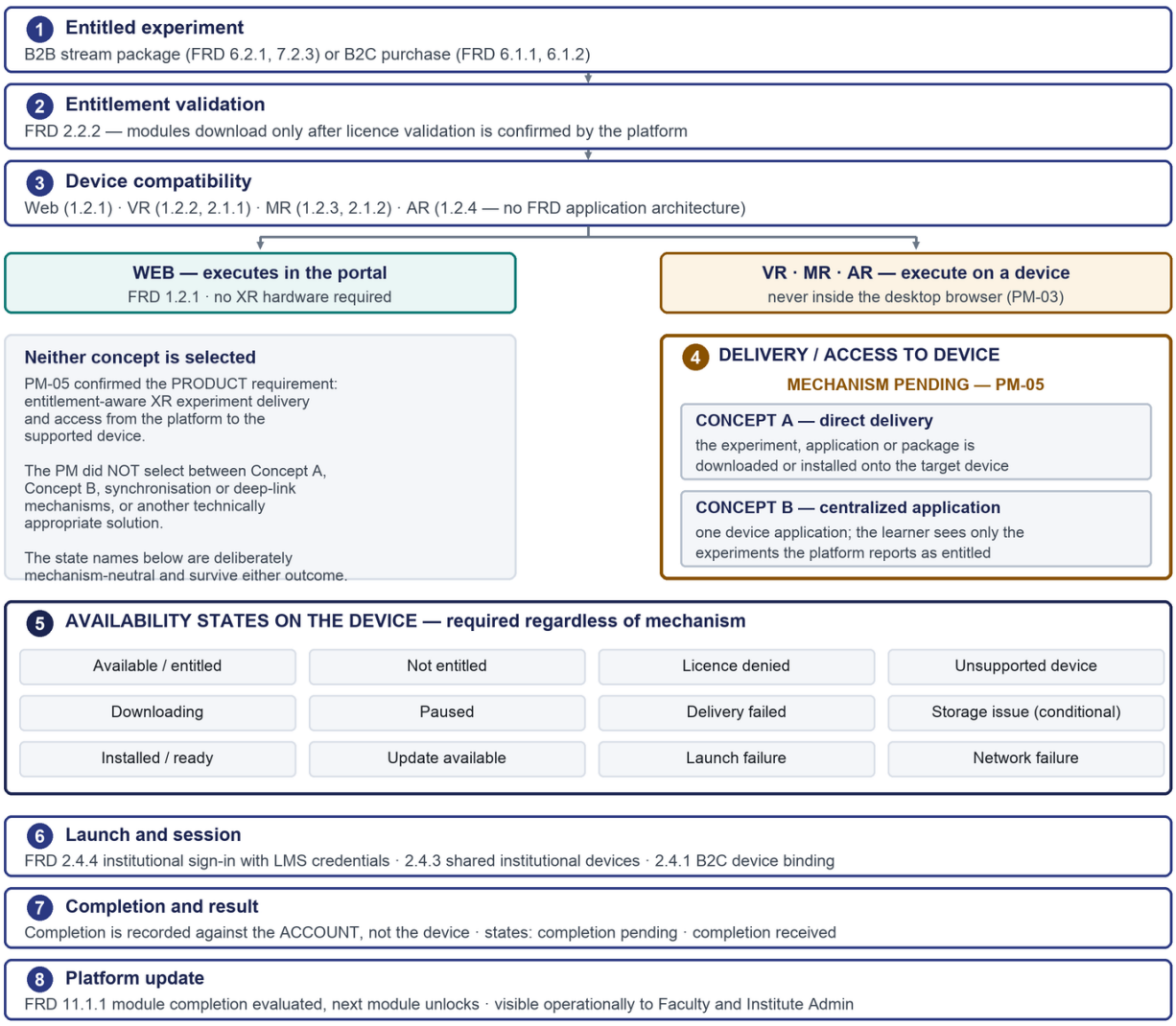


Figure 10 — FLOW 5 — XR entitlement and device-delivery lifecycle

Table 83 — FLOW 5 steps

Step	What happens	Requirement
1	The learner holds an entitlement, from an institutional stream package or from an individual purchase.	6.2.1, 7.2.3, 6.1.1, 6.1.2
2	The platform confirms a valid licence. Failure produces an access-restricted state.	2.2.2, 10.1.2
3	The target runtime is determined: web in the portal, or VR, MR or AR on a device. An incompatible or unavailable device produces an unsupported-device state.	1.2.1–1.2.4, 2.1.1, 2.1.2
4	For a device format, the entitled experiment becomes available on the supported device. **The mechanism is not selected.**	PM-05
5	The learner can perceive where the experiment is, through the availability states listed below.	2.3.1, 12.2.1, PM-05
6	The learner runs the experiment. An institutional learner signs into the device application with LMS credentials; a B2C licence is bound to the device.	2.4.4, 2.4.1, 2.4.3
7	Completion is recorded against the account rather than the device. The portal reflects account state; it does not observe the session.	11.1.1
8	Module completion is evaluated and the next module unlocks. Progress becomes visible to the learner and, operationally, within the tenant.	11.1.1, 3.2.2, 1.3.4

Table 84 — The two concepts recorded, neither selected

Concept	Description	Relationship to Platform FRD_v4
A — direct delivery	The experiment, application or package can be downloaded or installed onto the target device.	Partially supported. FRD 2.2.1 packages modules as bundles but describes them as dynamically downloaded by a master application, which is the other concept's framing. Whether an experiment can be installed without a container application is never stated.
B — centralized application	One application on the device contains or supports the experiments, and the learner sees only those for which the platform reports a valid entitlement.	Strongly supported for VR. FRD 2.1.1 describes exactly such a container, 2.2.1 has it downloading bundles, 2.3.3 updates modules without reinstalling it, and 2.4.4 has the learner signing into it. Whether MR (sideloaded) and AR (no architecture at all) follow the same pattern is not stated.
Other	Synchronisation mechanisms, deep-link handoff, or another technically appropriate solution.	Left explicitly open by the stakeholder.

The honest position on the two concepts

- FRD Section 2 leans toward the centralized-application concept for VR. That is a fact about the FRD, not a stakeholder selection.
- The FRD's language addresses VR only; MR uses a different distribution model, and AR has no application architecture at all. The direct-delivery concept remains explicitly on the table.
- No selection is made here, and every state name in this document is chosen to survive either outcome. “Ready on your device” survives both; “downloaded” presumes one concept and “unlocked in the app” presumes the other, so neither is used.

9.6 FLOW 6 — Institutional SSO

The learner-facing authentication path existed under the earlier interpretation. The institution-side configuration capability that FRD 4.2.2 actually requires did not.

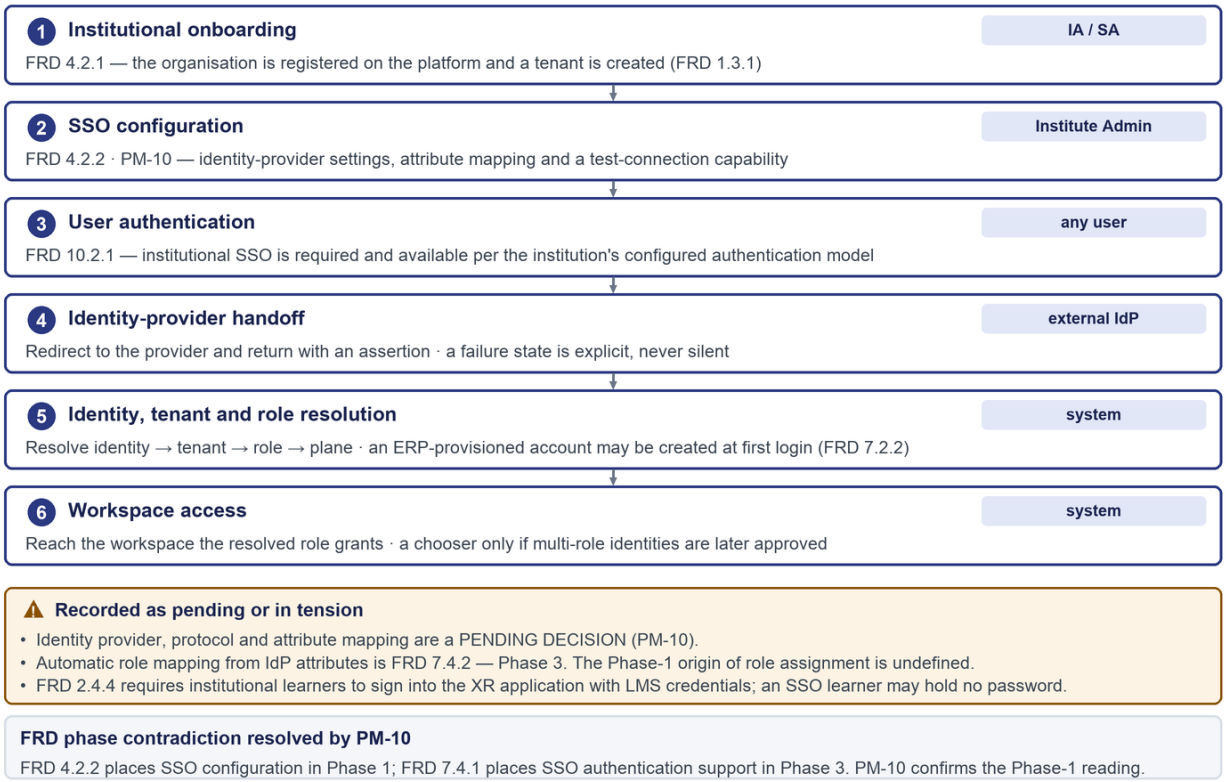


Figure 11 — FLOW 6 — institutional SSO, tenant and role resolution

Table 85 — FLOW 6 steps

Step	Actor	What happens	Requirement
1	Institute Admin / Super Admin	The organisation is registered on the platform and a tenant is created.	4.2.1, 1.3.1
2	Institute Admin	Configures single sign-on with a supported identity provider, sets provider settings, and tests the connection before enabling it.	4.2.2, PM-10
3	Any user	Signs in. Institutional SSO is required and is available according to the institution's configured authentication model; the relative prominence of authentication options is a design decision.	10.2.1, 4.2.2
4	Identity provider	The user is redirected to the provider and returns with an assertion. A failure produces an explicit failure state.	4.2.2, 7.4.1
5	Platform	Resolves identity, tenant, role and plane. An ERP-provisioned account may be created at first login.	7.2.2, PM-13
6	Platform	The user reaches the workspace their role resolves to. If multi-role identities are later approved, a workspace chooser is available as a conditional pattern; none is mandated.	8.1.x, QP-11

Recorded as pending or in tension

- Identity provider, protocol and attribute mapping are a PENDING DECISION by explicit stakeholder instruction (PM-10).
- Automatic role mapping from identity-provider attributes is FRD 7.4.2, which is Phase 3. The Phase-1 origin of institutional role assignment is therefore undefined and is recorded rather than invented.
- FRD 2.4.4 requires institutional learners to sign into the XR application with LMS credentials, while FRD 4.2.2 with PM-10 makes institutional SSO required. An SSO-authenticated learner may hold no password to type into a headset. Both requirements are Phase 1 and they collide.

9.7 FLOW 7 — ERP synchronisation

Every ERP capability has a learner-visible consequence that was already designed and an administrator-facing control surface that was not. The pattern matches entitlements: the effect existed, the cause did not.

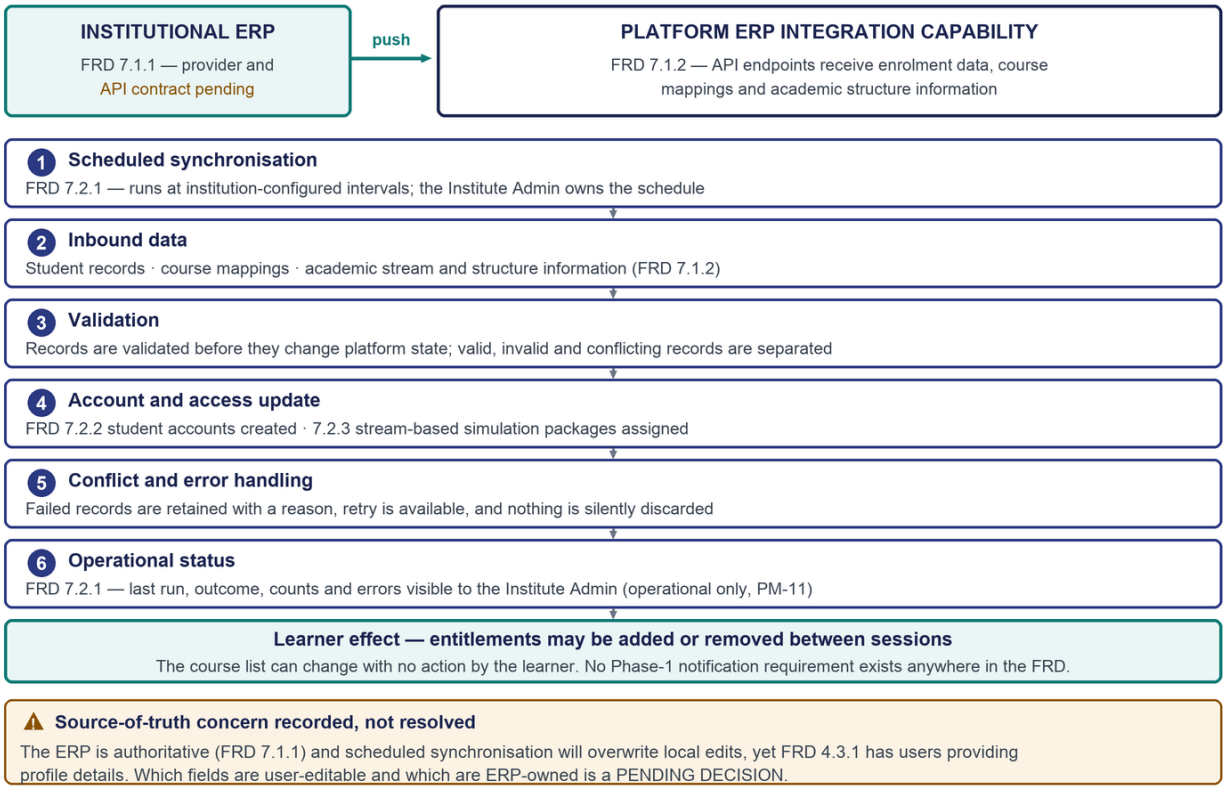


Figure 12 — FLOW 7 — ERP synchronisation

Table 86 — FLOW 7 steps

Step	What happens	Requirement
1	Scheduled synchronisation runs at the interval the Institute Admin has configured.	7.2.1
2	The ERP pushes student records, course mappings and academic structure information to the platform's API endpoints.	7.1.2
3	Records are validated before they change platform state. Valid, invalid and conflicting records are separated.	7.1.2
4	Student accounts are created automatically and stream-based simulation packages are assigned.	7.2.2, 7.2.3
5	Failed records are retained with a reason, retry is available, and nothing is silently discarded.	7.2.1
6	The last run, its outcome, its counts and its errors are visible to the Institute Admin as operational status.	7.2.1, PM-11

Source-of-truth concern recorded, not resolved

- The ERP is authoritative (FRD 7.1.1) and scheduled synchronisation will overwrite local edits, which is visible data loss from the user's point of view.
- FRD 4.3.1 nonetheless has users providing profile details, and it now applies to Faculty profiles as well as learner profiles.
- Which fields are user-editable and which are ERP-owned is a PENDING DECISION. The working position is that ERP-sourced fields are presented read-only with their provenance stated.
- Entitlements may be added or removed between sessions, so a learner's course list can change with no action by them. No Phase-1 notification requirement exists anywhere in the FRD.

9.8 FLOW 8 — Device binding and replacement

This is the platform's only Phase-1 two-role, cross-plane approval workflow outside content governance: a learner in the personal context requesting an action only a platform-wide Super Admin can grant. Neither end had any designed surface.

Table 87 — FLOW 8 steps

Step	Actor	What happens	Requirement
1	Platform	An individual licence is bound to a specific encrypted XR device identifier and a user account.	2.4.1
2	Learner	Requests a device replacement, which requires transferring the licence to a new device.	2.4.2
3	Super Admin	Reviews the request and approves the licence transfer.	2.4.2
4	Platform	The entitlement rebinds to the new device, and the learner can see that it has.	2.4.1, 2.4.2

- **B2B has no personal device binding.** Institutional deployments use shared devices (FRD 2.4.3), which is why this flow is B2C only.
- **Recorded gap.** The FRD describes no rejection path, no evidence requirement and no limit on replacement frequency. None is invented here.
- **Recorded gap.** Whether a mobile phone qualifies as an encrypted XR device identifier for the AR format is not stated.
- Device registration and remote device management are Phase 2 (FRD 2.5.x, 2.6.x) and must not appear in a Phase-1 administrative workspace.

9.9 FLOW 9 — Licence cap exhaustion

Table 88 — FLOW 9 steps

Step	Actor	What happens	Requirement
1	Institute Admin	Sees seat utilisation against the institutional cap as operational status, before exhaustion.	6.2.2, PM-11
2	Platform	The capped institutional licence is exhausted; further learner access is blocked.	6.2.2
3	Learner	Receives an access-restricted state explaining that access is institution-managed, directing them to their administrator, with no purchase prompt.	6.2.2, 1.1.2
4	Institute Admin	Acts to restore access. **How additional seats are acquired in Phase 1 is a PENDING DECISION**, because institutional billing is Phase 2.	6.2.2, 9.3.x

9.10 FLOW 10 — Version release to live content

Table 89 — FLOW 10 steps

Step	Actor	What happens	Requirement
1	Super Admin	Controls when a new module version is released to the production environment, and may hold a release.	2.3.2
2	Device application	Automatically downloads the updated module version when it becomes available.	2.3.1
3	Platform	The module updates independently of the device application, without requiring a full reinstall.	2.3.3
4	Learner	May see an update-available state on a delivered experiment. An unrequested update never blocks continuing.	2.3.1, PM-05

Recorded gap — mid-course content change

- Content can change part-way through a course. Phase 1 has no module lifecycle states, because FRD 1.5.1 is Phase 2, and no notification requirement, because none exists.
- The consequence is recorded rather than designed around: the platform can present an update-available state, but it cannot inform a learner who is not looking.

10. XR Delivery Requirements

What this chapter is and is not

- It defines the product and experience contract for getting an entitled experiment onto the device that runs it.
- It does NOT select a delivery mechanism, name an API, endpoint or protocol, specify a package format or file type, or invent a synchronisation or deep-link scheme. PM-05 explicitly declined to select, and this document holds that open.

10.1 The four runtime targets

Table 90 — Runtime targets and their application architecture

Format	Requirement	Runtime	Application architecture in Platform FRD_v4	Executes in the portal?
Web	1.2.1	Standard web browser, no XR hardware	Not applicable — runs in the portal itself	Yes
VR	1.2.2	VR headset, such as Meta Quest devices	2.1.1 — a centralized VR application acting as the container for downloading and launching XR simulation modules	No
MR	1.2.3	MR hardware, such as Ajna Lens	2.1.2 — a sideloaded application deployed on the device	No
AR	1.2.4	Android and iOS mobile devices	**None. No equivalent requirement exists anywhere in Section 2.**	No

This distribution matters for PM-03. “One central web portal” describes the control, entitlement and administration plane. It does not mean every XR runtime executes inside the browser: the FRD places VR, MR and AR runtimes outside it explicitly. **No interface in this platform may claim that a VR headset simulation runs inside the desktop browser.**

10.2 Supporting delivery requirements the FRD does define

Table 91 — Delivery-related Phase-1 requirements

Requirement	What it requires	Applies to
2.2.1	Modules packaged as encrypted asset bundles, dynamically downloaded by the master XR application	VR explicitly; others implied
2.2.2	Modules download only after licence validation is confirmed by the platform	All — the first entitlement gate
2.3.1	The XR application automatically downloads updated module versions when available	Device applications
2.3.2	Super Admin controls when versions release to production	Platform
2.3.3	Modules update independently of the master application, without a full reinstall	Device applications
2.4.1	Individual licences bound to an encrypted XR device identifier and a user account	B2C
2.4.2	Device replacement requires Super Admin approval before licence transfer	B2C
2.4.3	Institutional deployments allow multiple students on shared XR devices	B2B
2.4.4	Institutional students sign into the XR application with LMS credentials	B2B
10.1.1	All asset bundles encrypted before distribution	All
10.1.2	Modules require licence validation before execution	All — the second entitlement gate
12.2.1	Assets delivered through a content delivery network to reduce download latency	All

Entitlement-aware delivery is not a new requirement

- FRD 2.2.2 and 10.1.2 together already require it: modules download only after licence validation, and require licence validation before execution.
- What PM-05 adds is that this must be a designed, perceivable experience rather than only a restricted state.
- Entitlement is therefore checked twice, and both checks are visible. A learner can pass the browse-time check and fail the execution-time check, and that must not read as a defect.

10.3 The entitled-experiment lifecycle

Eight stages, holding regardless of which technical mechanism is eventually chosen.

Table 92 — Lifecycle stages

Stage	What happens	Requirement
1 — Entitled	The experiment is visible to the learner because an institutional assignment or an individual purchase created an entitlement.	6.2.1, 7.2.3, 6.1.1, 6.1.2, 9.2.2
2 — Validated	The platform confirms a valid licence for this account. Failure produces a licence-denied state.	2.2.2, 10.1.2
3 — Matched to a runtime	The required runtime is determined and the learner's access to it is established. No match produces an unsupported-device state.	1.2.1–1.2.4
4 — Delivered or made accessible	The entitled experiment becomes available on the supported device. **Mechanism pending.**	PM-05
5 — Perceivable	The learner can tell where the experiment is, through the states in section 10.4.	2.3.1, 12.2.1
6 — Launched	The experiment runs, in the portal for the web format or on the device otherwise.	1.2.1, 2.1.1, 2.1.2, 2.4.4
7 — Completed	Completion is recorded against the account, not the device. The portal reflects account state rather than observing the session.	11.1.1
8 — Reflected	Module completion is evaluated, the next module unlocks, and progress becomes visible within its permitted scope.	11.1.1, 3.2.2, 1.3.4

10.4 Required user and system states

Every state below is required regardless of mechanism. This is what allows detailed design of the learner-facing delivery experience to proceed while the mechanism decision remains open.

Table 93 — Required XR delivery states

State	Trigger	Requirement	What the user must understand
Available / entitled	Valid licence and a compatible runtime	2.2.2	This is yours, and here is how to reach it
Not entitled	No licence exists for this account	2.2.2	Why not, and what to do — a purchase path in the individual context, a direction to the institution in the institutional one
Licence denied	Validation fails at browse or at execution	2.2.2, 10.1.2	What happened and who to ask, without any internal authorization code
Licence cap reached	Institutional seat cap exhausted	6.2.2	Access is institution-managed; contact the administrator. No purchase prompt is offered
Unsupported device	The learner's device cannot run this format	1.2.2–1.2.4	The required device, stated honestly. **Never a non-functional launch control**
Downloading	Delivery in progress	12.2.1, 2.2.1	How far along it is — determinate progress where the mechanism can supply it, honest indeterminate treatment where it cannot
Paused	Delivery interrupted and resumable	Derived from 12.2.1	That it is paused rather than failed, and how to resume
Installed / ready	Available on the device	PM-05	That it can be run now, and on which device

State	Trigger	Requirement	What the user must understand
Update available	A new version has been released	2.3.1, 2.3.2	What changed where known. An unrequested update never blocks continuing
Delivery failed	Network or transport failure	Derived	What happened and what to do next, with retry available
Network failure	Connectivity lost during delivery or launch	Derived	That the failure is connectivity, and that retrying is safe
Storage issue	Insufficient device storage	Conditional	Held as a conditional state — not designed until the mechanism confirms it can occur
Launch failure	The experiment cannot start on the device	Derived from 10.1.2	That the failure is at launch rather than at entitlement, and what to try
Completion pending	The session finished but no completion has been recorded	11.1.1	That this is a normal waiting state, not an error
Completion received	The platform recorded the completion	11.1.1	That the module has progressed and what unlocked

10.5 Design rules for XR delivery

1. **Never select a mechanism.** Every delivery surface describes outcomes, never transport.
2. **Never show a non-functional control.** If the portal cannot launch it, the portal does not offer a launch control.
3. **Never invent device control.** No claimed live handoff, no remote install trigger, no invented push channel. PM-05 requires obtaining a mechanism, not inventing one.
4. **The portal reflects; it does not observe.** Completion is account state read by the portal, not an XR session watched by it.
5. **Waiting is designed.** Content delivery network latency and bundle size make preparation time real.
6. **Entitlement is checked twice and both checks are visible.**
7. **Every failure says what happened and what to do next**, with no internal authorization codes.
8. **AR is not assumed to behave like VR.** The recorded gap is flagged, not filled.
9. **State names are mechanism-neutral.** “Ready on your device” survives either concept; “downloaded” presumes one and “unlocked in the app” presumes the other, so neither is used.

10.6 Recorded gaps

Table 94 — XR delivery gaps carried forward

Gap	Detail	Classification
No AR application architecture	FRD 1.2.4 requires AR support; Section 2 defines an application architecture for VR and MR only. AR has no container, distribution channel, packaging requirement or update path. Neither delivery concept is grounded for AR, and mobile app-store review, update and entitlement constraints are unacknowledged by any requirement.	TECHNICAL-PENDING
Web simulation delivery path unstated	Whether web simulations use the same encrypted-bundle and licence-validation path as XR formats is never stated, which affects load time, the pre-launch experience and whether the execution-time gate at 10.1.2 applies in-browser.	TECHNICAL-PENDING
Package and submission format undefined	FRD 2.2.1 states only “encrypted asset bundles”. A vendor cannot be shown precisely what to upload, and a delivery state cannot indicate size or type.	TECHNICAL-PENDING
Device sign-in under SSO	FRD 2.4.4 requires LMS credentials at the device; FRD 4.2.2 with PM-10 requires institutional SSO. An SSO-authenticated learner may hold no password. Both are Phase 1 and they collide.	TECHNICAL-PENDING
Completion reconciliation	What happens when a completion is never recorded, and how quickly a recorded one reaches the portal, is undefined. Progression gates on recorded state, so a learner can be stuck through no fault of their own. Retry and refresh are offered; no manual override is provided, because an override would defeat the gate for everyone.	TECHNICAL-PENDING
MR sideloading authority	FRD 2.1.2 requires a sideloaded MR application. Sideloading is administratively gated by definition, and no requirement says who authorises or performs it.	PRODUCT-PENDING

11. Assessment Requirements

The governing position

- Assessment behaviour is configured through the portal and consumed by the relevant Web or XR application through flexible APIs (PM-08).
- For both B2B and B2C, assessment setup and configuration is owned by Faculty or the authorized institution side. The learner never authors or configures an assessment (PM-15).

11.1 Why this is a correction of interpretation, not of requirement

Platform FRD_v4 states **no question count, no duration, no score weighting, no pass mark and no attempt limit anywhere in its seventeen pages**. What it does state is configurability: Faculty upload question banks (5.1.1), questions *may* be randomized for each attempt (5.3.1), and Faculty *shall be able to enable* negative marking (5.3.2). Any fixed values that appeared in earlier prototype work filled a silence in the FRD; they were never FRD constants, and PM-08 confirms they must not be treated as platform constants.

11.2 Ownership

Table 95 — Assessment ownership

Context	Who owns configuration	Basis
B2B institutional learner	Faculty or the authorized institution side	PM-15; FRD 5.1.1, 5.1.2, 5.3.2 and 8.1.4 all name Faculty
B2C individual learner	Faculty or the authorized institution side — the same	PM-15, extending the reading to the case the FRD never addressed
Learner, either context	**Never authors or configures**	PM-15 — stated as a rule, not left as an absence

- **Institute Admin is not automatically granted assessment-authoring permission.** PM-15 explicitly declines to grant it, and the FRD names Faculty by name in every relevant requirement. Whether Institute Admin holds any assessment authority at all remains a PENDING DECISION.
- **For a pure B2C individual learner, which Faculty, institution or content owner supplies the assessment is explicitly deferred** as a content and implementation detail, and is explicitly not a blocker. Whoever supplies it acts on the authorized side, through the same portal surface, producing the same configuration.

11.3 Question types

Table 96 — Supported question types

Type	Requirement	Phase	Automatically scorable	Note
Multiple choice	5.2.1	1	Yes	Single-select or multi-select is not stated. Multi-select would require a partial-credit rule that no scoring model defines. PENDING DECISION.
True / False	5.2.2	1	Yes	No open questions.
Descriptive (short answer)	5.2.3	1	No	Grading owner, turnaround and rubric are undefined by any requirement. PENDING DECISION.
XR simulation performance scoring	5.4.1	2	—	Phase 2 in the original FRD. Out of Phase-1 scope.

No other question type exists in Platform FRD_v4, and none may be invented.

11.4 Question sourcing

Table 97 — Question sourcing requirements

Capability	Requirement	Owner	Status	Note
Structured spreadsheet upload	5.1.1	Faculty	UNCHANGED-FRD	The template structure is not specified by the FRD. Invalid rows must be reported individually rather than failing an entire upload.
AI question generation	5.1.2	Faculty	PRODUCT-PENDING	A Phase-1 Faculty authoring requirement, distinct from the Phase-3 learner-facing AI assistant at 5.5. Whether to build it in Phase 1 is a decision, not an inherited exclusion.
Human review of generated questions	Not in the FRD	Faculty	**Recommendation only**	Proposed in this draft as a quality control on 5.1.2. It is a recommendation, not an approved requirement, and must be approved separately before it is treated as one.

11.5 Configuration fields

Fields are separated by evidentiary strength. This separation prevents a plausible field from being promoted into an approved requirement.

11.5.1 Fields clearly implied by Platform FRD_v4

Table 98 — Configuration fields with explicit FRD basis

Field	Requirement basis	Type
Question types enabled	5.2.1, 5.2.2, 5.2.3	Multi-select across exactly the three Phase-1 types
Question selection and bank source	5.1.1, 5.1.2	Reference
Question randomization	5.3.1 — questions *may* be randomized for each attempt	Boolean
Negative marking enabled	5.3.2 — Faculty *shall be able to enable* it	Boolean, Faculty-owned
Assessment assignment target	8.1.4	Reference to a module, course or cohort
Attempt recording	11.2.2	Always on — recording is required, not optional
Score recording	11.2.1	Always on — recording is required, not optional

11.5.2 Fields required by the model, with no FRD value

These are the parameters an assessment cannot be rendered without. Each becomes a configuration field, and **none carries an approved default**. Any value shown anywhere is SAMPLE CONFIGURATION.

Table 99 — Configuration fields requiring a value decision

Field	Why it must exist	Status
Question count	An assessment must have a length	SAMPLE-CONFIGURATION — no approved value
Time limit	A timed attempt requires a budget	SAMPLE-CONFIGURATION — no approved value
Timing model	Total budget or per question — the behaviours differ	SAMPLE-CONFIGURATION — no approved value
Score for a correct answer	Weighting must exist if scoring exists	SAMPLE-CONFIGURATION — no approved value
Score for an incorrect answer	Required if negative marking is enabled under 5.3.2	SAMPLE-CONFIGURATION — no approved value
Score for an unanswered question	Must be defined for a complete scoring model	SAMPLE-CONFIGURATION — no approved value

Field	Why it must exist	Status
Score after a hint	Only if hints are in scope at all	**OPTIONAL / PRODUCT-PENDING** — no source defines hints
Pass threshold	Progression gates on passing	SAMPLE-CONFIGURATION — no approved value
Attempt limit	5.3.1's reference to “each attempt” confirms multiples exist; no limit is stated	SAMPLE-CONFIGURATION — no approved value
Score floor	Whether a reported score can fall below zero	SAMPLE-CONFIGURATION — no approved value
Hints	An optional capability that appeared in earlier prototype work	**OPTIONAL / PRODUCT-PENDING** — see the note below
Hint limit and hint scoring	Only if hints are in scope at all	**OPTIONAL / PRODUCT-PENDING** — no value, limit or scoring rule is assumed

How sample values must be handled

- No specification, screen, prototype or handoff may present any of these values as a platform default, a recommended value or an approved setting.
- Where an example is genuinely useful, it must be labelled SAMPLE CONFIGURATION at the point of use.

11.5.3 Questions requiring product clarification before they become fields

- Who grades descriptive answers, on what turnaround, and against what rubric — this can strand a learner mid-progression.
- Whether multiple-choice questions are single-select or multi-select, and if multi-select, what partial-credit rule applies.
- Whether an extended-time accommodation exists, and who grants it.
- Whether Institute Admin holds any assessment authority at all.
- Whether an assessment configuration is reusable across courses, cohorts or institutions.
- Whether a platform-level default configuration exists. **One is not invented here.**
- Whether a configuration may change mid-attempt or between attempts, which affects fairness and attempt history.

11.6 Result, visibility and progress

Table 100 — Result and progress requirements

Behaviour	Requirement	Note
Automatically scorable types are scored on submission	5.2.1, 5.2.2	Score is computed from the assessment's own configuration.
Descriptive answers enter an awaiting-evaluation state	5.2.3	An explicit, understandable waiting state. No grading owner is asserted.
The learner sees the score and the outcome	11.2.1	The score is computed from the assessment's own configuration.
Post-attempt feedback shown to the learner	**PRODUCT-PENDING**	**Answer-key visibility and post-attempt feedback policy are PRODUCT-PENDING.** No requirement states what feedback a learner may see. Learner-facing applications must receive only the assessment data permitted by the configured and approved feedback policy. See `QP-27`.
The score is recorded against the learner	11.2.1	The shape of a stored score depends on the configuration.
The attempt is recorded	11.2.2	Counting attempts is not the same as limiting them; any limit is configuration.
Module completion is evaluated	11.1.1	Feeds the approved progression and unlocks the next module.
Faculty sees learner results within the tenant	8.1.4, 11.2.1, 1.3.4	Operational status only in Phase 1; class-level reports are Phase 2.

11.7 Consequences for the interface

- **The pre-attempt disclosure must be generated from the configuration, not authored as fixed copy.** If a configuration disables negative marking, a hard-coded disclosure that mentions it would be untrue.
- **Layouts must tolerate variation.** Five questions or fifty; five minutes or none; a pass threshold of 40 per cent or 80. Fixed-count navigators and hard-coded maxima break.
- **Structural change to the existing learner assessment screens is none. The data-binding change is pervasive.**

11.8 The API contract

PM-08 confirms the requirement that a configured assessment is consumable by the relevant Web or XR application through flexible platform API integration. The contract itself is **TECHNICAL DETAIL TO BE DEFINED**: endpoint, method, transport, payload schema, field names, authentication, versioning, and whether results return synchronously or asynchronously are all undefined.

One constraint recorded now so it is not discovered late

- **Answer-key visibility and post-attempt feedback policy are PRODUCT-PENDING.** No requirement in Platform FRD_v4 and no stakeholder decision states what feedback a learner may see after an attempt.
- Whatever policy is approved, the contract must honour it: **learner-facing applications must receive only the assessment data permitted by the configured and approved feedback policy** — and a response body is one browser developer-tools tab away from being read, so payload scope and displayed scope must match.
- This is recorded so that the feedback policy is decided deliberately rather than settled by whatever the first payload happens to contain. See `QP-27`.

12. Learning Requirements

The approved progression

- Learning Resources → Simulation → Assessment → Completion / progression (PM-09).
- This approval covers the learner progression model only. It is not generalised into governance sequencing, entitlement rules, or any other role's workflow.

12.1 The content hierarchy

Table 101 — Content structure

Level	Definition	Requirement	Note
Certification program	One or more courses grouped into a structured learning pathway	3.3.1	**PM-deferred to Phase 2.** No Phase-1 certification experience is built. Whether the structural node is still modelled is not assumed either way.
Course	A collection of modules grouped around an engineering topic or learning objective	3.2.1	A structural grouping. Phase-1 purchase requirements explicitly define individual module and bundle purchasing; direct course purchase is not specified.
Module	The smallest learning unit, containing a single XR simulation experience	3.1.1	The atomic unit of the content model and the object every governance gate acts upon.
Module components	Theoretical explanations, simulation instructions and assessments	3.1.2	The FRD says modules *may* include these, so all three are optional and every renderer must tolerate their absence.

12.2 Sequencing and progression

Table 102 — Sequencing requirements

Behaviour	Requirement	Note
A course defines the sequence in which its modules must be completed	3.2.2	Enforced between modules. A later module is not reachable until its prerequisite is complete, and a locked item states why.
Within a module, components follow the approved order	PM-09, 3.1.2	Learning resources, then simulation, then assessment.
Module completion is evaluated against every required component	11.1.1	A module with no assessment or no resources completes on the components it actually has.
Completion is recorded against the account, not the device	11.1.1	A device-run simulation reports back to the account; the portal reflects account state rather than observing the session.
A standalone purchased module is not gated behind another module	**DESIGN-PENDING**	Derived rather than stated: FRD 3.2.2 defines sequencing within a course, and a standalone purchase has no course context. Module-internal ordering still applies.
Whether the sequence is per-course configurable	Not stated	**PENDING DECISION.** PM-09 approves the model, not its mutability. Neither configurable nor fixed is assumed.

12.3 Learning resources

- Modules may carry supporting learning material — theoretical explanations and simulation instructions (FRD 3.1.2).
- Where a resource is required for completion, that requirement must be visible before the learner begins, not discovered when completion fails.
- Where a resource is optional, it must be distinguishable from a required one.

- **Recorded open item:** ownership of video captions, transcripts and audio description is undefined (`QX-03`). The position taken in this draft is that a video without captions should not be a required component for completion, because that would make completion impossible for some learners. That is a design position, not an approved rule.

12.4 Access, entitlement and error states

Table 103 — Learning access states

State	Trigger	Requirement	Required behaviour
Available	Entitled and prerequisites met	8.1.5	Reachable, with the next action deterministic
Locked by sequence	A prerequisite module is incomplete	3.2.2	State which prerequisite blocks it and where to go
Not entitled	No entitlement exists	2.2.2	Individual context: a purchase path. Institutional context: a direction to the institution
Access restricted	Licence cap reached or licence invalid	6.2.2, 2.2.2	Explain the cause and the next step, with no internal codes. No purchase prompt is offered in the institutional context, because FRD 1.1.2 makes the institution the licence holder
Content unavailable	Content withdrawn, changed or failed to load	2.3.2	Explain the situation without implying learner error
Empty learning	No entitlement has been assigned or purchased	7.2.3, 9.2.2	Institutional: direct to the institution. Individual: offer discovery
Awaiting evaluation	A descriptive answer has not been graded	5.2.3	A normal waiting state, not an error. **Recorded risk: this can strand progression**

12.5 Resuming learning

A learner returning to the platform must be able to resume without reconstructing where they were. The deterministic next action follows directly from the sequencing requirements at 3.2.2 and from the module-internal order PM-09 approves: there is exactly one next thing to do, and it can be surfaced.

13. Commerce, Licensing and Entitlements

Licensing and entitlements is the platform's structural spine. Learning reads it, XR delivery validates against it twice, commerce creates it, ERP populates it, and tenancy scopes it. Two origins — institutional licensing and individual purchase — produce the same object, and that object is what a learner consumes.

13.1 B2C individual commerce

Table 104 — B2C commerce requirements

Capability	Requirement	Status	Note
Individual module purchase	6.1.1	UNCHANGED-FRD	A module is a purchasable product.
Bundle purchase	6.1.2	UNCHANGED-FRD	A bundle is a purchasable product. **Tension recorded:** bundle creation (6.3.1) and pricing configuration (6.3.2) are Phase 2, so Phase 1 sells bundles that no Phase-1 requirement lets anyone create.
Direct course purchase	Not specified	**NOT-SPECIFIED / PRODUCT-PENDING**	Phase-1 purchase requirements explicitly define individual module and bundle purchasing (6.1.1, 6.1.2). **Direct course purchase is not specified and is not included in the current design unless subsequently approved.** It is neither required nor prohibited by any source. See `QP-28`.
Product detail	6.1.1, 6.1.2	UNCHANGED-FRD	Access duration is shown only where authoritative data exists; it is never estimated.
Checkout	6.1.x	UNCHANGED-FRD	The purchase must complete. Whether checkout is single-item or basket-based is **DESIGN-PENDING** — no requirement specifies a cart model.
Indian payment gateway integration	9.1.1	TECHNICAL-PENDING	**No provider is selected.** Redirect or embedded checkout is undetermined.
International payment gateway integration	9.1.2	PRODUCT-PENDING	**No provider is selected.** Currency, tax treatment and price presentation are undefined and are not invented here.
Payment result handling	9.1.x	UNCHANGED-FRD	Success, decline and indeterminate outcomes are each designed states.
Automated licence activation	9.2.1	UNCHANGED-FRD	Activation is automatic on successful payment; no manual step is required of the learner.
Entitlement assignment	9.2.2	UNCHANGED-FRD	The purchase appears in the learner's entitled content immediately.
Personal device binding	2.4.1	PRODUCT-PENDING	Individual licences bind to an encrypted device identifier and an account. Whether a phone qualifies for the AR format is unstated.
Receipt or invoice	Not defined for B2C	PRODUCT-PENDING	GST-compliant institutional invoicing (9.3.1) is Phase 2 and institution-facing. What a B2C buyer receives is undefined.
Refund, cancellation and revocation	Not in the FRD	PRODUCT-PENDING	No rule is stated or implied anywhere. **None is invented here.**

13.2 B2B institutional licensing

Table 105 — B2B licensing requirements

Capability	Requirement	Status	Note
Stream-based institutional licensing	6.2.1	PRODUCT-PENDING	Institutions purchase packages mapped to academic streams. **How this is transacted in Phase 1 is undefined**, because institutional billing (9.3.1, 9.3.2) is Phase 2.
Capped institutional licence	6.2.2	UNCHANGED-FRD	Licences may be capped by maximum student count.
Seat and licence visibility	6.2.2	UNCHANGED-FRD	The Institute Admin must see seat utilisation before exhaustion, because the cap is the direct cause of a learner-facing access wall.
Stream to package mapping	6.2.1, 7.2.3	UNCHANGED-FRD	The Institute Admin maps academic streams to licensed packages; ERP supplies the stream data.

Capability	Requirement	Status	Note
Access assignment	8.1.3, 7.2.3	UNCHANGED-FRD	Entitlements are granted and revoked within the tenant.
Institutional entitlement	1.1.2, 6.2.1	UNCHANGED-FRD	Students and faculty access under a shared institutional subscription.
Shared device access	2.4.3	UNCHANGED-FRD	Institutional deployments use shared devices, which is why there is no personal device binding in this context.
Institutional invoicing and subscription billing	9.3.1, 9.3.2	FUTURE-PHASE	Phase 2 in the original FRD.
Licence grace period	6.5.2	FUTURE-PHASE	Phase 3 in the original FRD.

13.3 Entitlement validation

Table 106 — Where entitlement is enforced

Point	Requirement	Effect
At browse and before delivery	2.2.2	A module is not delivered to a device before the platform confirms a valid licence.
Before execution	10.1.2	A module does not execute without licence validation. A learner may pass the first check and fail this one.
Against an institutional seat cap	6.2.2	Access is refused when the cap is exhausted, with an explanation that directs the learner to their institution.
Against tenant boundaries	1.3.1, 1.3.4	An entitlement is meaningful only within its own tenant or its own personal context.

What this chapter does not decide

- No payment provider is selected, for either the domestic or the international requirement.
- No currency, tax, refund, cancellation or invoicing rule is stated. These are compliance-adjacent and remain PENDING DECISION.
- Direct course purchase is not specified by any requirement, is not included in the current design, and is not prohibited either. It is recorded as NOT-SPECIFIED / PRODUCT-PENDING (`QP-28`).
- How an institutional licence is transacted in Phase 1, before Phase-2 billing exists, is recorded as an open question rather than answered.

14. SSO, ERP and Integration Requirements

14.1 Institutional single sign-on

Institutional SSO is required (PM-10). This resolves a contradiction inside Platform FRD_v4, which places SSO configuration in Phase 1 at 4.2.2 and SSO authentication support in Phase 3 at 7.4.1. The Phase-1 reading is confirmed. The requirement is a **configuration capability**, not merely a login button.

Table 107 — SSO requirements

Capability	Requirement	Owner	Status
Institution configures single sign-on for its tenant	4.2.2	Institute Admin	UNCHANGED-FRD — the configuration surface is newly designed
Identity-provider settings	4.2.2	Institute Admin	Provider and protocol are a PENDING DECISION
Attribute mapping	4.2.2	Institute Admin	Mapping scheme is a PENDING DECISION
Test the connection before enabling	Derived from 4.2.2	Institute Admin	Required so a misconfiguration is found before it locks users out
Users authenticate through institutional SSO	4.2.2, 7.4.1	Learner, Faculty, Institute Admin	FRD-INTERNAL-CONFLICT resolved by PM-10 in favour of Phase 1. **SSO is required** and is available according to the institution's configured authentication model; it is not described as merely preferred
Identity, tenant, role and plane resolution	8.1.x, 1.3.1, PM-13	Platform	Newly specified — with one role there was nothing to resolve
Automatic role mapping from IdP attributes	7.4.2	Institute Admin	FUTURE-PHASE — Phase 3. Its absence leaves the Phase-1 role-assignment origin undefined
Failure and fallback states	Derived from 4.2.2, 10.2.1	Platform	An SSO failure is explicit and offers a route forward; it is never a silent dead end

Two recorded consequences

- The Phase-1 origin of institutional role assignment is undefined. FRD 7.4.2 (role mapping from identity-provider attributes) and FRD 8.3 (custom role creation) are both Phase 3. Super Admin assigns platform roles under 8.1.1; how a person becomes Faculty or Institute Admin within a tenant in Phase 1 is recorded as an open question.
- FRD 2.4.4 requires institutional students to sign into the XR application using LMS credentials. Under institutional SSO a learner may hold no password. Both requirements are Phase 1 and they collide; the collision is recorded and not designed around.

14.2 ERP integration

Table 108 — ERP requirements

Capability	Requirement	Owner	Status
Integration with academic ERP systems through standardized API integration	7.1.1	Institute Admin, Super Admin	TECHNICAL-PENDING — provider and contract undetermined. The systems named in the FRD are examples, not a selection
API endpoints receive enrolment, course-mapping and academic structure data	7.1.2	Platform	TECHNICAL-PENDING — the contract is undefined
Scheduled synchronisation at configurable intervals	7.2.1	Institute Admin	UNCHANGED-FRD — the configuration surface is newly designed
Automatic student account creation	7.2.2	Platform	UNCHANGED-FRD
Stream-based package assignment	7.2.3	Platform	UNCHANGED-FRD — the Institute Admin needs mapping visibility
Record validation before state change	Derived from 7.1.2	Platform	Valid, invalid and conflicting records are separated

Capability	Requirement	Owner	Status
Failed-record handling and retry	Derived from 7.2.1	Institute Admin	Failed records are retained with a reason; nothing is silently discarded
Synchronisation operational status	7.2.1	Institute Admin	Last run, outcome, counts and errors — operational only per PM-11
Real-time provisioning at first login	7.3.1, 7.3.2	Platform	FUTURE-PHASE — Phase 2

Source-of-truth concern

- The ERP is authoritative (7.1.1) and scheduled synchronisation (7.2.1) will overwrite local edits, which reads as data loss to the person whose edit disappeared.
- FRD 4.3.1 nonetheless has users providing profile details, and this now applies to Faculty profiles as well as learner profiles.
- Which fields are user-editable and which are ERP-owned is a PENDING DECISION. The working position is that ERP-sourced fields are read-only with their provenance stated.

14.3 Integration boundaries

Table 109 — Integration boundaries and their status

Integration	What crosses the boundary	Requirement	Contract status
XR application	Entitlement state outward; completion and result inward	2.1.1, 2.1.2, 2.2.2, 11.1.1, PM-05	PENDING — mechanism deliberately unselected
Assessment consumption	Assessment configuration outward; attempt result inward	PM-08	PENDING — schema, transport and versioning undefined
Payment gateway	Transaction request outward; payment outcome inward	9.1.1, 9.1.2	PENDING — no provider selected
Identity provider	Authentication request outward; assertion inward	4.2.2, 7.4.1	PENDING — no provider or protocol selected
ERP	Enrolment, course-mapping and academic structure data inward	7.1.1, 7.1.2	PENDING — no provider or contract selected
Object storage and content delivery	Encrypted asset bundles outward to devices	10.1.1, 12.1.2, 12.2.1	PENDING — technology unselected
E-mail and notification	Outbound messages	No Phase-1 requirement	**Not justified for Phase 1.** In-workflow status is specified instead

Every integration boundary above is a **contract to be defined**, not a chosen product. This document names no provider, protocol, endpoint or payload format for any of them.

15. Reporting and Analytics

The governing position

- Build core functionality and core operations first. Detailed reporting and analytics parameters are added incrementally once the underlying functionality and the available data are understood (PM-11).
- This agrees with the FRD's own phasing rather than deviating from it. Every Phase-1 analytics requirement in the FRD is a recording requirement; every dashboard requirement is Phase 2 or Phase 3.

15.1 The operational / analytical test

Applied to every proposed reporting element in Phase 1.

Table 110 — What qualifies as Phase-1 reporting

Question the element answers	Classification	Phase 1?	Basis
What is the state of the thing I operate right now?	Operational	Yes	PM-11 permits operational summaries and status
Which of my learners are blocked today?	Operational	Yes	8.1.4 Faculty role definition
How many seats of our licence are in use?	Operational	Yes	6.2.2 capped licence
What is in my review queue?	Operational	Yes	1.4.2, 1.4.3 governance gates
How is this cohort performing over the semester?	Analytical	No	11.3.1 — Phase 2
What is our module adoption trend?	Analytical	No	11.3.2 — Phase 2
Where do learners make mistakes inside a simulation?	Analytical	No	11.4.1, 11.4.2 — Phase 3

15.2 Phase-1 recording requirements

Table 111 — What Phase 1 records

What is recorded	Requirement	Visible to
Module completion status for each learner	11.1.1	The learner for their own record; Faculty and Institute Admin as tenant-bounded operational status
Time spent interacting with simulations	11.1.2	The learner; Faculty and Institute Admin within the tenant. Presentation granularity is a PENDING DECISION
Assessment scores associated with the learner	11.2.1	The learner; Faculty within the tenant
Number of attempts for each simulation or assessment	11.2.2	The learner as attempt history; Faculty within the tenant

15.3 Phase-1 operational status by role

Table 112 — Permitted Phase-1 operational status

Role	What they see	Requirement	Bound
Learner	Own progress: completion, time spent, scores, attempt history	8.1.5, 11.1.x, 11.2.x	Own records only
Faculty	Operational learner status: who has completed, who is blocked, what attempts exist	8.1.4	Own tenant only (1.3.4)
Institute Admin	Institutional operational status: seat utilisation, active learners, ERP synchronisation state, SSO configuration state	8.1.3, 6.2.2, 7.2.1, 4.2.2	Own tenant only (1.3.4)
Content Manager	Content pipeline status: review queue, returned submissions, content awaiting publication	8.1.2, 1.4.3	Central content plane
Vendor	Own submission status and review history	1.4.1, PM-14	Own submissions only
Academic Reviewer	Validation queue status and own decision history	1.4.2, PM-14	Central content plane
Super Admin	Cross-tenant operational oversight, publication queue, pending device replacements, institution status	8.1.1, 1.3.5	Platform-wide, bounded to governance and operational oversight

15.4 What is deliberately not specified

- **No detailed KPI dashboards are invented.** Advanced reporting parameters are incremental by explicit instruction.
- **Report formulas, filters, aggregations and metric definitions remain PENDING DECISION.** No metric is defined here that has not been approved.
- **Detailed institutional analytics follow the original phase model** — institutional usage dashboards and licence utilisation tracking are Phase 2, class-level and adoption reports are Phase 2, and XR interaction analytics are Phase 3 — unless a later decision changes that.
- **Every reporting surface in every phase remains bounded by tenant scoping.** An institution's reporting sees only that institution's data, and the only exception is the Super Admin governance access the FRD grants.

16. Non-Functional Requirements

How this chapter is written

- Platform FRD_v4 leaves most non-functional characteristics high-level or unstated. Where no approved target exists, this chapter records the requirement category and labels the target PRODUCT-PENDING or TECHNICAL-PENDING.
- **No numeric target is invented.** A fabricated availability figure, response-time budget or retention period would read as an approved commitment and would be wrong.

Table 113 — Non-functional requirement categories

Category	Requirement position	FRD basis	Target status
Accessibility	The platform shall meet a stated accessibility conformance level across all seven workspaces, and the level shall be stated rather than implied. Known non-conformances shall be recorded rather than omitted.	Not stated in the FRD	PRODUCT-PENDING — conformance level to be approved
Responsive behaviour	The platform shall define which viewport classes each workspace supports. Administrative workflows are desktop-typical, but that must be decided per workspace rather than inherited from the learner surface.	Not stated	PRODUCT-PENDING
Supported browser and device policy	The platform shall state which browsers, operating systems and device classes are supported, and how an unsupported combination is handled.	1.2.1, 1.2.4 imply but do not state	PRODUCT-PENDING
Performance	The platform shall define response-time expectations for interactive operations and for asset delivery. Delivery latency is explicitly acknowledged as real.	12.2.1 acknowledges latency	TECHNICAL-PENDING — no numeric target approved
Availability	The platform shall define an availability target and a maintenance-window policy.	Not stated	TECHNICAL-PENDING — no numeric target approved
Scalability	The platform shall scale with platform traffic and shall support additional regional data instances.	12.3.1, 12.3.2 (Phase 2), 1.3.3	TECHNICAL-PENDING
Security	The platform shall enforce secure authentication, session expiry, asset encryption, licence validation before execution, and periodic vulnerability assessment and penetration testing.	10.1.1, 10.1.2, 10.2.1, 10.2.2, 10.3.1, 10.3.2, 12.1.2	Requirements stated; policy detail PENDING
Authentication policy	The platform shall define password policy, multi-factor behaviour, lockout behaviour and reset behaviour.	Absent from the FRD entirely	PRODUCT-PENDING — none invented here
Privacy	The platform shall define what personal data it holds, on what basis, and how tenant and B2C separation is preserved in practice.	1.3.1, 1.3.2, 10.4.1 (Phase 2)	PRODUCT-PENDING
Auditability	Content governance decisions, comments, returns and resubmissions shall be recorded and traceable. Audit outside content governance is undefined.	PM-14; nothing broader in the FRD	Content governance stated; broader audit PRODUCT-PENDING
Backup and recovery	The platform shall define backup frequency, retention and recovery objectives.	Not stated	TECHNICAL-PENDING — no target approved
Monitoring and observability	The platform shall define what is monitored, what is alerted on, and who receives alerts. Integration failures — ERP, SSO, payment, delivery — are the operationally significant events.	Not stated	TECHNICAL-PENDING
Maintainability	Shared capabilities shall be defined once and consumed by many roles rather than reimplemented per workspace.	Derived from the capability model	Stated as a design constraint
Testability	Every requirement in this document carries at least one testable acceptance statement, and every state that a user can reach shall be reachable in test.	Derived	Stated as a design constraint
Documentation	The platform shall maintain requirement, architecture, integration-contract and operational documentation, and shall record which decisions remain open.	Derived	Stated as a process constraint
Data retention	The platform shall implement configurable data retention policies for user data and analytics records.	10.4.2 — Phase 2	FUTURE-PHASE; periods not stated

Category	Requirement position	FRD basis	Target status
File and package size limits	The platform shall define acceptable submission and asset-bundle sizes and how an oversized submission is handled.	2.2.1 states only “encrypted asset bundles”	TECHNICAL-PENDING — depends on the package format decision
Environment separation	The platform shall maintain a staging environment distinct from production, since the governance chain depends on one.	1.4.1 requires a staging environment; 1.4.4 requires a production environment	Requirement stated; topology TECHNICAL-PENDING

Why the pending labels matter here

- An unapproved number in a non-functional requirement is more dangerous than an acknowledged gap: it is quoted, designed against, and eventually contracted against.
- Each PENDING entry above names a decision that should be taken and recorded, with an owner, before build begins.

17. UI/UX Requirements

Scope of this chapter

- This chapter records interface requirements at the level a functional requirement document should carry: navigation model, shared surfaces, required states and process.
- It deliberately does not become a screen-by-screen design document. The detailed screen inventory produced by the full-platform reframe is referenced as a design-planning artefact, not reproduced here.

17.1 The approved design process

PM-02 sets the order of work, and it is recorded here as a requirement on the process rather than on the product.

1. Create the complete platform sitemap.
2. Create the role and permission model.
3. Create workflows for all roles.
4. Identify the most important end-to-end flows.
5. Take those flows into detailed interface design.

Detailed interface work does not begin for a flow before its sitemap position, its permission set and its workflow have been recorded and reviewed. Which flows are taken first is a stakeholder selection that has not yet been made.

17.2 Navigation and shell

Table 114 — Navigation requirements

Requirement	Basis	Detail
One shared shell across all seven workspaces	PM-03	The same header, account menu, authentication behaviour, component set and state grammar. What differs is navigation content and permitted actions, never the chrome.
Single sign-in entry point for all roles	**DESIGN-PENDING**	A design position, not a confirmed requirement. FRD 10.2.1 requires secure authentication and PM-03 requires one central platform; neither specifies the sign-in model. See `QX-10`.
Role resolved from the authenticated identity	**DESIGN-PENDING**	A design position taken because a pre-login role choice would expose the role structure. No source requires or forbids either model. See `QX-10`.
Role-based navigation	8.1.1–8.1.5, PM-07	Each workspace exposes only its own capabilities. Out-of-phase capabilities are absent, not shown disabled.
Workspace chooser	**Conditional / PRODUCT-PENDING**	Available as a conditional design pattern **only if** multi-role identities are later approved. Whether they exist in Phase 1 is not stated by any source, so no chooser is mandated. See `QP-11`.
Desktop-first administrative workflows	Derived	Administrative work is desktop-typical. This must be decided per workspace rather than inherited from the learner surface — recorded as an open design question.
Responsive foundations	Derived	Layouts are built so that a later responsive pass is possible without restructuring. The responsive pass itself is a separate scope decision.
Prohibitions expressed positively	1.4.5	A vendor sees where a submission is, never a disabled publish control, because a disabled control implies a permission that will never exist.
Shared surfaces are shared components	Derived	One module viewer consumed by five roles in permission-filtered contexts, one state grammar, one shell — not per-role rebuilds.
Tenant boundaries are structural	1.3.1, 1.3.4	No tenant switcher exists for tenant-plane roles. Only Super Admin crosses tenants, through an explicit institutional context.

17.3 Required interface states

Every state below must exist as a designed state rather than as an absence. Two rules apply to all of them: each says **what happened** and **what to do next**, and none exposes an internal authorization code, identifier or stack trace.

Table 115 — Required states across the platform

State	Where it applies	Requirement basis
Loading and processing	Every surface that waits, especially asset and device preparation	12.2.1
Empty	Every list, queue and collection, with a context-appropriate next action	Derived
Success	Every state-changing action, so the user knows it took effect	Derived
Validation error	Every form, at field level rather than only at form level	Derived
System error	Any unexpected failure	Derived
Permission denied	Any attempt to reach something the role may not reach	8.1.x, 1.4.5
Unsupported device	Simulation access where the runtime is unavailable	1.2.2–1.2.4
Expired or invalid licence	Access and launch attempts	2.2.2, 10.1.2
Licence cap reached	Institutional access when seats are exhausted	6.2.2
Failed payment	Purchase, including the indeterminate outcome	9.1.1, 9.1.2
Changes required	A returned submission, carrying reviewer comments and a stated reason	PM-14
Resubmitted	A corrected submission that has re-entered review	PM-14
Approved / validated	Each governance gate decision	1.4.2, 1.4.3
Published	Content released to production	1.4.4
Delivery and download states	Available, downloading, paused, failed, installed, update available	PM-05, 2.3.1, 12.2.1
Session expiry	Every workspace, protecting unsaved learner and administrative work	10.2.2
Awaiting evaluation	A submitted descriptive answer	5.2.3
Synchronisation status	ERP configuration and history	7.2.1

17.4 Accessibility

- The platform shall meet a **stated** accessibility conformance level, and the level shall be recorded rather than implied.
- Known non-conformances shall be recorded openly. A conformance claim that omits a known gap is worse than no claim.
- Meaning shall never be carried by colour alone. Every classification, state and status in this document and in the interface carries a text label.
- Ownership of video captions, transcripts and audio description is undefined and is recorded as an open question. A video without captions must not be a required component for completion.
- The reflow behaviour required for small viewports is a known outstanding item that now applies across all seven workspaces rather than one.

17.5 Relationship to the screen inventory

The full-platform reframe produced a screen inventory covering nine workspace areas. It is a **design-planning artefact**, not a requirement, and it is referenced rather than reproduced here. Its role is to show that every Phase-1 requirement has somewhere to live and to support the selection of highest-priority flows for detailed design. Nothing in this document depends on it.

18. Acceptance Criteria

Status of these criteria

- These are the conditions that must hold for the corresponding requirements to be considered met. They are written to be testable.
- **No implementation test has been run and none is claimed to have passed.** The platform described in this document is not implemented. These criteria describe what will be verified, not what has been verified.

Criteria are grouped by capability area and cover the happy path, validation, empty states, error states, permission states, tenant boundaries, audit and history states, retry and recovery, entitlement states, unsupported devices, content rejection and resubmission, configurable assessment, single sign-on, XR delivery, and payment and entitlement activation.

18.1 Authentication, roles and tenancy

Table 116 — Authentication, roles and tenancy

ID	Scenario	Criterion
AC-01	Happy path — sign-in and workspace resolution	Given a provisioned user with exactly one role, when they authenticate successfully, then they land directly in that role's workspace and see only its navigation.
AC-02	Permission state — role boundary	Given a user in one role, when they attempt to reach a capability belonging to another role, then a permission-denied state explains the refusal in role-appropriate language and exposes no internal code.
AC-03	Tenant boundary	Given a Faculty member or Institute Admin of institution A, when they search, filter or report on any surface, then no record belonging to institution B is returned in any state.
AC-04	Cross-tenant governance	Given a Super Admin, when they view institutional data, then it is presented within its institutional context rather than aggregated across institutions, and no other role can reach more than one tenant.
AC-05	Central plane isolation	Given a Vendor, Academic Reviewer or Content Manager, when they use any surface, then no tenant data is reachable and no tenant switcher exists.
AC-06	SSO configuration	Given an Institute Admin configuring single sign-on, when they run a connection test, then the outcome is reported explicitly, and single sign-on can be enabled only after a successful test.
AC-07	SSO failure	Given an institutional user authenticating through single sign-on, when the identity provider returns a failure, then an explicit failure state states what happened and offers a route forward.
AC-08	Session expiry with unsaved work	Given a user with unsaved work — a learner mid-attempt, an Institute Admin mid-configuration or a Vendor mid-upload — when the session approaches expiry, then they are warned before it expires and the unsaved work is not silently discarded.

18.2 Content governance

Table 117 — Content governance

ID	Scenario	Criterion
AC-09	Happy path — submission to publication	Given a Vendor submission, when the Academic Reviewer validates it and the Content Manager approves readiness, then it appears in the Super Admin publication queue and can be published.
AC-10	Validation — return requires a reason	Given a reviewer recording a changes-required decision, when no reason is supplied, then the decision cannot be recorded.
AC-11	Return and resubmission	Given a returned submission, when the Vendor opens it, then the reviewer's comments and reason are visible and the build, assets and metadata become editable; and when it is resubmitted, it re-enters review.
AC-12	No shortcut to publication	Given a submission that has not recorded approval at both review gates, when the Super Admin views the publication queue, then it is absent from that queue in every state.
AC-13	Vendor prohibition	Given any submission in any state, when the Vendor views it, then no publication control is present, enabled or disabled, anywhere in the workspace.
AC-14	Audit and history state	Given any submission, when an authorised role views its review history, then every decision, its author, its timestamp and its comments are visible in order.
AC-15	Empty state — review queue	Given a reviewer with nothing awaiting them, when they open their queue, then an empty state confirms there is nothing to review rather than presenting an ambiguous blank list.

18.3 Entitlement, licensing and payment

Table 118 — Entitlement, licensing and payment

ID	Scenario	Criterion
AC-16	Entitlement state — entitled	Given an entitled module, when the learner opens it, then it is reachable and the next action is unambiguous.
AC-17	Entitlement state — not entitled	Given a module for which no entitlement exists, when the learner attempts access, then an access-restricted state explains the cause and the next step: a purchase path in the individual context, a direction to the institution in the institutional context.
AC-18	Licence cap	Given an institutional licence at its seat cap, when a further learner attempts access, then they are refused with an explanation directing them to their institution and no purchase prompt; and the Institute Admin could see seat utilisation before this occurred.
AC-19	Payment and entitlement activation	Given a successful individual payment, when it completes, then the licence activates automatically, the entitlement is assigned, and the purchased content is reachable without any further learner action.
AC-20	Payment failure and recovery	Given a declined or indeterminate payment outcome, when the learner returns to the platform, then the state is stated plainly, no entitlement has been created for an unpaid transaction, and a retry path is available.
AC-21	Double validation	Given a learner who passed the browse-time licence check, when execution-time validation fails, then the resulting state explains that the failure is at launch rather than at entitlement, and does not read as a defect.

18.4 XR delivery

Table 119 — XR delivery

ID	Scenario	Criterion
AC-22	Delivery — entitled and compatible	Given an entitled experiment and a compatible supported device, when the learner requests access, then the experiment becomes available on that device and the learner can perceive its state throughout.
AC-23	Unsupported device	Given an entitled experiment whose runtime the learner cannot reach, when they open it, then the required device is named honestly and no non-functional launch control is offered.
AC-24	Delivery failure and retry	Given a delivery that fails, when the learner views the experiment, then the state states what happened and what to do next, and a retry is available.
AC-25	Update available	Given a released new version of a delivered module, when the learner views it, then an update-available state is shown and an unrequested update does not block continuing.
AC-26	Completion recorded to the account	Given a simulation completed on a device, when the platform records completion, then it is recorded against the account rather than the device, and module progression is evaluated.
AC-27	Completion pending	Given a session that has finished but whose completion has not been recorded, when the learner views the module, then a completion-pending state is presented as a normal waiting state with a refresh available, and no manual completion override is offered.
AC-28	Device replacement	Given a learner requesting a device replacement, when Super Admin approves it, then the licence transfers to the new device and the learner can see that it has; and before approval the request shows as awaiting approval.

18.5 Assessment

Table 120 — Assessment

ID	Scenario	Criterion
AC-29	Configuration-driven disclosure	Given an assessment with a specific configuration, when a learner opens its overview, then every disclosed value — scoring, penalties, timing, attempts and pass threshold — is generated from that configuration and none is fixed copy.
AC-30	Configuration variation	Given two assessments with materially different configurations, when each is rendered, then both render correctly without layout failure, and neither displays a value belonging to the other.
AC-31	Learner authoring prohibition	Given any assessment in any state, when a learner views it, then no interface offers any capability to create, edit or configure it.
AC-32	Answer key prohibition	Given a completed attempt, when the learner reviews it, then the feedback shown matches the approved feedback policy, and the payload the client receives carries no assessment data beyond what that policy permits. **The policy itself is PRODUCT-PENDING (`QP-27`); this criterion becomes testable once it is approved.**
AC-33	Question bank upload validation	Given a structured question-bank upload containing invalid rows, when it is processed, then valid rows are accepted and each invalid row is reported individually with its reason.
AC-34	Descriptive answer	Given a submitted descriptive answer, when scoring completes for the automatically scorable questions, then the descriptive answer enters an explicit awaiting-evaluation state that the learner can understand.
AC-35	Result and progress	Given a completed attempt, when it is submitted, then the score and the attempt are recorded, module completion is evaluated, and the result is visible to the learner and, within the tenant, to Faculty.

18.6 Learning and progression

Table 121 — Learning and progression

ID	Scenario	Criterion
AC-36	Sequencing	Given a course with a defined module sequence, when a learner opens it, then a module whose prerequisite is incomplete is not reachable and states which prerequisite blocks it.
AC-37	Module completion	Given a module whose required components are all complete, when the last one completes, then the module is recorded complete and the next module becomes reachable.
AC-38	Optional components	Given a module with no learning resources, or no assessment, or neither, when the learner opens it, then it renders correctly and its completion model reflects only the components it has.
AC-39	Resume	Given a returning learner with work in progress, when they arrive, then a single unambiguous next action is offered without requiring them to reconstruct where they were.
AC-40	Empty entitlement	Given a learner with no entitled content, when they open their learning area, then an empty state explains why and what to do — direct to the institution in the institutional context, offer discovery in the individual context.

18.7 Integration and operations

Table 122 — Integration and operations

ID	Scenario	Criterion
AC-41	ERP synchronisation — happy path	Given a configured ERP integration, when a scheduled synchronisation runs successfully, then student accounts and stream-based entitlements are updated and the Institute Admin can see the run, its outcome and its counts.
AC-42	ERP synchronisation — failure and retry	Given a synchronisation run containing failed records, when it completes, then failed records are retained with a reason, nothing is silently discarded, and a retry is available.
AC-43	Entitlement change between sessions	Given a synchronisation that removes an entitlement, when the learner next signs in, then the content is no longer offered and the change does not present as a system error.
AC-44	Operational status boundary	Given any Phase-1 reporting surface, when a role views it, then it answers a current-state operational question, is bounded to that role's permitted scope, and presents no cohort-over-time analytical view.

19. Questions and Decision Register

19.1 Resolved stakeholder questions

Nineteen questions have been answered directly by the Project Manager across the two confirmation sessions. **None of these is shown as open anywhere in this document.** Any document still treating one of them as open is out of date.

Table 123 — Resolved stakeholder questions

Question	Resolution	Authority
What is the requirement and design scope?	All Phase-1 requirements and all Phase-1 roles.	PM-01
What order must the design work follow?	Sitemap, then permissions, then all-role workflows, then priority flows into detailed design.	PM-02
Is this one platform or several products?	One central portal with role-based access and workspaces.	PM-03
Is it B2B, B2C, or both?	Both. Neither is sequenced ahead of the other.	PM-04
Must entitled experiments reach the device?	Yes — confirmed as a product requirement. The mechanism is deliberately unselected.	PM-05
Do the XR simulations already exist?	Yes. The team is not assumed to build every simulation from scratch.	PM-06
Is Vendor a real role with a login?	Yes — an actual platform login and workspace.	PM-07
Is Academic Reviewer a real role with a login?	Yes — an actual platform login and workspace.	PM-07
Are assessment rules fixed or configurable?	Configurable through the portal, consumed by applications through flexible APIs.	PM-08
Are the earlier fixed assessment values platform constants?	No. They are sample configuration only.	PM-08
Is the proposed learning sequence acceptable?	Yes — resources, then simulation, then assessment, then completion.	PM-09
Is institutional SSO required or deferrable?	Required. It stays in the architecture and in the institutional authentication experience.	PM-10
How much reporting is needed now?	Core operations first; analytics added incrementally.	PM-11
Is certification in Phase 1?	No — Phase 2. Recorded as a deviation from the FRD as written.	PM-12
Where do Vendor, Academic Reviewer and Content Manager sit in the tenancy model?	The central content plane. Super Admin alone retains cross-tenant access.	PM-13
Is there a reject, revise and resubmit path in the governance chain?	Yes — approve or changes required, with comments, correction, resubmission and review history.	PM-14
Is there an audit trail for governance decisions?	Yes, for the content governance chain. Review history and status traceability are required.	PM-14
Who configures assessments for a B2C individual learner?	Faculty or the authorized institution side — the same as for B2B.	PM-15
May a learner author or configure an assessment?	Never, in either market context.	PM-15

19.2 Open questions — summary

Table 124 — Open questions by class and priority

Class	Blocker	High	Medium	Low	Total
Product	0	3	14	7	24
Technical	0	4	9	2	15
Data / Integration	0	2	6	2	10
Design	0	3	6	1	10
Total open	**0**	**12**	**35**	**12**	**59**
Future phase — recorded so their absence is understood as scope	—	—	—	—	19
Closed by stakeholder decision	—	—	—	—	19

No blocking questions

- No open question blocks the platform architecture, the role model, the capability map or the workflow set. That is stated deliberately: calling something a blocker when work can continue is as damaging as missing a real one.
- Several questions do block **development** of a specific capability. Those are marked individually in the register.

19.3 High-priority open questions

These twelve change platform structure, authority, or a legal or compliance position. The complete register, including medium and low priority items, is at Appendix B.

Table 125 — High-priority open questions

ID	Class	Question	Current safe assumption	Owner
QP-02	Product	Who grades descriptive answers, on what turnaround, and against what rubric?	The grading queue is named and positioned but explicitly unowned and undesigned.	Project Manager with academic stakeholder
QP-04	Product	How is an institutional licence transacted in Phase 1, given that institutional billing is Phase 2?	Licences are shown as existing. No in-platform purchase flow is invented.	Project Manager with commercial owner
QP-27	Product	What post-attempt feedback may a learner see, and is answer-key visibility ever permitted?	No feedback policy is asserted in either direction. Learner-facing applications receive only the assessment data the configured and approved policy permits, and that policy is undefined.	Project Manager with academic stakeholder
QT-01	Technical	Which XR delivery mechanism is used?	The delivery contract is mechanism-neutral by design and survives any outcome. **Pending by explicit instruction.**	Team Lead with Project Manager
QT-02	Technical	There is no AR application architecture in Platform FRD_v4. What is it?	Flagged, not filled. **AR is explicitly not assumed to behave like VR.**	Team Lead
QT-03	Technical	What is the assessment configuration API contract?	Abstract. One constraint is recorded now: a learner-facing payload must carry only the assessment data the approved feedback policy permits, and that policy is itself open (QP-27).	Team Lead
QT-04	Technical	How does an SSO-authenticated learner sign into the XR application, which requires LMS credentials?	Recorded as a collision. **Not designed around.**	Team Lead with Project Manager
QD-01	Data / Integration	Which profile fields are user-editable and which are ERP-owned?	ERP-sourced fields are presented read-only with their provenance stated.	Team Lead with Institute stakeholder

ID	Class	Question	Current safe assumption	Owner
QD-02	Data / Integration	What is the Phase-1 origin of role assignment?	Super Admin assigns platform roles under 8.1.1. The institutional role-assignment origin is left open rather than invented.	Project Manager with Team Lead
QX-01	Design	Does the desktop-first viewport decision extend to all six new workspaces?	Not extended automatically. Administrative work is desktop-typical, but this must be decided rather than inherited.	Design lead with Project Manager
QX-02	Design	How is the outstanding reflow non-conformance handled now that it spans all workspaces?	Stated plainly as a known non-conformance; layouts are built so a responsive pass is possible without restructuring.	Design lead with Project Manager
QX-03	Design	Who owns video captions, transcripts and audio description?	Slots are specified; the content is unowned. **A video without captions must not be a required component for completion.**	Project Manager with academic stakeholder

Four of these are pending by explicit instruction, not by omission

- QT-01 — the XR delivery mechanism (PM-05).
- QD-04 — the SSO provider, protocol and attribute mapping (PM-10).
- QP-25 — which institution supplies a pure-B2C assessment (PM-15).
- The detailed reporting metrics and KPI definitions (PM-11).

The single highest-value product question outstanding is QP-02, ownership of descriptive-answer grading, because it is the only open item that can strand a learner mid-progression through no fault of their own.

20. Dependency Register

Each dependency below is something outside this document that Phase-1 delivery relies on. The final column answers the question that matters most for sequencing: **can the platform architecture and interface design proceed without it?** In almost every case the answer is yes, and the dependency blocks build rather than design.

Table 126 — Dependency register

Dependency	Owner	Current status	Impact if unresolved	Required decision	Can Phase-1 architecture and design proceed?
Existing XR simulation inventory	Programme / content owner	Confirmed available (PM-06); the inventory itself is not yet catalogued	Determines what can be published, entitled and delivered at launch	Catalogue the existing builds, formats, versions and device compatibility	Yes
XR build and package ownership	Vendor organisations / programme owner	Not established	Determines who corrects a returned submission and who maintains a published module	Confirm ownership and support contacts per simulation	Yes
XR integration interfaces	Team Lead / vendor engineering	Undefined	The delivery mechanism, the completion return path and the assessment consumption path all depend on it	Define the interface contract once the delivery mechanism is selected	Yes — architecture can proceed; delivery build cannot
Identity provider	Institution / Team Lead	Not selected — PENDING by PM-10	Institutional authentication and the Phase-1 role-assignment origin	Select the provider and protocol, and define attribute mapping	Yes
SSO configuration model	Team Lead with institution stakeholder	Requirement confirmed; configuration schema undefined	The Institute Admin configuration surface and the tenant authentication path	Define the configuration fields once the protocol is selected	Yes
ERP provider and APIs	Institution / Team Lead	Not selected; the FRD names examples only	Automated provisioning, stream mapping and entitlement assignment for every institutional learner	Select the provider per institution and define the integration contract	Yes
Payment gateway	Commercial owner with Team Lead	Not selected	All B2C purchase, activation and entitlement assignment	Select domestic and international providers and decide redirect or embedded checkout	Yes
Cloud platform and environments	Team Lead	Not selected; the FRD names providers as examples	Hosting, scaling, storage, and the separation of staging from production the governance chain requires	Select the platform and define the environment topology	Yes — no architectural statement here depends on it
Object storage and content distribution	Team Lead	Concept confirmed by the FRD; technology not selected	Encrypted asset storage and acceptable delivery latency	Select the storage and distribution technology	Yes
E-mail and notification service	Team Lead with Project Manager	No Phase-1 requirement exists	Would carry any outbound message; currently nothing requires one	Decide whether any Phase-1 outbound message is required at all	Yes
Content sources and metadata schema	Content Manager with Team Lead	Schema not enumerated in the FRD	What the content readiness gate actually reviews	Define the module metadata schema and the compatibility criteria	Yes
XR device availability	Institution / programme owner	Not established	Whether institutional learners can reach VR and MR formats at all	Confirm device fleets per institution and per format	Yes
Subject-matter reviewers and validation criteria	Academic stakeholder	Reviewers confirmed as a role (PM-07); the criteria they validate against are undefined	The academic validation gate cannot operate without a rubric	Define the engineering curriculum standards used at the validation gate	Yes — the gate is designed; it cannot operate
Analytics and reporting definitions	Project Manager	Deferred by PM-11	What any reporting surface beyond operational status can contain	Define metrics, formulas and filters incrementally once core operations exist	Yes
Technology stack	Team Lead	Not selected	Build, but not architecture — this document is technology-neutral throughout	Select the stack before build begins	Yes
Accessibility conformance target	Project Manager with design lead	Not stated by any requirement	What the platform commits to, and how known non-conformances are recorded	Approve a conformance level and a position on the outstanding reflow item	Yes

The two dependencies that block operation rather than build

- **Subject-matter reviewers and validation criteria.** The academic validation gate is fully designed, but it cannot operate until someone owns and states the engineering curriculum standards it validates against.
- **Existing XR simulation inventory.** The platform can be built to ingest, govern and deliver content, but it has nothing to publish until the available simulations are catalogued with their formats, versions and device compatibility.

21. Summary of Changes from Platform FRD_v4

What this chapter does and does not assert

- It records what changed relative to Platform FRD_v4 and relative to the earlier narrower interpretation of it.
- It does not assert that Platform FRD_v4 was wrong. Most entries below are clarifications or additions; only two are deviations from the FRD as written, and both are labelled as such.

Table 127 — Summary of changes from Platform FRD_v4

Area	Previous position	Updated position	Source	Effect	Status
Scope interpretation	The detailed Phase-1 scope was read as the Student/Learner desktop experience.	Phase 1 covers all Phase-1 requirements and all Phase-1 roles.	PM-01	Design scope moves from one workspace to seven. Requirement breadth is unchanged.	Confirmed
Phase-1 platform scope	Requirement coverage was measured against one role's surfaces.	All 78 Phase-1 requirements are re-audited with an owning role assigned to each.	PM-01	Eighteen Phase-1 requirements had no designed surface anywhere; each now has an owner and a place.	Confirmed
Role model	Five FRD-defined roles, of which only the Student role was in design scope.	Seven confirmed roles, each with a workspace.	PM-07 with PM-01	Vendor and Academic Reviewer are promoted from FRD actors to first-class roles. Recorded as an extension of the FRD.	Confirmed
Vendor workspace	Vendor was an external actor with no login, no permission set and no surface.	Vendor has an authenticated workspace covering submission, correction, resubmission and status.	PM-07 with PM-14	The content governance chain gains its first actor. FRD 1.4.5's prohibition becomes a designed, positively expressed constraint.	Confirmed
Academic Reviewer workspace	The academic team was an external actor with no login and no permission set.	Academic Reviewer has an authenticated workspace covering validation, request-for-changes and decision history.	PM-07 with PM-14	The gate that governs every module on the platform becomes a designed decision surface. Its rubric remains unowned.	Confirmed
Tenancy model	Vendor, Academic Reviewer and Content Manager were unplaced in the FRD's isolation model.	A confirmed two-plane model: central content and governance plane, and institution tenant planes, with Super Admin spanning both.	PM-13	Resolves the largest structural unknown. Fills an FRD silence without contradicting any FRD text.	Confirmed
B2B and B2C	A B2B-first sequencing had entered the work as a working interpretation.	Both are Phase 1 and neither is sequenced ahead of the other.	PM-04	Any market sequencing returns to pending. Both are represented throughout the architecture.	Confirmed
XR access model	XR formats were surfaced as device guidance only.	Entitlement-aware delivery and access from the platform to the supported device, with a full state model.	PM-05	The guidance-only model was incomplete against requirements the FRD already contained at 2.2.2 and 10.1.2.	Product requirement confirmed; mechanism PENDING
XR simulation availability	It was unclear whether the platform team must originate simulations.	The required simulations are available; the platform ingests, governs and delivers them.	PM-06	Simulation authoring is not a Phase-1 platform surface. Technical discovery of the inventory is a dependency.	Confirmed
Assessment model	Specific assessment values were treated as platform constants.	A configuration-driven model. Every value is a configuration field; earlier values are sample configuration.	PM-08	Structural change to the learner assessment screens is none; the data-binding change is pervasive.	Confirmed
Assessment ownership	Ownership for a B2C learner was undefined, with three defensible readings.	Faculty or the authorized institution side owns configuration for both B2B and B2C. The learner never authors or configures.	PM-15	Removes assessment ownership as a B2B/B2C divergence. The learner-side prohibition becomes a stated rule.	Confirmed
Learning progression	The progression was a design proposal.	Learning resources, then simulation, then assessment, then completion — approved.	PM-09	The learner sequencing design is retained. The approval covers the learner model only.	Confirmed
Content governance chain	A forward-only chain: submit, validate, approve, publish.	The same chain with return loops at both review gates, comments and reasons, correction, resubmission and review history.	PM-14	Recorded as an extension of the FRD, which contains no rejection, revision, resubmission or history requirement. A gate that can only approve is not a gate.	Confirmed
SSO	The FRD contradicted itself: configuration in Phase 1, authentication support in Phase 3.	Institutional SSO is required, and the institution-side configuration capability is in scope.	PM-10	Resolves an internal FRD conflict and surfaces a configuration capability that had never been designed.	Confirmed; provider PENDING
Reporting strategy	Reporting risked being inferred as large administrative dashboards.	Core operations first; analytics added incrementally. Phase 1 delivers operational status and the FRD's recording requirements.	PM-11	Agrees with the FRD's own phasing. No analytics dashboard is designed, and reporting is not removed from the long-term set.	Confirmed; metrics PENDING

Area	Previous position	Updated position	Source	Effect	Status
Certification	Certification structure and completion criteria were Phase 1 in the FRD.	Certification is a Phase-2 activity.	PM-12	Recorded as a deviation from the FRD as written. The FRD Phase-1 count remains 78; the working set becomes 76 plus 2 deferred.	Confirmed
Architecture documentation	No consolidated platform architecture existed.	System context, role and workspace architecture, two-plane model, capability map, logical service architecture and security boundaries.	PM-02 with PM-13	Technology-neutral throughout. No stack, provider or deployment topology is selected.	New in this draft
Workflow documentation	Workflows existed for one role's journeys.	Ten cross-role end-to-end workflows covering all seven roles.	PM-02	Each workflow states what must happen; pending mechanisms are drawn as abstract steps rather than invented.	New in this draft
Acceptance criteria	No consolidated acceptance criteria existed.	Forty-four testable criteria across seven capability areas, covering happy path, validation, empty, error, permission, tenant, audit, retry, entitlement, device, rejection, assessment, SSO and payment states.	Derived	No implementation test is claimed to have passed. These describe what will be verified.	New in this draft
Open-question register	Questions were held against a single-role scope.	A platform-scope register: 59 open questions with priority, owner and blocking status, 19 future-phase items, and 19 closed by stakeholder decision.	PM-13, PM-14, PM-15 and the re-audit	No open question blocks the architecture. Closed questions are not shown as open.	New in this draft
Dependency register	Dependencies were implicit.	Sixteen dependencies with owner, status, impact, required decision and whether Phase-1 design can proceed without them.	Derived	Almost every dependency blocks build rather than design. Two block operation rather than build.	New in this draft
Non-functional requirements	The FRD leaves most non-functional characteristics high-level or unstated.	Eighteen requirement categories recorded, each with an explicit target status.	Derived	**No numeric target is invented.** Where no approved target exists, the category is labelled pending.	New in this draft

21.1 Deviations and extensions on record

Platform FRD_v4 is unchanged and must remain unchanged. The following are the points at which this draft departs from, extends, or resolves a contradiction in the FRD as written.

Table 128 — Deviation, extension and conflict register

Ref	Item	Type	Record
DV-1	Certification (3.3.1, 3.3.2) moved from Phase 1 to Phase 2	DEVIATION from the FRD	PM-12. The original phase is preserved in every table. The FRD Phase-1 count remains 78 as a document fact.
EX-1	Vendor and Academic Reviewer promoted to roles beyond the FRD's five	EXTENSION of the FRD	PM-07. FRD 8.3, the FRD's own mechanism for extending the role model, is Phase 3 and cannot justify this within Phase 1.
EX-2	Rejection, resubmission and review history added to the governance chain	EXTENSION of the FRD	PM-14. FRD 1.4.1–1.4.4 describe a forward-only chain with no such requirement. A gate that can only approve is not a gate.
EX-3	Two-plane placement of Vendor, Academic Reviewer and Content Manager	FILLS AN FRD SILENCE	PM-13. The FRD isolates tenants and grants cross-tenant access to Super Admin only, but never places these three roles. No FRD text is contradicted.
CF-1	SSO phase contradiction — 4.2.2 is Phase 1, 7.4.1 is Phase 3	FRD INTERNAL CONFLICT — resolved	PM-10 confirms the Phase-1 reading. 7.4.1 carries a Phase-1 working position while its original phase is preserved.
CF-2	Institute Admin course structures — 8.1.3 is Phase 1, 3.4.1 and 3.4.2 are Phase 2	FRD INTERNAL TENSION — recorded	Working interpretation: in Phase 1 the Institute Admin views, maps and assigns structure arriving from ERP, and does not author courses. Carried as an open question.
CF-3	Faculty performance monitoring — 8.1.4 is Phase 1, 11.3.1 is Phase 2	FRD INTERNAL TENSION — recorded	Working interpretation per PM-11: Phase-1 Faculty receives operational learner status, not aggregated analytics.
GP-1	No AR application architecture exists in FRD Section 2	FRD GAP	1.2.4 requires AR support; Section 2 defines application architecture for VR and MR only. Flagged, not filled.
GP-2	AI question generation (5.1.2) is a Phase-1 Faculty requirement	SCOPE CATCH	It had been treated as excluded on the basis of the Phase-3 learner-facing AI assistant at 5.5 — a different requirement, phase and user. Reinstated as PRODUCT-PENDING.
GP-3	Device binding and replacement (2.4.1, 2.4.2) had no surface anywhere	FRD GAP — now addressed	Both are Phase-1 requirements. The two-role approval workflow is specified here; the rejection path the FRD omits is not invented.
GP-4	No audit trail outside content governance	FRD GAP	PM-14 covers the content chain only. Licensing, entitlement, configuration and device-replacement audit remain undefined and are not invented.

Appendix A. Traceability Matrix

Every requirement identifier in Platform FRD_v4 appears below. The matrix maps each to its updated position, classification, phase, owning role, capability domain, governing stakeholder decision, source document from the full-platform re-audit, workflow, open question and acceptance criterion.

The matrix was generated mechanically from the requirement register rather than assembled by hand, and the verification results are recorded in Appendix C.

Table 129 — Traceability matrix — all 131 Platform FRD_v4 requirements

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
1.1.1	Multi-Tenant SaaS Platform	UNCHANGED-FRD	1 → 1	All seven	TEN, CFG	PM-03	01, 04	All	—	A single authenticated portal resolves the signed-in identity to exactly one workspace per held role.
1.1.2	B2B Institutional Learning Environment	UNCHANGED-FRD	1 → 1	Learner, Faculty, Institute Admin, Super Admin	TEN, ENT	PM-04	01, 06	FLOW 2	—	A faculty member and a student of the same institution both access the platform under the institution's licence, within one tenant.
1.1.3	B2C Individual Learning Environment	UNCHANGED-FRD	1 → 1	Learner, Super Admin	COM, ENT	PM-04	01, 06	FLOW 3	—	An individual learner registers, purchases a module and reaches it without any institutional affiliation.
1.1.4	Centralized Content Ownership	UNCHANGED-FRD	1 → 1	Super Admin, Content Manager, Institute Admin, Vendor	GOV, SEC	PM-06	01, 03	FLOW 1	—	No institutional role is offered any control that edits, exports or redistributes module content.
1.2.1	Web Simulation Support	UNCHANGED-FRD	1 → 1	Learner	XRD, LRN	PM-05	01, 07	FLOW 5	QT-06	An entitled web simulation launches inside the portal with determinate asset loading and no XR hardware.
1.2.2	VR Simulation Support	TECHNICAL-PENDING	1 → 1	Learner, Institute Admin, Super Admin	XRD	PM-05	01, 07	FLOW 5	QT-01	An entitled VR simulation reaches the learner's supported headset and its state is visible in the portal, without the portal claiming to launch it.
1.2.3	MR Simulation Support	TECHNICAL-PENDING	1 → 1	Learner, Institute Admin, Super Admin	XRD	PM-05	01, 07	FLOW 5	QP-14	An entitled MR simulation is presented with an honest device requirement and an accurate availability state.
1.2.4	AR Simulation Support	TECHNICAL-PENDING	1 → 1	Learner	XRD	PM-05	01, 07	FLOW 5	QT-02	An entitled AR simulation is presented with an honest device requirement; no delivery behaviour is asserted that the FRD does not define.
1.3.1	Tenant-Wise Database Isolation	UNCHANGED-FRD	1 → 1	Institute Admin, Super Admin	TEN, SEC	PM-01, PM-13	01, 02, 04	FLOW 2	—	No tenant-plane role can see, name or reach data belonging to another institution, and no tenant switcher exists for them.
1.3.2	B2C Database Isolation	UNCHANGED-FRD	1 → 1	Super Admin	TEN, SEC	PM-04, PM-13	01, 03	FLOW 3	QD-05	A B2C account's data is held apart from every institutional tenant environment.
1.3.3	B2C Database Expansion	UNCHANGED-FRD	1 → 1	No role-facing surface	TEN	—	01	—	—	The data architecture permits additional regional B2C instances without a change to the role or permission model.
1.3.4	Tenant-Scoped Analytics	UNCHANGED-FRD	1 → 1	Institute Admin, Faculty, Super Admin	RPT, TEN	PM-11	01, 02, 03	FLOW 2	—	Every Faculty and Institute Admin reporting surface returns only that institution's data, with no cross-tenant aggregate anywhere.
1.3.5	Cross-Tenant Governance Access	UNCHANGED-FRD	1 → 1	Super Admin	TEN, SEC, RPT	PM-01, PM-07, PM-13	01, 02, 04	FLOW 2	—	Only Super Admin can reach more than one tenant, and that access is presented as governance and operational oversight rather than unrestricted data access.
1.4.1	Vendor Module Submission	PM-CLARIFIED	1 → 1	Vendor, Content Manager	GOV	PM-07, PM-13, PM-14	01, 02, 04, 05	FLOW 1	QT-05	An authorised vendor creates a submission, uploads a build and assets, supplies metadata and submits it; the submission appears in the Academic Reviewer's queue.
1.4.2	Academic Validation by BITS Pilani	PM-CLARIFIED	1 → 1	Academic Reviewer	GOV	PM-07, PM-13, PM-14	01, 02, 04, 05	FLOW 1	QD-08	An Academic Reviewer records either academic validation or a request for changes with a stated reason; no module advances past this gate without a recorded decision.
1.4.3	Content Manager Approval	PM-CLARIFIED	1 → 1	Content Manager	GOV, CAT	PM-01, PM-13, PM-14	01, 02, 04, 05	FLOW 1	QD-07	A Content Manager reviews metadata, documentation and platform compatibility, reads the academic validation record read-only, and records either approval or a request for changes.

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
1.4.4	Production Publishing by Super Admin	UNCHANGED-FRD	1 → 1	Super Admin	GOV	PM-01, PM-14	01, 02, 04, 05	FLOW 1	—	No role other than Super Admin can publish, and a module reaches the publish queue only after both prior gates have recorded approval.
1.4.5	Restricted Vendor Publishing	UNCHANGED-FRD	1 → 1	Vendor, Super Admin	GOV, SEC	PM-07	01, 02, 04	FLOW 1	—	No publish action exists in the Vendor workspace in any state, and no Vendor-initiated path reaches production.
1.5.1	Module Lifecycle States	FUTURE-PHASE	2 → 2	Content Manager, Super Admin	CAT	—	01, 03	—	QF-09	Not required in Phase 1.
1.5.2	Academic Version Locking	FUTURE-PHASE	2 → 2	Institute Admin	CAT, TEN	—	01	—	QF-09	Not required in Phase 1.
1.6.1	Version Tracking	FUTURE-PHASE	2 → 2	Content Manager, Super Admin	CAT	—	01	—	QF-09	Not required in Phase 1.
1.6.2	Version Rollback	FUTURE-PHASE	2 → 2	Super Admin	CAT	—	01	—	QF-09	Not required in Phase 1.
1.7.1	Regional CDN Distribution	FUTURE-PHASE	3 → 3	No role-facing surface	XRD	—	01	—	—	Not required in Phase 1.
1.7.2	Multi-Region Database Replication	FUTURE-PHASE	3 → 3	No role-facing surface	TEN	—	01	—	—	Not required in Phase 1.
2.1.1	Master VR Application	TECHNICAL-PENDING	1 → 1	Learner, Institute Admin, Super Admin	XRD	PM-05	01, 07	FLOW 5	QT-01	The learner-facing delivery experience is described in mechanism-neutral terms that remain correct under either concept.
2.1.2	MR Device Application	TECHNICAL-PENDING	1 → 1	Learner, Institute Admin	XRD	PM-05	01, 07	FLOW 5	QP-14	MR access is presented as an institutionally provisioned route, with no self-service sideloading offered to a learner.
2.2.1	XR Asset Bundle Packaging	TECHNICAL-PENDING	1 → 1	No role-facing surface	XRD, SEC	PM-05	01, 07	FLOW 1, FLOW 5	QT-05	Delivery and upload surfaces describe the artefact generically until a format is defined; no format is asserted.
2.2.2	License-Based Module Access	UNCHANGED-FRD	1 → 1	Learner, Super Admin	XRD, ENT	PM-05	01, 07	FLOW 5	—	No module is delivered to a device before the platform confirms a valid licence, and a failed validation produces an explicit, code-free explanation.
2.3.1	Automatic Module Updates	TECHNICAL-PENDING	1 → 1	Learner	XRD	PM-05	01, 07	FLOW 5	QT-01	An updated module produces a visible 'update available' state; an unrequested update never blocks a learner from continuing.
2.3.2	Controlled Update Releases	UNCHANGED-FRD	1 → 1	Super Admin	GOV, CFG	PM-01	01, 02, 04	FLOW 5	—	A new module version reaches production only when Super Admin releases it; holding a release is an available and visible outcome.
2.3.3	Module Update Without Application Reinstallation	TECHNICAL-PENDING	1 → 1	No role-facing surface	XRD	PM-05	01, 07	FLOW 5	QT-01	The delivery model chosen must permit module-level update without reinstalling the device application.
2.4.1	B2C Device Binding	PRODUCT-PENDING	1 → 1	Learner, Super Admin	ENT, XRD	PM-05	01, 02, 07	FLOW 5, FLOW 8	QT-02	A B2C learner can see which device a licence is bound to, and the binding is enforced at access and at execution.
2.4.2	Device Replacement Authorization	PRODUCT-PENDING	1 → 1	Learner, Super Admin	ENT, XRD	PM-01, PM-07	01, 02, 07	FLOW 8	—	A learner submits a replacement request; the licence transfers only after Super Admin approval, and the request's state is visible to the learner throughout.
2.4.3	Institutional Device Access	UNCHANGED-FRD	1 → 1	Learner, Institute Admin	XRD, TEN	PM-05	01, 06, 07	FLOW 5	—	An institutional learner reaches a simulation on a shared device without any personal device binding being asserted.
2.4.4	Session-Based Device Login	FRD-INTERNAL-CONFLICT	1 → 1	Learner, Institute Admin	IAM, XRD	PM-10	01, 07	FLOW 5, FLOW 6	QT-04	The device sign-in route for an institutional learner is stated only once a mechanism compatible with institutional SSO is defined.
2.5.1	Device Registration	FUTURE-PHASE	2 → 2	Institute Admin	XRD	—	01, 02	—	QF-07	Not required in Phase 1.

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
2.5.2	Device Metadata Tracking	FUTURE-PHASE	2 → 2	Institute Admin	XRD	—	01	—	QF-07	Not required in Phase 1.
2.6.1	Remote Module Control	FUTURE-PHASE	2 → 2	Institute Admin, Super Admin	XRD	—	01, 02	—	QF-08	Not required in Phase 1.
2.6.2	Remote Application Updates	FUTURE-PHASE	2 → 2	Super Admin	XRD	—	01	—	QF-08	Not required in Phase 1.
2.7.1	Simulation Interaction Logs	FUTURE-PHASE	3 → 3	Faculty, Super Admin	RPT	—	01	—	QF-18	Not required in Phase 1.
2.7.2	Simulation Error Tracking	FUTURE-PHASE	3 → 3	Faculty, Super Admin	RPT	—	01	—	QF-18	Not required in Phase 1.
3.1.1	Simulation Module Definition	UNCHANGED-FRD	1 → 1	Learner, Faculty, Content Manager	CAT, LRN	PM-09	01, 03	FLOW 2, FLOW 3	—	Every module presents exactly one simulation experience; no module offers a choice between simulations.
3.1.2	Module Learning Components	UNCHANGED-FRD	1 → 1	Learner, Faculty, Content Manager	CAT, LRN, ASM	PM-09	01, 03	FLOW 2, FLOW 3	—	A module with no resources, no assessment, or neither, renders correctly and its completion model adjusts accordingly.
3.2.1	Course Composition	UNCHANGED-FRD	1 → 1	Learner, Faculty, Institute Admin	CAT, LRN	PM-09	01, 03	FLOW 2	—	A course lists its modules in their defined order. No purchase capability is asserted for a course, and none is prohibited.
3.2.2	Course Module Sequencing	UNCHANGED-FRD	1 → 1	Learner, Faculty	LRN, PRG	PM-09	01, 02, 03	FLOW 2	QP-10	A learner cannot open a later module in a course before the prerequisite module is complete, and the reason is stated on the locked item.
3.3.1	Certification Program Structure	PM-DEFERRED	1 → 2	No Phase-1 role-facing surface	CAT	PM-12	01, 03	—	QP-24, QF-01	No certification structure is presented in any Phase-1 workspace.
3.3.2	Certification Completion Criteria	PM-DEFERRED	1 → 2	No Phase-1 role-facing surface	CAT, PRG	PM-12	01, 03	—	QF-01	No certificate artefact, issuance flow or eligibility surface is built in Phase 1.
3.4.1	Institutional Course Creation	FUTURE-PHASE	2 → 2	Institute Admin	CAT, TEN	—	01, 02	—	QF-03	Not required in Phase 1; in Phase 1 institutional course structure arrives through ERP.
3.4.2	Course Module Mapping	FUTURE-PHASE	2 → 2	Institute Admin	CAT, TEN	—	01, 02	—	QF-03	Not required in Phase 1.
3.4.3	Course Stream Assignment	FUTURE-PHASE	2 → 2	Institute Admin	CAT, TEN	—	01	—	—	Not required in Phase 1.
3.5.1	Student Certification Progress	FUTURE-PHASE	3 → 3	Learner	PRG	PM-12	01	—	QF-02	Not required in Phase 1.
3.5.2	Certification Eligibility Reports	FUTURE-PHASE	3 → 3	Faculty, Institute Admin	RPT	PM-12	01	—	QF-02	Not required in Phase 1.
4.1.1	Email Registration	UNCHANGED-FRD	1 → 1	Learner	IAM	PM-04	01, 04	FLOW 3	—	Self-registration is offered only in the B2C context and never creates any role other than Learner.
4.1.2	Social Login	PRODUCT-PENDING	1 → 1	Learner	IAM	PM-04	01, 03	FLOW 3	QP-19	Social sign-in is offered only in the B2C context; no provider is presented as confirmed until one is approved.
4.2.1	Institutional Account Creation	UNCHANGED-FRD	1 → 1	Institute Admin, Super Admin	IAM, TEN	PM-01	01, 04	FLOW 2, FLOW 6	—	An institution can be registered and a tenant created, after which institutional configuration becomes available.
4.2.2	SSO Integration	FRD-INTERNAL-CONFLICT	1 → 1	Institute Admin, Super Admin	IAM, TEN, CFG	PM-10	01, 02, 04, 05	FLOW 6	QD-04	An Institute Admin can configure, test and enable institutional SSO for their own tenant, and a failed test is reported explicitly.
4.3.1	Profile Information Collection	PRODUCT-PENDING	1 → 1	Learner, Faculty	IAM	PM-01	01, 02	FLOW 7	QD-01	ERP-sourced fields are presented read-only with their provenance stated; user-editable fields are not silently overwritten without notice.
4.4.1	Learning Progress Tracking	FUTURE-PHASE	2 → 2	Learner, Faculty, Institute Admin	PRG	PM-11	01	—	—	Not required in Phase 1 beyond the recording required by 11.1.1 and 11.1.2.

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
4.4.2	Assessment Performance Tracking	FUTURE-PHASE	2 → 2	Learner, Faculty	PRG, ASM	PM-11	01	—	—	Not required in Phase 1 beyond the recording required by 11.2.1.
4.5.1	Personalized Module Recommendations	FUTURE-PHASE	2 → 2	Learner	LRN	—	01	—	QF-17	Not required in Phase 1.
4.5.2	Certification Path Recommendations	FUTURE-PHASE	2 → 2	Learner	LRN	PM-12	01	—	QF-17	Not required in Phase 1.
5.1.1	Question Bank Upload	UNCHANGED-FRD	1 → 1	Faculty	ASM	PM-08, PM-15	01, 02, 08	FLOW 4	—	A Faculty member uploads a structured question bank; invalid rows are reported individually rather than failing the whole upload silently.
5.1.2	AI Question Generation	PRODUCT-PENDING	1 → 1	Faculty	ASM	PM-08	01, 03, 08	FLOW 4	—	Generated questions enter the question bank in a reviewable state; no generated question reaches a learner without the Faculty owner accepting it.
5.2.1	Multiple Choice Questions	UNCHANGED-FRD	1 → 1	Learner, Faculty	ASM	PM-08	01, 08	FLOW 4	QP-12	An MCQ renders, is answerable and is scored automatically according to the assessment configuration.
5.2.2	True/False Questions	UNCHANGED-FRD	1 → 1	Learner, Faculty	ASM	PM-08	01, 08	FLOW 4	—	A true/false question renders, is answerable and is scored automatically.
5.2.3	Descriptive Questions	PRODUCT-PENDING	1 → 1	Learner, Faculty	ASM	PM-08	01, 02, 08	FLOW 4	QP-02	A submitted descriptive answer enters an explicit awaiting-evaluation state that the learner can understand; no grading owner is asserted.
5.3.1	Randomized Question Order	UNCHANGED-FRD	1 → 1	Faculty	ASM	PM-08	01, 08	FLOW 4	—	Randomization is an explicit setting whose current value is disclosed to the learner before an attempt begins.
5.3.2	Negative Marking	UNCHANGED-FRD	1 → 1	Faculty, Learner	ASM	PM-08, PM-15	01, 02, 08	FLOW 4	—	Negative marking is enabled or disabled per assessment, and the pre-attempt disclosure states the configured behaviour rather than fixed copy.
5.4.1	Simulation Performance Scoring	FUTURE-PHASE	2 → 2	Learner, Faculty	ASM, PRG	—	01, 08	—	QF-13	Not required in Phase 1.
5.4.2	Simulation Attempt Tracking	FUTURE-PHASE	2 → 2	Learner, Faculty	PRG	—	01	—	QF-13	Not required in Phase 1.
5.5.1	Concept Explanation Support	FUTURE-PHASE	3 → 3	Learner	LRN	—	01, 08	—	QF-16	Not required in Phase 1.
5.5.2	Practice Question Generation	FUTURE-PHASE	3 → 3	Learner	ASM	—	01, 08	—	QF-16	Not required in Phase 1.
6.1.1	Individual Module Purchase	UNCHANGED-FRD	1 → 1	Learner, Super Admin	COM, ENT	PM-04	01, 06	FLOW 3	—	An individual learner can purchase a single module and reach it immediately after activation.
6.1.2	Bundle Purchases	UNCHANGED-FRD	1 → 1	Learner, Super Admin	COM, ENT	PM-04	01, 02, 06	FLOW 3	QP-09	A bundle purchase entitles the learner to every module the bundle contains, each reachable individually.
6.2.1	Stream-Based Licensing	PRODUCT-PENDING	1 → 1	Institute Admin, Super Admin	ENT, TEN	PM-01, PM-04	01, 02, 05	FLOW 2, FLOW 9	QP-04	An institutional licence exists against a tenant and maps to academic streams; no in-platform purchase flow is asserted until the transaction mechanism is decided.
6.2.2	Capped Institutional License	UNCHANGED-FRD	1 → 1	Institute Admin, Super Admin	ENT, TEN	PM-01	01, 02	FLOW 2, FLOW 9	—	Seat utilisation against the cap is visible to the Institute Admin before exhaustion, and an exhausted cap produces an explicit learner-facing explanation without a purchase prompt.
6.3.1	Bundle Creation	FUTURE-PHASE	2 → 2	Super Admin	COM	—	01, 02	—	QP-09	Not required in Phase 1.
6.3.2	Pricing Configuration	FUTURE-PHASE	2 → 2	Super Admin	COM	—	01, 02	—	QP-09	Not required in Phase 1.
6.4.1	Coupon Generation	FUTURE-PHASE	2 → 2	Super Admin	COM	—	01, 02	—	QF-10	Not required in Phase 1.
6.4.2	Coupon Expiry Configuration	FUTURE-PHASE	2 → 2	Super Admin	COM	—	01	—	QF-10	Not required in Phase 1.

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
6.5.1	License Activation	FUTURE-PHASE	3 → 3	Learner	ENT	—	01	FLOW 3	—	Phase-1 behaviour is governed by 9.2.1.
6.5.2	Grace Period	FUTURE-PHASE	3 → 3	Institute Admin, Super Admin	ENT	—	01	—	—	Not required in Phase 1.
7.1.1	Supported ERP Systems	TECHNICAL-PENDING	1 → 1	Institute Admin, Super Admin	ERP	PM-01	01, 02, 05	FLOW 7	QD-03	ERP integration is configured per tenant against a provider-neutral connection model; no provider is presented as selected.
7.1.2	ERP API Integration	TECHNICAL-PENDING	1 → 1	Institute Admin, Super Admin	ERP	PM-01	01, 05	FLOW 7	QD-03	Inbound ERP data is accepted, validated and reflected in tenant state, with the outcome of each run visible to the Institute Admin.
7.2.1	Scheduled ERP Synchronization	UNCHANGED-FRD	1 → 1	Institute Admin	ERP, CFG	PM-01	01, 02, 05	FLOW 7	QD-06	An Institute Admin sets the synchronisation interval and can see the last run, its outcome, its counts and its errors.
7.2.2	ERP-Based Account Creation	UNCHANGED-FRD	1 → 1	Learner, Institute Admin	ERP, IAM	PM-01	01, 05	FLOW 6, FLOW 7	—	A student record received from ERP produces an account the learner can sign into, with entitlements already in place.
7.2.3	Stream-Based Access Assignment	UNCHANGED-FRD	1 → 1	Learner, Institute Admin, Faculty	ERP, ENT	PM-01	01, 02, 05	FLOW 2, FLOW 7	—	A student in a mapped stream receives the mapped package automatically; a student in an unmapped stream sees an empty entitlement state that explains what to do.
7.3.1	First Login Provisioning	FUTURE-PHASE	2 → 2	Learner, Institute Admin	ERP	—	01	—	QF-12	Not required in Phase 1.
7.3.2	Dynamic Entitlement Assignment	FUTURE-PHASE	2 → 2	Learner, Institute Admin	ERP, ENT	—	01	—	QF-12	Not required in Phase 1.
7.4.1	SSO Authentication Support	FRD-INTERNAL-CONFLICT	3 → 1	Learner, Faculty, Institute Admin	IAM	PM-10	01, 05	FLOW 6	QD-04	Institutional SSO is an available authentication path for a configured tenant, and a failed assertion produces an explicit failure state.
7.4.2	Role-Based Access Mapping	FUTURE-PHASE	3 → 3	Institute Admin, Super Admin	IAM	PM-10	01, 02	FLOW 6	QD-02	Not required in Phase 1; Phase-1 role assignment origin is recorded as pending.
8.1.1	Super Admin Role	UNCHANGED-FRD	1 → 1	Super Admin	CFG, GOV, ENT, RPT	PM-01, PM-11	01, 02, 04	FLOW 1, FLOW 8	QP-04	A Super Admin can configure the platform, publish modules, manage licences and see cross-tenant operational status, with no detailed analytics dashboard asserted.
8.1.2	Content Manager Role	UNCHANGED-FRD	1 → 1	Content Manager	GOV, CAT	PM-01, PM-13	01, 02, 04	FLOW 1	—	A Content Manager can review and edit module metadata and hand cleared content to Super Admin for publication, without any publish capability.
8.1.3	Institute Admin Role	FRD-INTERNAL-CONFLICT	1 → 1	Institute Admin	TEN, IAM, CAT, RPT	PM-01, PM-11	01, 02, 04	FLOW 2, FLOW 7	QP-07	An Institute Admin manages tenant configuration, student access and operational institutional status, with no course-authoring capability offered.
8.1.4	Faculty Role	FRD-INTERNAL-CONFLICT	1 → 1	Faculty	LRN, ASM, RPT	PM-01, PM-08, PM-11, PM-15	01, 02, 04, 08	FLOW 2, FLOW 4	QP-08	A Faculty member assigns content, configures assessments, and sees who has completed, who is blocked and what attempts exist — without any aggregated performance report.
8.1.5	Student Role	UNCHANGED-FRD	1 → 1	Learner	LRN, XRD, PRG	PM-01, PM-09	01, 02, 04, 09	FLOW 2, FLOW 3	—	A learner reaches only entitled modules, completes them in the approved sequence, and sees their own progress and no one else's.
8.2.1	Institutional Usage Monitoring	FUTURE-PHASE	2 → 2	Institute Admin	RPT	PM-11	01, 02	—	QF-06	Not required in Phase 1.
8.2.2	License Utilization Tracking	FUTURE-PHASE	2 → 2	Institute Admin, Super Admin	RPT, ENT	PM-11	01, 02	—	QF-06	Not required in Phase 1 beyond operational seat utilisation under 6.2.2.

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
8.3.1	Custom Role Creation	FUTURE-PHASE	3 → 3	Super Admin	CFG, IAM	PM-07	01, 02	—	QF-14	Not required in Phase 1; no role may be added in Phase 1 without an explicit stakeholder confirmation.
8.3.2	Permission-Level Access Control	FUTURE-PHASE	3 → 3	Super Admin	CFG, IAM	—	01, 02	—	QF-14	Not required in Phase 1.
9.1.1	Indian Payment Gateway Support	TECHNICAL-PENDING	1 → 1	Learner	COM	PM-04	01, 03	FLOW 3	QT-11	A domestic purchase completes through a provider-neutral payment step whose success, decline and indeterminate outcomes are all designed.
9.1.2	International Payment Gateway Support	PRODUCT-PENDING	1 → 1	Learner	COM	PM-04	01, 03	FLOW 3	QD-10, QP-18	An international purchase completes without the interface asserting any currency, tax or refund rule that has not been approved.
9.2.1	Automated License Activation	UNCHANGED-FRD	1 → 1	Learner, Super Admin	ENT, COM	PM-04	01, 03	FLOW 3	—	A successful payment activates the licence and links it to the account without any further learner action.
9.2.2	Entitlement Assignment	UNCHANGED-FRD	1 → 1	Learner	ENT	PM-04	01, 03	FLOW 3	QP-17	A purchased module appears in My Learning immediately after activation and is reachable from there.
9.3.1	GST-Compliant Invoice Generation	FUTURE-PHASE	2 → 2	Institute Admin, Super Admin	COM	—	01, 02	—	QP-04	Not required in Phase 1.
9.3.2	Institutional Subscription Billing	FUTURE-PHASE	2 → 2	Institute Admin, Super Admin	COM	—	01, 02	—	QP-04	Not required in Phase 1.
9.4.1	Payment Status Tracking	FUTURE-PHASE	3 → 3	Super Admin	COM	—	01	—	—	Not required in Phase 1.
9.4.2	Grace Period Notifications	FUTURE-PHASE	3 → 3	Institute Admin	NOT	—	01, 03	—	—	Not required in Phase 1.
10.1.1	XR Asset Encryption	UNCHANGED-FRD	1 → 1	No role-facing surface	SEC, XRD	PM-05	01, 03	FLOW 5	—	No workspace offers a route to obtain an unprotected asset bundle.
10.1.2	Secure Module Authorization	UNCHANGED-FRD	1 → 1	Learner	SEC, ENT, XRD	PM-05	01, 07	FLOW 5	—	A module refuses to execute without a valid licence, and the resulting state explains what happened and what to do next without internal codes.
10.2.1	Secure Login Authentication	PRODUCT-PENDING	1 → 1	All seven	SEC, IAM	PM-10	01, 02	FLOW 6	QT-14	No workspace is reachable without authentication, and the authentication policy is stated once and applied to every role.
10.2.2	Session Management	UNCHANGED-FRD	1 → 1	All seven	SEC, IAM	PM-01	01, 02, 03	—	QT-13	An expiring session warns before it expires and does not silently discard unsaved administrative or learner work.
10.3.1	VAPT Compliance	UNCHANGED-FRD	1 → 1	No role-facing surface	SEC	—	01	—	—	A VAPT schedule exists and is owned before production release.
10.3.2	Security Patch Management	UNCHANGED-FRD	1 → 1	No role-facing surface	SEC	—	01	—	—	A patch process exists with a stated owner and response expectation before production release.
10.4.1	Data Access Controls	FUTURE-PHASE	2 → 2	Super Admin	SEC	—	01	—	QF-19	Not required in Phase 1 beyond the enforcement required by 1.3.x and 8.1.x.
10.4.2	Data Retention Policies	FUTURE-PHASE	2 → 2	Super Admin	SEC	—	01	—	QF-19	Not required in Phase 1.
11.1.1	Module Completion Tracking	UNCHANGED-FRD	1 → 1	Learner, Faculty, Institute Admin	PRG	PM-09, PM-11	01, 03	FLOW 2, FLOW 5	QT-07	Completing every required component of a module records module completion against the learner's account and unlocks the next module.
11.1.2	Time Spent Tracking	PRODUCT-PENDING	1 → 1	Learner, Faculty, Institute Admin	PRG	PM-11	01, 03	FLOW 2	QP-11	Time spent is recorded per learner and is visible within the tenant boundary, without any aggregated analytical view.

FRD ID	Updated requirement	Status / classification	Phase FRD → v5	Role(s)	Capability	PM decision	Re-audit source	Workflow	Open question	Acceptance criterion
11.2.1	Assessment Score Tracking	UNCHANGED-FRD	1 → 1	Learner, Faculty	PRG, ASM	PM-08, PM-11	01, 08	FLOW 4	—	A completed attempt stores a score against the learner and is visible to the learner and, within the tenant, to Faculty.
11.2.2	Attempt Tracking	UNCHANGED-FRD	1 → 1	Learner, Faculty	PRG, ASM	PM-08	01, 08	FLOW 4	—	Every attempt is recorded and retrievable as attempt history, independently of any configured limit.
11.3.1	Class-Level Performance Reports	FUTURE-PHASE	2 → 2	Faculty	RPT	PM-11	01, 02	—	QF-04	Not required in Phase 1.
11.3.2	Institutional Adoption Reports	FUTURE-PHASE	2 → 2	Institute Admin	RPT	PM-11	01, 02	—	QF-05	Not required in Phase 1.
11.4.1	Simulation Heatmaps	FUTURE-PHASE	3 → 3	Faculty, Super Admin	RPT	PM-11	01, 02	—	QF-18	Not required in Phase 1.
11.4.2	Decision Path Analysis	FUTURE-PHASE	3 → 3	Faculty, Super Admin	RPT	PM-11	01, 02	—	QF-18	Not required in Phase 1.
12.1.1	Cloud Infrastructure Hosting	TECHNICAL-PENDING	1 → 1	No role-facing surface	CFG	—	01, 03	—	QT-12	A hosting decision is recorded before build, and no architectural statement in this document depends on which provider is chosen.
12.1.2	Encrypted Asset Storage	UNCHANGED-FRD	1 → 1	No role-facing surface	SEC, XRD	—	01	—	—	Asset bundles are stored encrypted at rest, with no route to an unprotected copy from any workspace.
12.2.1	CDN-Based Asset Delivery	UNCHANGED-FRD	1 → 1	Learner	XRD	PM-05	01, 07	FLOW 5	—	Asset loading and device preparation present honest progress; an indeterminate wait is presented as such rather than as a stalled interface.
12.3.1	Dynamic Server Scaling	FUTURE-PHASE	2 → 2	No role-facing surface	CFG	—	01	—	—	Not required in Phase 1.
12.3.2	Load Balancing	FUTURE-PHASE	2 → 2	No role-facing surface	CFG	—	01	—	—	Not required in Phase 1.
12.4.1	Regional CDN Nodes	FUTURE-PHASE	3 → 3	No role-facing surface	XRD	—	01	—	—	Not required in Phase 1.
12.4.2	Multi-Region Database Replication	FUTURE-PHASE	3 → 3	No role-facing surface	TEN	—	01	—	—	Not required in Phase 1.

Appendix B. Complete Open-Question Register

The complete register. High-priority items also appear in section 19.3. Closed questions appear in section 19.1 and are not repeated here.

B.1 Product questions

Table 130 — Product open questions

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QP-02	Who grades descriptive answers, on what turnaround, and against what rubric?	HIGH	5.2.3; compounded by 3.2.2 and 11.1.1	Faculty, Learner; FLOW 4	The grading queue is named and positioned but explicitly unowned and undesigned.	Project Manager with academic stakeholder	Blocks development of the grading capability; does not block design of the rest
QP-04	How is an institutional licence transacted in Phase 1, given that institutional billing is Phase 2?	HIGH	6.2.1 against 9.3.1 and 9.3.2	Institute Admin, Super Admin; FLOW 2, FLOW 9	Licences are shown as existing. No in-platform purchase flow is invented.	Project Manager with commercial owner	Blocks development of the institutional purchase path; does not block design
QP-07	Who creates and manages Faculty accounts?	MEDIUM	8.1.3 names only student access; 7.2.2 names only student records	Institute Admin, Super Admin; FLOW 2	The faculty-management surface is drawn as pending rather than assigned to a role.	Project Manager	Blocks development of faculty provisioning; does not block design
QP-08	Does Institute Admin hold any assessment authority at all?	MEDIUM	5.1.1, 5.3.2, 8.1.4 against 8.1.3	Institute Admin, Faculty; FLOW 4	None is granted. PM-15 explicitly declines to grant it, and Faculty is the requirement-named role.	Project Manager	Neither
QP-09	Who creates bundles and sets prices in Phase 1, given that both are Phase 2?	MEDIUM	6.1.2 against 6.3.1 and 6.3.2	Super Admin, Learner; FLOW 3	No Phase-1 bundle-authoring or pricing surface is invented.	Project Manager with commercial owner	Blocks development of B2C bundle sales; does not block design
QP-10	Is the learning sequence per-course configurable, or a platform constant?	MEDIUM	3.2.2; PM-09 approves the model, not its mutability	Faculty, Learner; FLOW 2, FLOW 3	Treated as a platform model. No configuration surface is drawn.	Project Manager	Neither
QP-11	Do multi-role identities exist in Phase 1?	MEDIUM	8.1.x; 8.3 custom roles is Phase 3	All seven; FLOW 6	A workspace chooser renders only if such identities exist, and costs nothing if they do not.	Project Manager	Neither
QP-12	Are multiple-choice questions single-select or multi-select?	MEDIUM	5.2.1	Faculty, Learner; FLOW 4	Single-select. Multi-select would require a partial-credit rule that no scoring model defines.	Project Manager with academic stakeholder	Blocks development of multi-select scoring; does not block design
QP-13	Is there an extended-time accommodation for assessments, and who grants it?	MEDIUM	5.2.x, 5.3.x	Faculty, Learner; FLOW 4	Improved by PM-08 — it becomes an ordinary configuration field rather than a special case.	Project Manager with academic stakeholder	Neither
QP-14	Who authorises and performs MR sideloading?	MEDIUM	2.1.2	Institute Admin, Super Admin; FLOW 5	Plausibly Institute Admin, but not assigned. No sideloading capability is offered to a learner.	Team Lead with Project Manager	Blocks development of the MR provisioning path; does not block design
QP-15	Is academic validation shown to learners or buyers as a trust signal?	MEDIUM	1.4.2	Learner; FLOW 1, FLOW 3	Not shown. Adding it unasked would invent a product feature.	Project Manager	Neither
QP-16	Are assessment configurations reusable across courses, cohorts or institutions?	MEDIUM	PM-08	Faculty; FLOW 4	Not addressed. No reuse mechanism is drawn.	Project Manager	Neither
QP-17	What receipt or invoice does a B2C buyer receive?	MEDIUM	9.2.1; 9.3.1 is Phase 2 and institution-facing	Learner; FLOW 3	A purchase record is provided; the document format is pending.	Commercial owner with Project Manager	Blocks development of receipt generation; does not block design
QP-18	What are the refund, cancellation and revocation rules?	MEDIUM	9.1.x, 9.2.x	Learner, Super Admin; FLOW 3	None stated or implied. **No rule is invented.**	Commercial and legal owner	Blocks development of refund handling; does not block design
QP-19	Which social login providers are approved?	LOW	4.1.2	Learner; FLOW 3	Any provider shown is a clearly labelled sample.	Project Manager	Neither

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QP-20	Are assessment hints in scope at all, and if so what are their scoring and limit rules?	MEDIUM	No FRD requirement defines hints. PM-08 confirms configurable assessment behaviour and does not mention hints	Faculty, Learner; FLOW 4	Hints are treated as OPTIONAL and unapproved. No hint behaviour, hint scoring or hint limit is assumed in either direction.	Project Manager with academic stakeholder	Neither
QP-21	Is there a platform-level default assessment configuration?	LOW	PM-08	Faculty; FLOW 4	**None is invented.**	Project Manager	Neither
QP-22	May an assessment configuration change between attempts?	LOW	PM-08, 11.2.2	Faculty, Learner; FLOW 4	Not addressed. Affects fairness and the meaning of attempt history.	Project Manager with academic stakeholder	Neither
QP-23	Is there a real help destination behind the consistent help entry?	LOW	Not in the FRD	All seven	The location is consistent; the destination is unconfirmed.	Project Manager	Neither
QP-24	Is the structural certification node modelled even though the experience is deferred?	LOW	3.3.1; PM-12	Content model; no Phase-1 workflow	**Not assumed either way.** PM-12 defers the experience, not necessarily the data model.	Project Manager	Neither
QP-25	For a pure B2C individual learner, which Faculty, institution or content owner supplies the assessment?	LOW	PM-15	Faculty, Learner; FLOW 4	Explicitly deferred by PM-15 as a content and implementation detail, and explicitly not a blocker. Whoever supplies it acts on the authorized side and produces the same configuration.	Project Manager with content operations	Neither
QP-27	What post-attempt feedback may a learner see, and is answer-key visibility ever permitted?	HIGH	5.2.1–5.2.3, 11.2.1; no requirement states a feedback policy	Faculty, Learner; FLOW 4	No feedback policy is asserted in either direction. Learner-facing applications receive only the assessment data the configured and approved policy permits, and that policy is undefined.	Project Manager with academic stakeholder	Blocks development of the result surface and the assessment API payload; does not block design
QP-28	Is direct course purchase in scope, given that Phase-1 purchase requirements name only modules and bundles?	MEDIUM	6.1.1, 6.1.2 against 3.2.1	Learner, Super Admin; FLOW 3	Not specified and not included in the current design. It is neither required nor prohibited, and no course-purchase surface is drawn.	Project Manager with commercial owner	Neither
QP-26	May Super Admin also return a submission at the publication gate?	LOW	1.4.4; PM-14 names Academic Reviewer and Content Manager only	Super Admin; FLOW 1	**Not invented.** No return capability is granted at the publication gate.	Project Manager	Neither

B.2 Technical questions

Table 131 — Technical open questions

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QT-01	Which XR delivery mechanism is used?	HIGH	2.1.1, 2.2.1, 2.2.2, 2.3.x; PM-05	Learner, Super Admin; FLOW 5	The delivery contract is mechanism-neutral by design and survives any outcome. **Pending by explicit instruction.**	Team Lead with Project Manager	Blocks development of delivery; does not block design
QT-02	There is no AR application architecture in Platform FRD_v4. What is it?	HIGH	1.2.4 against Section 2, which covers VR and MR only	Learner; FLOW 5	Flagged, not filled. **AR is explicitly not assumed to behave like VR.**	Team Lead	Blocks development of AR delivery; does not block design of the other formats

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QT-03	What is the assessment configuration API contract?	HIGH	PM-08	Faculty, Learner, external applications; FLOW 4	Abstract. One constraint is recorded now: a learner-facing payload must carry only the assessment data the approved feedback policy permits, and that policy is itself open (QP-27).	Team Lead	Blocks development of application consumption; does not block design
QT-04	How does an SSO-authenticated learner sign into the XR application, which requires LMS credentials?	HIGH	2.4.4 against 4.2.2 and PM-10 — both Phase 1	Learner, Institute Admin; FLOW 5, FLOW 6	Recorded as a collision. **Not designed around.**	Team Lead with Project Manager	Blocks development of institutional device sign-in; does not block design
QT-05	What is the XR package and submission format?	MEDIUM	2.2.1 states only “encrypted asset bundles”	Vendor, Content Manager; FLOW 1, FLOW 5	The upload surface is specified generically and the format is marked pending.	Team Lead	Blocks development of submission handling; does not block design
QT-06	Do web simulations use the same encrypted-bundle and licence-validation path as XR formats?	MEDIUM	1.2.1 against 2.2.1 and 2.2.2	Learner; FLOW 5	Unstated. Affects load time, pre-launch experience and whether the execution-time gate applies in-browser.	Team Lead	Neither
QT-07	What is the completion reconciliation behaviour when a completion is never recorded?	MEDIUM	11.1.1; PM-05	Learner, Faculty; FLOW 5	Retry and refresh are offered. **No manual override**, because an override would defeat the progression gate for everyone.	Team Lead	Blocks development of reconciliation; does not block design
QT-08	How quickly does a device completion reach the portal?	MEDIUM	11.1.1, 12.2.1	Learner; FLOW 5	Read-on-navigation plus an explicit refresh. Cheap to remove if a push mechanism exists.	Team Lead	Neither
QT-09	Is the assessment timer server-authoritative, and how often are answers persisted?	MEDIUM	PM-08, 5.2.x	Learner; FLOW 4	Server-authoritative is assumed for design purposes and is unconfirmed.	Team Lead	Blocks development of the attempt runtime; does not block design
QT-10	How is video resource coverage measured?	MEDIUM	3.1.2	Learner; FLOW 2, FLOW 3	Coverage rather than furthest playhead, because furthest playhead defeats the completion gate.	Team Lead	Neither
QT-11	Which payment gateway, and redirect or embedded checkout?	MEDIUM	9.1.1, 9.1.2	Learner; FLOW 3	A generic secure redirect is assumed for design purposes. **No provider is named.**	Commercial owner with Team Lead	Blocks development of payment; does not block design
QT-12	What is the final technology stack and hosting platform?	MEDIUM	12.1.1 names providers as examples	All; platform-wide	No architectural statement in this document depends on the answer.	Team Lead	Blocks build; does not block design
QT-13	Are session-expiry durations per-tenant configurable?	MEDIUM	10.2.2	All seven	A single platform-wide behaviour is assumed.	Team Lead with Project Manager	Neither
QT-14	What are the password policy, multi-factor and lockout requirements?	LOW	10.2.1; absent from the FRD entirely	All seven; FLOW 6	A design-supplied minimum is assumed and is labelled as such. **No policy is invented as approved.**	Team Lead with Project Manager	Blocks development of authentication policy; does not block design
QT-15	Is a storage-constraint state technically reachable on target devices?	LOW	PM-05	Learner; FLOW 5	Held as a conditional state. Not designed until the delivery mechanism confirms it can occur.	Team Lead	Neither

B.3 Data / Integration questions

Table 132 — Data / Integration open questions

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QD-01	Which profile fields are user-editable and which are ERP-owned?	HIGH	4.3.1 against 7.1.1 and 7.2.1	Learner, Faculty, Institute Admin; FLOW 7	ERP-sourced fields are presented read-only with their provenance stated.	Team Lead with Institute stakeholder	Blocks development of profile editing; does not block design
QD-02	What is the Phase-1 origin of role assignment?	HIGH	7.4.2 and 8.3 are both Phase 3	All seven; FLOW 6	Super Admin assigns platform roles under 8.1.1. The institutional role-assignment origin is left open rather than invented.	Project Manager with Team Lead	Blocks development of role provisioning; does not block design
QD-03	Which ERP providers, and what is the integration contract?	MEDIUM	7.1.1, 7.1.2	Institute Admin; FLOW 7	Drawn generically. The systems named in the FRD are examples, not a selection.	Team Lead with Institute stakeholder	Blocks development of ERP integration; does not block design
QD-04	Which identity provider, protocol and attribute mapping for SSO?	MEDIUM	4.2.2, 7.4.1, 7.4.2; PM-10	Institute Admin, Learner, Faculty; FLOW 6	Drawn generically. **Pending by explicit instruction.**	Team Lead with Institute stakeholder	Blocks development of SSO; does not block design
QD-05	Can one person hold both a B2B and a B2C identity, and do they link?	MEDIUM	1.3.1, 1.3.2	Learner; FLOW 3, FLOW 6	One account per context, with an explicit collision state. No identity linking is invented.	Project Manager	Neither
QD-06	Is the learner informed when synchronisation adds or removes entitlements?	MEDIUM	7.2.1, 7.2.3	Learner, Institute Admin; FLOW 7	No Phase-1 notification requirement exists. In-workflow state is shown; no push notification is designed.	Project Manager	Neither
QD-07	What module metadata schema does the content readiness gate review?	MEDIUM	1.4.3	Content Manager, Vendor; FLOW 1	Never enumerated in the FRD. The review surface is specified; the schema is pending.	Team Lead with Content Manager	Blocks development of metadata validation; does not block design
QD-08	What curriculum standards does academic validation test against?	MEDIUM	1.4.2	Academic Reviewer; FLOW 1	Drawn as pending. **This is content someone must own**, not a design question.	Academic stakeholder	Blocks operation of the validation gate; does not block design
QD-09	What are the localization, currency and regional formatting requirements?	LOW	9.1.2 implies international reach; no internationalisation requirement exists	Learner; FLOW 3	A single language and currency with expansion tolerance in layout.	Project Manager	Neither
QD-10	What are the currency, tax treatment and price presentation rules?	LOW	9.1.2; 9.3.1 is Phase 2	Learner; FLOW 3	Neutral presentation. **No tax or legal rule is invented.**	Commercial and legal owner	Blocks development of pricing display; does not block design

B.4 Design questions

Table 133 — Design open questions

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QX-01	Does the desktop-first viewport decision extend to all six new workspaces?	HIGH	Derived; no FRD requirement	All seven	Not extended automatically. Administrative work is desktop-typical, but this must be decided rather than inherited.	Design lead with Project Manager	Blocks detailed design of the new workspaces
QX-02	How is the outstanding reflow non-conformance handled now that it spans all workspaces?	HIGH	Accessibility; no FRD requirement	All seven	Stated plainly as a known non-conformance; layouts are built so a responsive pass is possible without restructuring.	Design lead with Project Manager	Blocks an accessibility conformance claim; does not block design
QX-03	Who owns video captions, transcripts and audio description?	HIGH	3.1.2; 1.4.1 and 1.4.3 are candidate homes	Content Manager, Vendor, Learner; FLOW 1	Slots are specified; the content is unowned. **A video without captions must not be a required component for completion.**	Project Manager with academic stakeholder	Blocks accessible content delivery; does not block design

ID	Question	Priority	Affected requirement	Affected role / workflow	Current safe assumption	Owner	Blocks design, development or neither
QX-04	Is one queue component sufficient for the four governance roles?	MEDIUM	1.4.1–1.4.4	Vendor, Academic Reviewer, Content Manager, Super Admin; FLOW 1	One component with variants.	Design lead	Neither
QX-05	Is one configuration-form pattern sufficient for SSO, ERP, assessment and platform settings?	MEDIUM	4.2.2, 7.2.1, PM-08, 8.1.1	Institute Admin, Faculty, Super Admin	One pattern, validated during detailed design.	Design lead	Neither
QX-06	Does the shared shell need per-workspace visual differentiation?	MEDIUM	PM-03	All seven	Navigation content only. Differentiated chrome fragments the platform into apparent separate products.	Design lead	Neither
QX-07	Is a vector logo and a reversed variant available?	MEDIUM	Branding	All seven	Outstanding. It constrains the header lockup and any dark presentation.	Project Manager	Blocks final visual design; does not block structure
QX-08	How is gate position best expressed visually across the governance chain?	MEDIUM	1.4.1–1.4.4; PM-14	Four governance roles; FLOW 1	A status component with four positions plus the returned state.	Design lead	Neither
QX-10	Is a single sign-in entry point for all roles the approved model, or are separate entry points or a pre-login role choice acceptable?	MEDIUM	10.2.1; PM-03 confirms one central platform with role-based access and does not specify the sign-in model	All seven; FLOW 6	One sign-in entry point with role resolved from the authenticated identity is a design position, not a confirmed requirement. No source requires or forbids any alternative.	Design lead with Project Manager	Neither
QX-09	Should a B2C learner see any indication that assessment behaviour differs from an institutional learner's?	LOW	PM-15	Learner; FLOW 4	No difference is asserted. PM-15 makes ownership identical on both sides.	Design lead with Project Manager	Neither

B.5 Future-phase items

Recorded so that their absence from Phase 1 is understood as scope rather than oversight.

Table 134 — Future-phase items

ID	Item	Requirement	Note
QF-01	Certification programs and completion criteria	3.3.1, 3.3.2	PM-deferred to Phase 2 — **the FRD places these in Phase 1**
QF-02	Certification progress tracking and eligibility reports	3.5.1, 3.5.2	Phase 3 in the original FRD
QF-03	In-platform institutional course creation and module mapping	3.4.1, 3.4.2	Phase 2 — creates tension T-1 with Phase-1 requirement 8.1.3
QF-04	Class-level performance reports	11.3.1	Phase 2 — creates tension T-2 with Phase-1 requirement 8.1.4
QF-05	Institutional adoption reports	11.3.2	Phase 2 in the original FRD
QF-06	Institutional usage dashboard and licence utilisation tracking	8.2.1, 8.2.2	Phase 2 in the original FRD
QF-07	XR device registration and metadata tracking	2.5.1, 2.5.2	Phase 2 — keeps device screens out of Phase 1 administrative workspaces
QF-08	Remote device configuration and application updates	2.6.1, 2.6.2	Phase 2 in the original FRD
QF-09	Module lifecycle states and version management	1.5.1, 1.5.2, 1.6.1, 1.6.2	Phase 2 — Phase-1 governance runs on gate position instead
QF-10	Bundle creation, pricing configuration and coupons	6.3.1, 6.3.2, 6.4.1, 6.4.2	Phase 2 — creates the Phase-1 question recorded as QP-09
QF-11	Institutional invoicing and subscription billing	9.3.1, 9.3.2	Phase 2 — creates the Phase-1 question recorded as QP-04

ID	Item	Requirement	Note
QF-12	Real-time ERP provisioning at first login	7.3.1, 7.3.2	Phase 2 in the original FRD
QF-13	XR simulation performance scoring and attempt tracking	5.4.1, 5.4.2	Phase 2 in the original FRD
QF-14	Custom role creation and permission-level access control	8.3.1, 8.3.2	Phase 3 — **the reason the seven-role set must be fixed in Phase 1**
QF-15	SSO role mapping from identity-provider attributes	7.4.2	Phase 3 — creates the question recorded as QD-02
QF-16	AI learning assistant for learners	5.5.1, 5.5.2	Phase 3 — **distinct from requirement 5.1.2, which is Phase 1**
QF-17	Recommendation engine	4.5.1, 4.5.2	Phase 2 — the only consumer of the Phase-1 learning-preferences field
QF-18	XR telemetry, heatmaps and decision-path analysis	2.7.1, 2.7.2, 11.4.1, 11.4.2	Phase 3 in the original FRD
QF-19	Data protection compliance and retention policies	10.4.1, 10.4.2	Phase 2 in the original FRD

Appendix C. Verification of This Draft

The checks below were run mechanically against the requirement register that generates this document, not by manual counting. They verify internal completeness and consistency; they do not verify the platform, which is not implemented.

Table 135 — Verification results

Check	Expected	Result
Every Platform FRD_v4 requirement identifier is present	131	131 — PASS
No duplicate requirement identifier	0 duplicates	0 — PASS
Original Phase-1 requirement count preserved	78	78 — PASS
Original Phase-2 requirement count preserved	33	33 — PASS
Original Phase-3 requirement count preserved	20	20 — PASS
PM-deferred requirements out of the Phase-1 working set	2 (3.3.1, 3.3.2)	2 — PASS
Phase-1 working set (76 from the FRD plus requirement 7.4.1)	77	77 — PASS
All seven confirmed roles have a role chapter	7	7 — PASS
PM-01 through PM-15 are all recorded	15	15 — PASS
Added requirements carry V5-ADD identifiers	all	10 — PASS
Clarified requirements carry V5-CLR identifiers	all	12 — PASS
No Review-005 decision identifier is reused as a requirement identifier	0	0 — PASS
Questions closed by stakeholder decision are not shown as open	0 reopened	0 — PASS
Open questions carried in the register	59	59 — PASS
High-priority open questions	12	12 — PASS
Future-phase items recorded	19	19 — PASS
Resolved stakeholder questions recorded	19	19 — PASS
Dependencies recorded with owner, status and impact	16	16 — PASS
Acceptance criteria recorded	44	44 — PASS
Architecture and workflow diagrams embedded in this document	12	24 — **FAIL**
Diagram image files present on disk (12 master + 3 compact-edition)	15	15 — PASS
Every requirement carries an acceptance statement	131	131 — PASS
Every requirement carries a classification label	131	131 — PASS

What these checks do not establish

- They do not establish that this draft is approved. It is not.
- They do not establish that any implementation test has passed. No implementation exists.
- They do not establish that the open questions have been answered. Fifty-nine remain open by design.